

Woodford-Barnett Resource Potential Study Final Report

Detailed Analysis of Source Rock and Oil samples, GIS Database and
Regional Interpretation of the Resource Potential of the
Woodford-Barnett Formations of the Midland Basin

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Final Report
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Introduction

Introduction

This report documents and characterizes the hydrocarbon potential of sediments in the Woodford-Barnett (Devonian-Mississippian) stratigraphy of the Midland Basin/Central Basin Platform (CBP) area. It builds on the findings from two earlier works by GeoMark which include a study of the Woodford sediments and oils and an oil study including over 150 oils from all sources in the larger, overall Permian Basin (PB).

Although the Woodford has long been recognized as a source for both conventional and unconventional exploration targets in the Permian Basin, the role of the Barnett has been less well understood. However, its source potential in the Fort Worth Basin for both oil and gas has been well recognized and documented. Successful, recent unconventional exploration of the Barnett, notably on the Central Basin Platform (CBP), has begun to make it an exploration focus.

The current study is more detailed than our earlier work. Although limited geographically to the Midland Basin and CBP, it is expanded stratigraphically to include the Upper and Lower Barnett Formations and also the Mississippi Lime which separates the Woodford and Barnett Formations, and, of course, the Woodford itself. The same formations in the Delaware Basin will be worked in a separate study that is underway. Due to the importance of Ward and Winkler Counties in the current unconventional exploration of these formations, all wells in both counties will be included and integrated with the data in both the Midland and Delaware Basin studies.

Introduction

Many more additional samples and more detailed sediment work with total organic carbon (TOC) and Rock-Eval™ (RE) pyrolysis performed pre and post sediment extraction are included in this study. Also included are characterizations of several whole extracts by gas chromatography, extract SARA separation and subsequent analysis of saturate and aromatic fractions by gas chromatography-mass spectrometry (gc-ms) and carbon isotopic analysis. Finally, detailed oil characterizations and where possible, their direct correlation with sediment extracts, are included.

The report begins with a summary of conclusions followed by a description of the study sediments and oils database. Following that is the stratigraphic and mapping approach. The sediments start with a discussion of the organic richness, kerogen quality, hydrocarbon potential and thermal maturity as described by total organic carbon (TOC), Rock-Eval™ (RE) and maturity parameters. The sediment maps follow that discussion, and following the maps, is their interpretation. Finally, the oils and extracts are discussed followed by interpretation/integration with sediments.

Conclusions

The Woodford and the Lower Barnett have good to excellent unconventional source potential on the basis of organic richness, kerogen quality, and thermal maturity.

The Upper Barnett and Mississippi Lime have little to no source potential and would likely not make good unconventional, self sourcing targets.

The areas of highest potential for the Lower Barnett and Woodford are:

- Central to eastern Andrews/Ector counties possibly extending south and north into Crane and Gaines counties, respectively.
- Martin/Midland/Upton counties, however, thermal maturity and therefore hydrocarbon phase is a consideration as maturity is more elevated than on the CBP.
- Ward/Winkler county areas on the CBP and to the west towards the Delaware Basin, however, the area from immature/low mature to overmature may be quite limited due to the rate that burial depth increases east to west. Maturity must be carefully evaluated.

The possibility of immature/low mature targets on the CBP and in the northern Midland may be improved by migration of more mature hydrocarbons from the deeper more mature basin.

Study Deliverables

The deliverables for the study include:

- Interpretive Woodford-Barnett Report in Adobe Acrobat™ (this document)
- Interpretive report on Oils in the Permian Basin in Acrobat™
- ArcGIS ARCMAP™ project
- Folio of GIS based Maps (hard copy and Acrobat™) plus ESRI ARCMAP™ shapefiles and raster images for the folio maps
- 3-year subscription to raw sediment and oil data for study samples through GeoMark's online database (rfdbase.com)
- Interpretive sediment and oil data (in Excel™)

Woodford-Barnett Study Database

Sediments

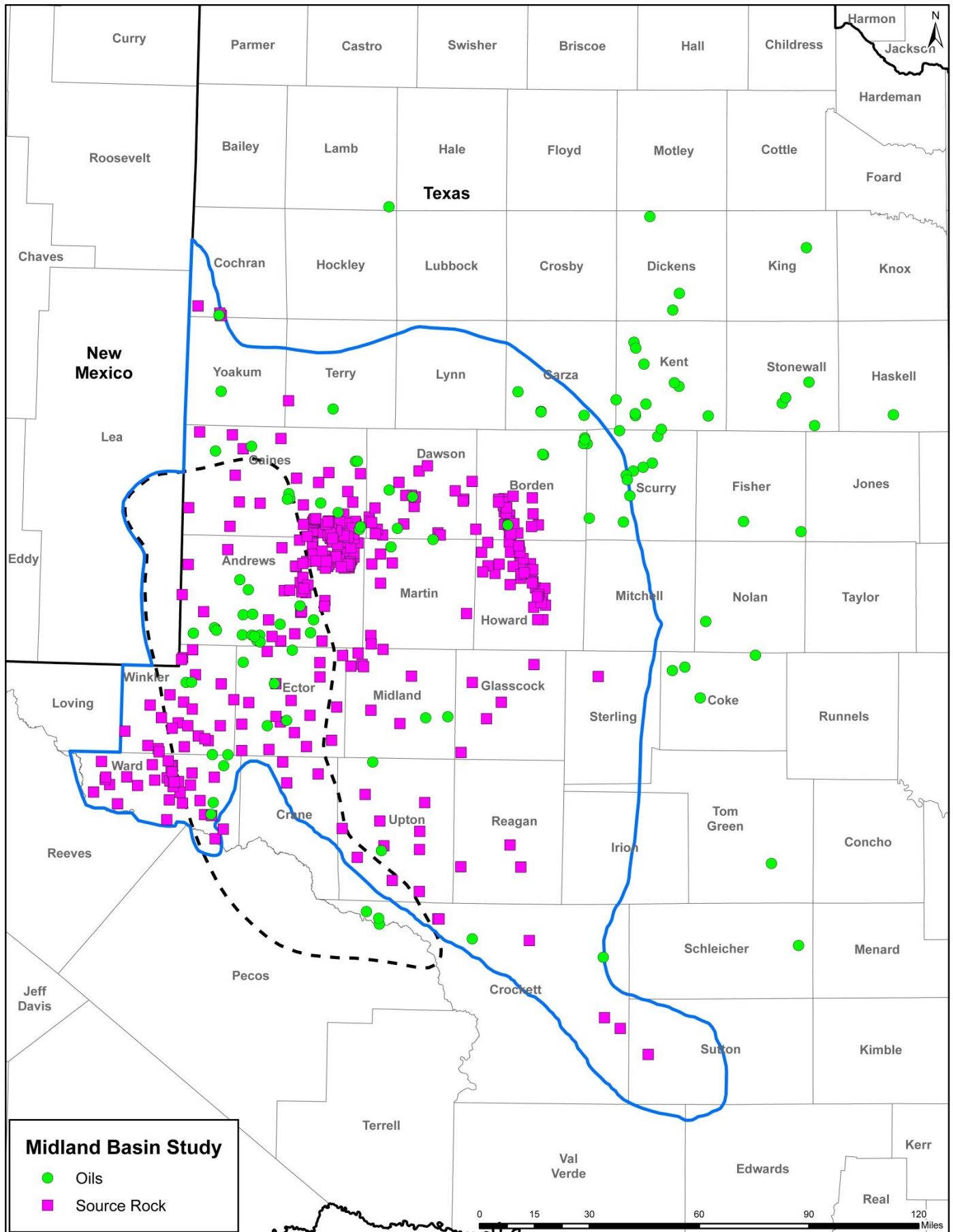
- 2200+ Sediment Samples for TOC
- 1700+ Sediment Samples for RE
- 700+ Extracted and Rerun for TOC/RE
- 325 Total Extract Gas Chromatography
- 200 Biomarker and Saturate/Aromatic Isotopes

Oils

- >100 Oils from Vertical Completions
- Whole-Oil Chromatography
- SARA
- Saturate and Aromatic Fraction Isotopes
- Saturate and Aromatic Fractions GC-MS (biomarkers)
- Multivariate Statistics

See database map on slide 9

Woodford-Barnett Study Database



Stratigraphy and Mapping

Stratigraphy and Mapping

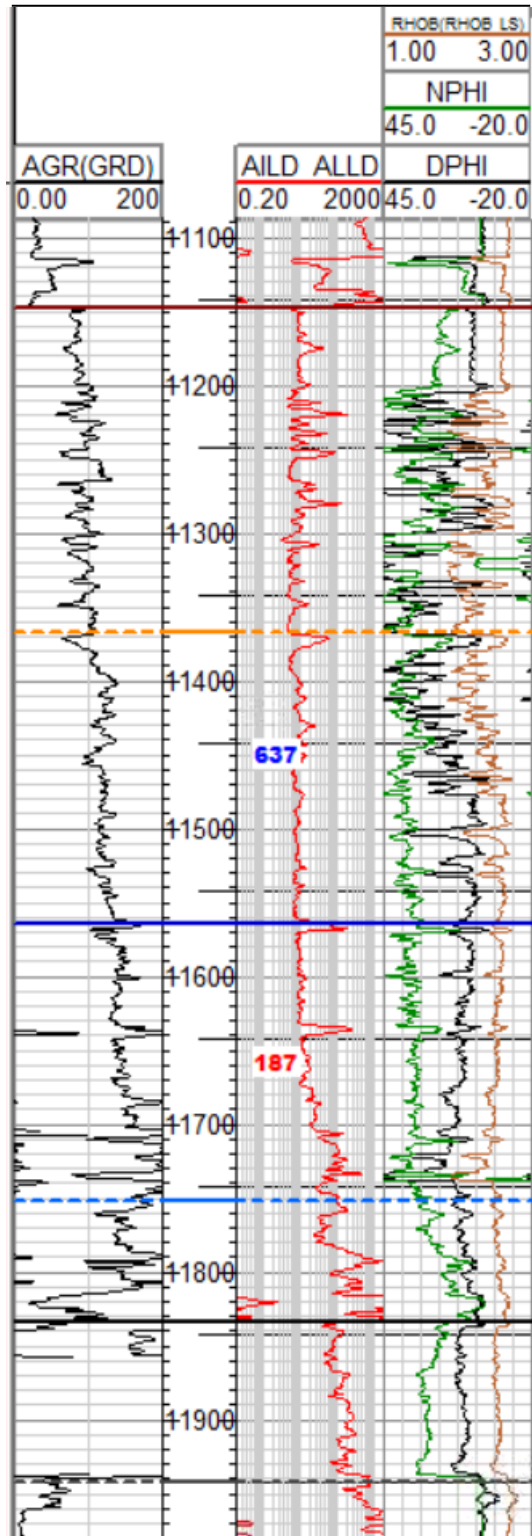
With the extension of this study up to the younger Mississippian Barnett Formation, a stratigraphic correlation was done using primarily gamma ray, sonic/neutron and resistivity logs to insure consistent stratigraphic picks for the four study horizons were made across the basin. Slide 11 shows a set of type logs with the picks and the table in slides 13-14 shows the tops for the wells.

Based on the data in the table, present day top depth and isopach maps for the Woodford, Mississippi Lime, Lower and Upper Barnett are shown in slides 22-23, 30-31, 37-38, 44-45, respectively. Although the Central Basin Platform is shown for reference on all the maps, it is important to remember that all the study formations were deposited in the paleo Tobosa Basin, prior to the uplift that created the CBP and differentiated the Midland and Delaware Basins. As seen from the isopachs, the horizon thicks generally occur on the CBP and to the western part of the AOI, notably in Ward/Winkler counties. This is at least part of the reason for the significant exploration focus for this area.

Woodford-Barnett Stratigraphy

Midland Basin

Bradford 1-32
42329312430000



Top Barnett

Lower Barnett

Mississippi Lime

Woodford

Base Woodford

Woodford-Barnett Maps

Mapped Parameters

- Average TOCpd Unextracted
 - Average TOCpd Extracted
- Maximum TOCpd Unextracted
 - Maximum TOCpd Extracted
 - Hlpd Unextracted
 - Hlpd Extracted
- %VRET(Tmax) Unextracted
 - %VRET(Tmax) Extracted
- %Measured Ro Woodford Formation only
 - %Carbonate Unextracted

Mapped Formations

- Woodford
- Mississippi Lime
 - Lower Barnett
 - Upper Barnett

Study Tops

API	Well	County	Latitude	Longitude	Top Upper Barnett	Top Lower Barnett	Top Miss Lime (Base Barnett)	Top Wood (Base Miss Lime)	Top Devon (Base Woodford)
42003301340000	Butler 1	Andrews	32.520800	-102.360690	11433	11800	11997	12436	12560
42003025440000	Fasken David #1K	Andrews	32.1126108	-102.4011887	10217	10702	10903	11169	11250
42003005360000	Fasken Est 1	Andrews	32.274150	-102.389910	11396	11900	12110	12430	12510
42003107750000	Fasken Op Area #1-C	Andrews	32.2504943	-102.3924488	11562	11818	12022	12338	12383
42033006960000	GB Jones Etal 1	Andrews	32.28523	-103.05771				8768	9203
42003017230000	Oldham Laura C 1	Andrews	32.219270	-102.955520	8553	8704	8740	9195	9632
42135037900000	Univ AA-1 1	Andrews	32.46313	-102.25883					9415
42003300900000	University 5-3	Andrews	32.429813	-102.387487	11403	11770	11981	12410	12490
42003100400000	University 8-2 2	Andrews	32.409229	-102.318028	11492	11811	12029	12408	12500
42003102260000	University E1	Andrews	32.419310	-102.285060	11330	11668	11858	12232	12322
42003070690000	University U #1	Andrews	32.225	-102.50298	10101	10435	10598	11056	11115
42003001750000	WF Cowden 8	Andrews	32.127168	-102.6344218	9180	9394	9627	10044	10272
42003009410000	Clayton A 1	Borden	32.674420	-101.585900	9067	9189	9300	9620	9667
42003016680000	Clayton-Johnson 1	Borden	32.693190	-101.426840	8517	8647	8767	8968	9005
42003300910000	TJ Good 1	Borden	32.548790	-101.526120	9196	9429	9523	9741	9840
42165032730000	Walker Paul 1	Cochran	33.43166	-103.01615				11632	11700
42003021840000	Waddell W N 926	Crane	31.5847768	-102.407311	8953	9299	9517	9547	9681
42033100230000	McMullan AH 1	Crockett	30.631350	-101.073120	NP	NP	NP	8927	8955
42003304620000	Schlinke Ruth 1	Crockett	31.016519	-101.837667	9581	9714	9837	9927	10020
42165011570000	Te Glasscock 1	Crockett	30.5887605	-101.0005753				8910	8950
42003071850000	Boatright 1	Dawson	32.815110	-101.921820	10572	10795	10991	11290	11350
42003100280000	Kent CSL 1	Dawson	32.542300	-102.160780	11200	11642	11754	12110	12240
42003012820000	Krezer 2	Dawson	32.540840	-101.856500	10523	10934	11011	11339	11435
42003077010000	Moore Rc 1	Dawson	32.691098	-101.976163	10743	11095	11245	11618	11677
42003102650000	RA Carr 1	Dawson	32.6975411	-101.9946261	10881	11273	11488	NR	
42003024500000	Smith 1	Dawson	32.755620	-101.710900	9646	9840	9928	10250	10260
42003010040000	United Presbyn-Byrd 1	Dawson	32.696148	-102.020988	10887	11226	11406	11750	11830
42003102350000	Addis Frank V 1	Ector	31.816640	-102.449990	10766	11166	11496	11700	11810
42003052500000	Barrow Augusta JVDA 22-1	Ector	31.970306	-102.407451	10944	11388	11629	11765	11850
42003071000000	Edward Ranch 32-1	Ector	31.692502	-102.464196	10176	10575	10895	10982	11110
42003046340000	Fasken '16' 1	Ector	32.057152	-102.303822	11158	11638	11857	12048	12120
42003317640000	JC Connell-Trahan 1	Ector	31.677800	-102.637580	6530	6708	6976	7257	7455
42003041640000	JE Parker B 1	Ector	31.809800	-102.608200	9281	9525	9773	10078	10314
42003045620000	Superior-Ratliff 1	Ector	32.027323	-102.409036	10456	10991	11192	11435	11530
42003072440000	WF Cowden 2	Ector	31.874920	-102.539960	10074	10414	10771	11020	11127
42227316630000	DT Boyd 1	Gaines	32.6662705	-102.2686462					12507
42003381830000	Garland A 1	Gaines	32.652190	-102.500440	11605	11956	12182	12675	12765
42033001960000	Kay Kimbell 1	Gaines	32.7419834	-102.4307321	12033	12495	12738	13192	13280
42003001900000	Mrs WB Dunn 1	Gaines	32.782330	-102.383530	11900	12395	12636	13101	13205
42227002030000	Ms Doss A 1	Gaines	32.6547724	-102.703011					12003
42003044410000	Proctor 9	Gaines	32.763730	-102.822510	9585	10235	10449	11006	11064
42003025310000	Books E C 1	Glasscock	31.813972	-101.629981	10188	10516	10602	10680	10720
42003070870000	Brunson Glen 1	Glasscock	31.957010	-101.697910	10124	10410	10520	10638	10690
42003009970000	Bryans M V `A` 7	Glasscock	31.678833	-101.746613	10716	11095	11234	11250	11320
42003103690000	Glass 216 1	Glasscock	32.031600	-101.412200	9539	9592	9597	9680	9740
42003108820000	Gist SO A 1	Howard	32.397360	-101.655570	9819	10112	10154	10414	10521
42003017930000	TXL D 1	Howard	32.209920	-101.396700	9196	9326	9403	9567	9640
42003045500000	Breedlove `A` 1	Martin	32.462143	-102.135054	11176	11631	11692	12010	12100
42003104490000	Courtney Cowden 1	Martin	32.104819	-102.171349	10896	11360	11555	11710	11783
42003002970000	Mabee Ranch 318-2	Martin	32.346227	-102.132706	11456	11815	11977	12250	12330
42003020680000	Bradford 32-1	Midland	31.790285	-102.031924	11115	11565	11752	11858	11945
42003061150000	Butler 5-22	Midland	32.022969	-102.212327	11164	11703	11893	12003	12100
42003046320000	Casselmann 4-1	Midland	32.068023	-102.231697	11249	11800	12035	12178	12256
42003001280000	Getty Fasken 1-19	Midland	32.015324	-102.264473	11290	11863	12029	12190	12255
42003010070000	McKee Ruby 1	Midland	31.979435	-101.981583	10928	11283	11499	11600	11660
42003306420000	Roy Parks 2	Midland	31.841130	-102.167750	10638	11233	11403	11458	11561
42003065440000	Scharbauer 3-27	Midland	32.014342	-102.205342	11073	11626	11778	11875	11965
42003077020000	Scharbauer L-1	Midland	32.083810	-102.116020	10809	11242	11440	11565	11652

Study Tops

API	Well	County	Latitude	Longitude	Top Upper Barnett	Top Lower Barnett	Top Miss Lime (Base Barnett)	Top Wood (Base Miss Lime)	Top Devon (Base Woodford)
42003030770000	Pembrook 1-10	Reagan	31.226130	-101.463460	10026	10131	10231	10333	10372
42003037200000	University 'C' 1	Reagan	31.313933	-101.514887	10000	10196	10369	10386	10432
42495055170000	Gb Hill Etal 1	Sutton	30.4862756	-100.8716498					8913
42033003820000	Crown 1	Terry	33.063614	-102.579308	12070	12374	12573	13162	13209
42003103790000	JP Rankin 1	Upton	31.505980	-102.187990	10818	11244	11453	11460	11565
42003077750000	King 1	Upton	31.255320	-102.219960	8917	9306	9395	9397	9540
42003016690000	Rush 1	Upton	31.166430	-102.057430	8532	8786	9006	9011	9133
42033004440000	1 HTCHNGS-SEALY N B #1	Ward	31.6005952	-103.0896304	12702	13244	13635	14149	14644
42003108230000	708 JV-D PYOTE #1	Ward	31.54391	-103.09327	11834	12196	12406	12911	13444
42003304270000	7108 JV-S WARD #1Y	Ward	31.5483	-103.076735	11419	11692	11905	12388	12833
42033100540000	7202 JV-S SEALY #1	Ward	31.5798803	-103.0840596	12106	12606	12948	13444	13906
42003002000000	7203 JV-S AMOCO #1X	Ward	31.5425436	-103.0533768	11650	11863	12113	12658	13126
42495300460000	AF Bray E 1	Ward	31.35782	-102.84096				6875	7121
42033011100000	Bass-Williams 1	Ward	31.52347	-103.37249	15739	16131	16420	16675	17240
42003110720000	Caprito #1	Ward	31.5560213	-103.1067256	11690	12003	12267	12756	13228
42003007910000	Dobbins WT 1	Ward	31.4569791	-103.0326317	11343	11427	11585	12268	12701
42003024510000	EW Estes 166	Ward	31.469003	-102.951725	9780	9986	10047	10669	10974
42033300290000	Geo Sealy 56 #1	Ward	31.605950	-103.083405	12867	13522	13839	14417	14895
42033009040000	Greer-McGinleas Unit 1	Ward	31.61413	-103.41261	16986	17411	17680	18014	18602
42003304810000	GWO 319	Ward	31.577700	-102.991920	10538	10983	11266	11916	12460
42003107650000	GWO 880	Ward	31.681310	-103.030580	11561	11856	12178	12695	13350
42003050680000	Herd J H 1	Ward	31.50004	-103.0451	11058	11151	11269	11863	12328
42003071220000	Hill Unit #1	Ward	31.4690744	-103.0939718	10181	10382	10448	10931	11298
42033100010000	Hutchings-Sealy #1	Ward	31.5972874	-103.1144747	13286	13758	14188	14658	15250
42003026760000	JF York E 1	Ward	31.4138300	-102.9294300	8969	9069	9147	9777	10112
42003014560000	JV-S PYOTE `3-20` #3	Ward	31.5405982	-103.0733696	10884	11112	11384	11922	12423
42495301520000	Lynch 1	Ward	31.3190200	-102.8778600			6292	6866	7050
42003047430000	Monroe GU 1	Ward	31.552700	-103.391100	10358	10747	11108	11791	12168
42033008510000	P Walker Et Al 2	Ward	31.55601	-103.29363	15246	15521	15768	16049	16623
42033102410000	Sealy Geo Sec 66 #1	Ward	31.6237885	-103.1033081	13071	13671	14065	14551	15054
42003046240000	St. University DM 1A	Ward	31.529663	-102.997450	10221	10527	10682	11304	11716
42033006360000	University Pyote Un 1	Ward	31.54594	-103.16385	13763	13987	14102	14583	15044
42495107190000	Campbell Seth C 1	Winkler	31.85852	-103.03873		7830	7978	8626	
42003103130000	CORNELL HOGG ET AL #1	Winkler	31.7651868	-103.0169141	8224	8458	8608	9127	9669
42495302440000	EE Brown R/A E 14	Winkler	31.77547	-103.0616		7803	7855	8358	8888
42003012800000	Elizabeth Hardie C1	Winkler	31.9684675	-102.9867548	8326	8595	8723	9223	9798
42003025610000	Goff 1	Winkler	32.03463	-103.04846	10221	10508	10712	11248	11871
42003012590000	Hendrick Oper Area 1	Winkler	31.84321	-103.19385	10735	10985	11186	11611	12201
42003033350000	Hendrick Unit D 1-A	Winkler	31.7944453	-103.140142	7541	8082	8220	8694	9258
42033304760000	Hutchings Sealy 2	Winkler	31.6587551	-103.1470719	13400	13939	14438	14905	15483
42033006540000	Hutchings-Sealy 1	Winkler	31.67105	-103.15175	13429	14103	14548	14985	15547
42003101610000	Lineberry 1	Winkler	32.027490	-103.053425	9927	10166	10388	10960	11521
42003028180000	McCutcheon 1	Winkler	31.88654	-103.10323	7071	7180	7273	7546	8061
42003041600000	RM Evans 1	Winkler	32.067205	-103.002826	9606	9789	10128	10572	11183
42003021990000	SEALY-SMITH FDN A #22	Winkler	31.715041	-102.936301	8537	8790	8910	9576	10048
42003018360000	Tobe Morton 1	Winkler	31.753508	-103.089004	9219	9442	9619	10075	10503
42033005700000	University `39-21` 1	Winkler	31.6826585	-103.1995946	14997	15584	16016	16500	17111
42033100070000	University Unsec 11 1	Winkler	31.737	-103.30803	16066	16654	17088	17565	18257
42003026300000	Waddell 1	Winkler	31.9346207	-102.8662047	6931	7653	7867	8570	8960
42003019710000	WD Blue Est 1	Winkler	31.872190	-102.805640	7229	7395	7518	7928	8291
42003038470000	WS JASPER #1	Winkler	31.831716	-102.997985	8188	8299	8416	8999	9546

Sediment Richness: TOC, Kerogen Quality and Thermal Maturity

Assessing optimal locations for exploring unconventional targets requires integration of geochemical parameters pertaining to source richness, quality and thermal maturity. The overall present day (pd) richness is illustrated by TOC (wt.%) maps. In an attempt to mitigate issues of cuttings homogenization and dilution, both average TOC_{pd} and maximum TOC_{pd} are mapped.

Kerogen quality and hydrocarbon potential are commonly evaluated by the Rock-Eval™ hydrogen index (HI) parameter which is defined as the S2 peak normalized to TOC. Since this parameter decreases with hydrocarbon generation, its present day value is also a measure of thermal maturity. However, there are complications with the straight forward use of HI due to its co-dependence on kerogen type and thermal maturity. For example, comparison of HI_{pd} values across an AOI in an effort to evaluate kerogen quality can be misleading if there is also maturity variation across the area. Conversely, of course, the evaluation of thermal maturity using HI_{pd} also requires a fairly homogenous kerogen type. Variability of both parameters, kerogen quality and maturity, is generally inevitable to some extent, in regional work.

The original potential of a source or H_{lo}, is the best indicator of source quality. However, since H_{lo} drops with hydrocarbon generation, its true value can only be measured in immature / very low maturity sediments in the AOI and again, quality may vary within the AOI.

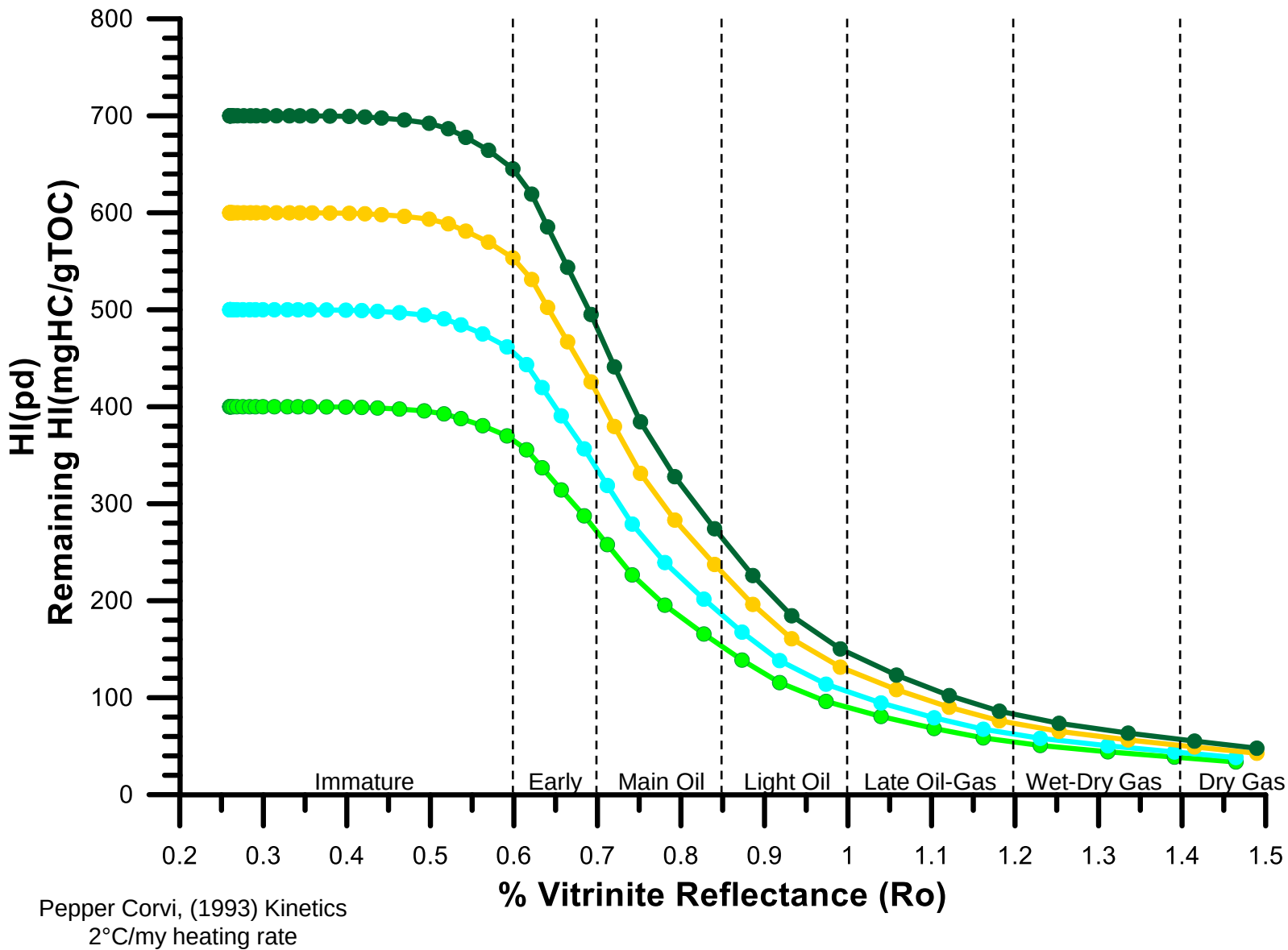
Sediment Richness: TOC, Kerogen Quality and Thermal Maturity

Oil prone sources generally have Hlo values that range from about 400-700 mg HC/gTOC. A good way to visualize the effect of maturity and kerogen quality on HI is shown in slide 17, a plot of Hl_{pd} vs %Ro for a family of marine oil prone kerogens ranging in initial quality (Hlo) from 400-700mgHC/gTOC. The plots clearly show the greatest variability with the various Hlo values occurs at early to oil window maturities from about 0.60% - 1.1%. At high maturities, low Hl_{pd} values, the kerogens become almost indistinguishable.

Clearly, the most effective use of Hl_{pd} as an exploration tool requires its interpretation in context with additional maturity parameters as used in this study (discussed below). Two sets of Hl_{pd} maps are made for this study. One is colored/contoured to represent source/kerogen quality and the other is colored/contoured to represent thermal maturity. Clearly interpretive assumptions about Hlo have been made to make the maturity contoured maps. These assumptions have been based on the use of additional maturity tools as well as the author's experience in this and other unconventional plays.

The table in slide 17 shows the HI assumptions made for each of the mapped formations for the given maturity ranges chosen for the study. The Hl_{pd} contour values have been varied on the study maps colored/contoured for maturity for the various formations, to match the values indicated for the maturity levels shown in slide 17.

Ro vs HI for Marine Shales (“Type II”)

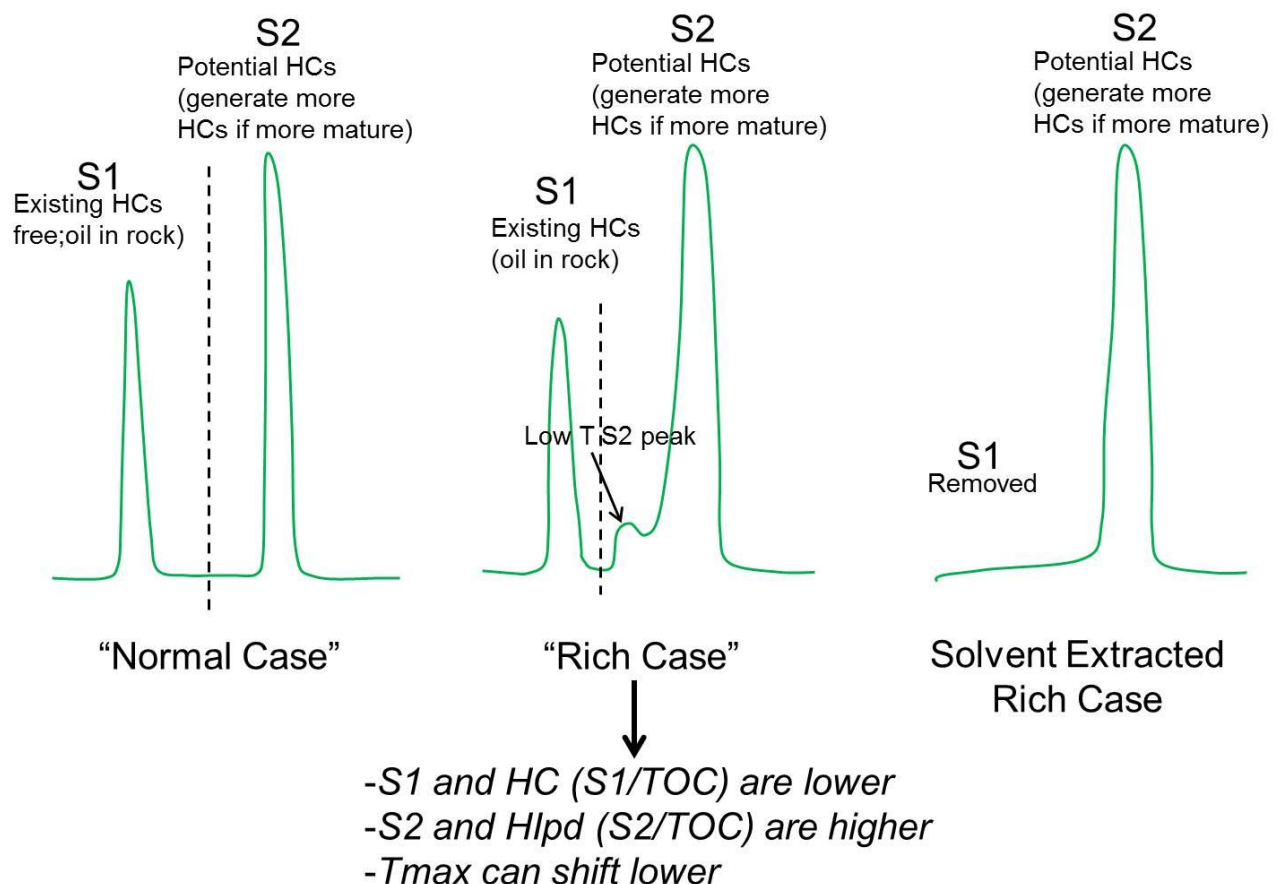


Ro Interval Mapped	<0.6	0.6-0.7	0.7-0.85	0.85-1.0	1.0-1.2	1.2-1.4	1.4+
Hydrocarbon Generation	Immature	Early Mature	Main Oil	Light Oil	Late Oil-Gas	Wet gas-Dry gas	Dry Gas
	HIpd	HIpd	HIpd	HIpd	HIpd	HIpd	HIpd
Upper Barnett (Hio=400)	>370	370-275	275-158	158-90	90-61	61-41	<41
Lower Barnett (Hio=600)	>555	555-418	418-230	230-130	130-75	75-50	<50
Mississippi Lime (Hio=400)	>370	370-275	275-158	158-90	90-61	61-41	<41
Woodford (Hio=700)	>645	645-480	480-262	262-145	145-75	75-60	<60

Sediment Richness: TOC, Kerogen Quality and Thermal Maturity

One other Hlpd interpretive complication is an analytical one caused by the presence of free, migrated and/or *insitu* generated hydrocarbons in a sample. Frequently, these hydrocarbons co-elute during the Rock-Eval™ analysis with the S₂ peak causing a broader, larger S₂ peak and, therefore, elevating the Hlpd value. Samples then appear less mature, with a higher Hlpd than really exists. TOC and Rock-Eval™ measurements on both unextracted and solvent extracted (“removing the free hydrocarbons”) samples are very helpful to understand the extent of this issue (see below). Maps showing unextracted and extracted Hlpd values are made for this study.

Unextracted vs. Extracted



Sediment Richness: TOC, Kerogen Quality and Thermal Maturity

As mentioned above, independent thermal maturity evaluation greatly helps with HI interpretation. For this study, two parameters, vitrinite reflectance equivalence %VRET, derived from the Rock-Eval Tmax value, and measured vitrinite reflectance (Woodford Formation only) are used.

%VRET is calculated, using the equation first published by (Jarvie et al., 2001). While kerogen type can affect the Tmax value, disregarding samples with anomalously high or low Tmax values and/or very low S₂ values, significantly increases the interpretive reliability of the %Roeq values. For this study, average values for %VRET (unextracted and extracted) are interpreted, calculated and mapped.

Measured vitrinite reflectance values for the Woodford Formation are also used in this study. Although measurements are not available for the younger formations, the Woodford vitrinite map represents a maximum (oldest formation) proxy for their maturity as the maturity gradient in the Midland Basin is not steep.

Finally, although not a typical geochemical parameter, %carbonate can often be a useful parameter to integrate with geochemistry to hi-grade exploration potential. Often higher carbonate values are associated with better fracking tendencies. However, some carbonate layers can also be frac barriers absorbing the energy and stopping fracture propagation. Maps of % carbonate are included in this study.

Sediment Richness: TOC, Kerogen Quality and Thermal Maturity

As follows in slides 24-26, 32-34, 39-41, and 46-48 are unextracted and extracted maps of the following parameters: average TOC_{pd}, TOC_{pd} Maximum, H_l_{pd}-contoured to indicate kerogen quality, H_l_{pd} contoured to indicate thermal maturity (see slide 17 for H_l_o assumptions, and %vitrinite reflectance equivalence calculated from T_{max} (%VRET).

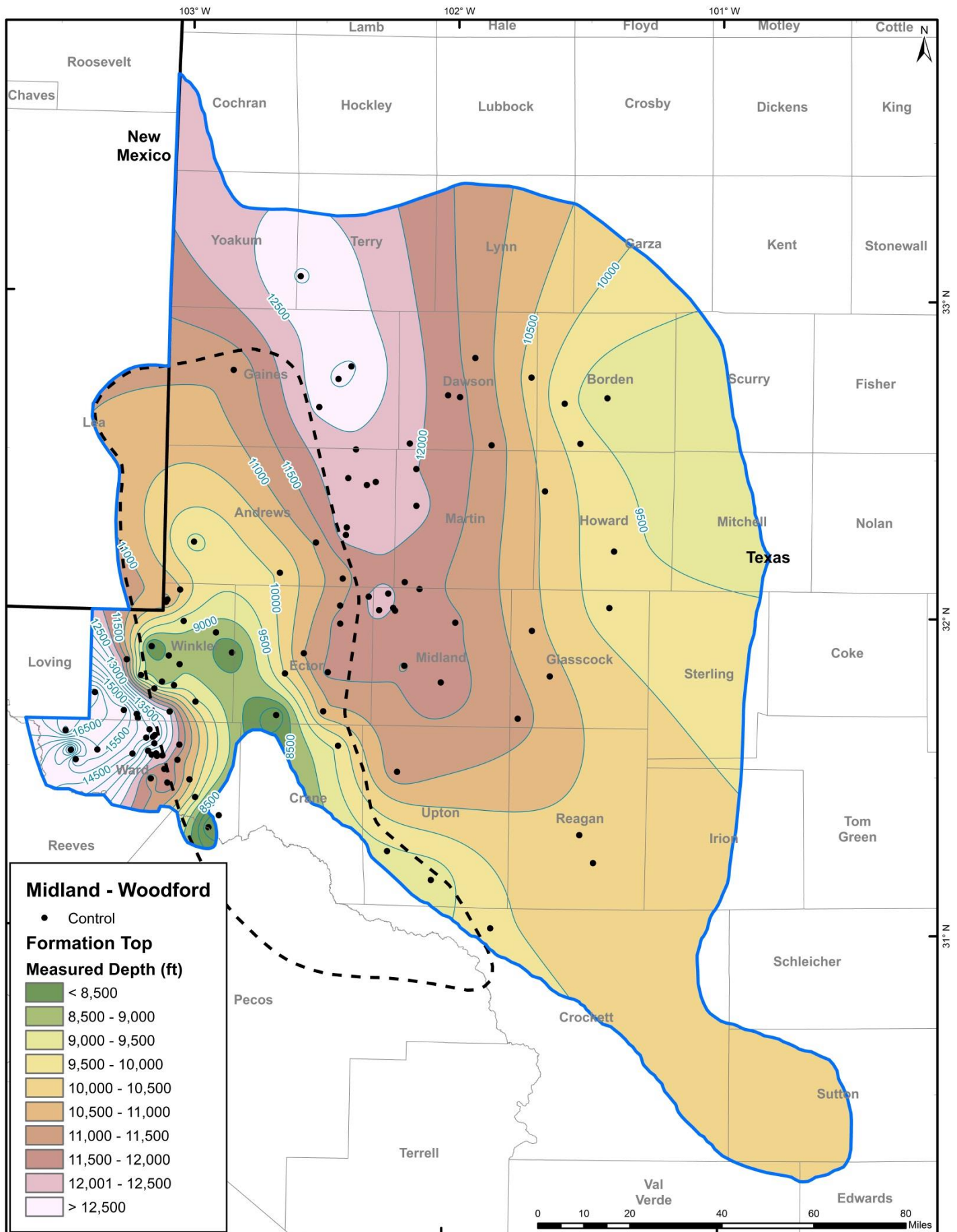
Slide 27 is the unextracted measured vitrinite reflectance (%R_o) for the Woodford formation which is the only formation that has measured vitrinite data. Slides 28, 35, 42, and 49 are the unextracted %carbonate maps for the Woodford, Mississippi Lime, Lower Barnett, and Upper Barnett, respectively.

Maps of the various parameters for the four horizons are shown starting with the Woodford, in deposition order. Each section of maps begins with the present day formation top depth map followed by its isopach and then the geochemical parameters.

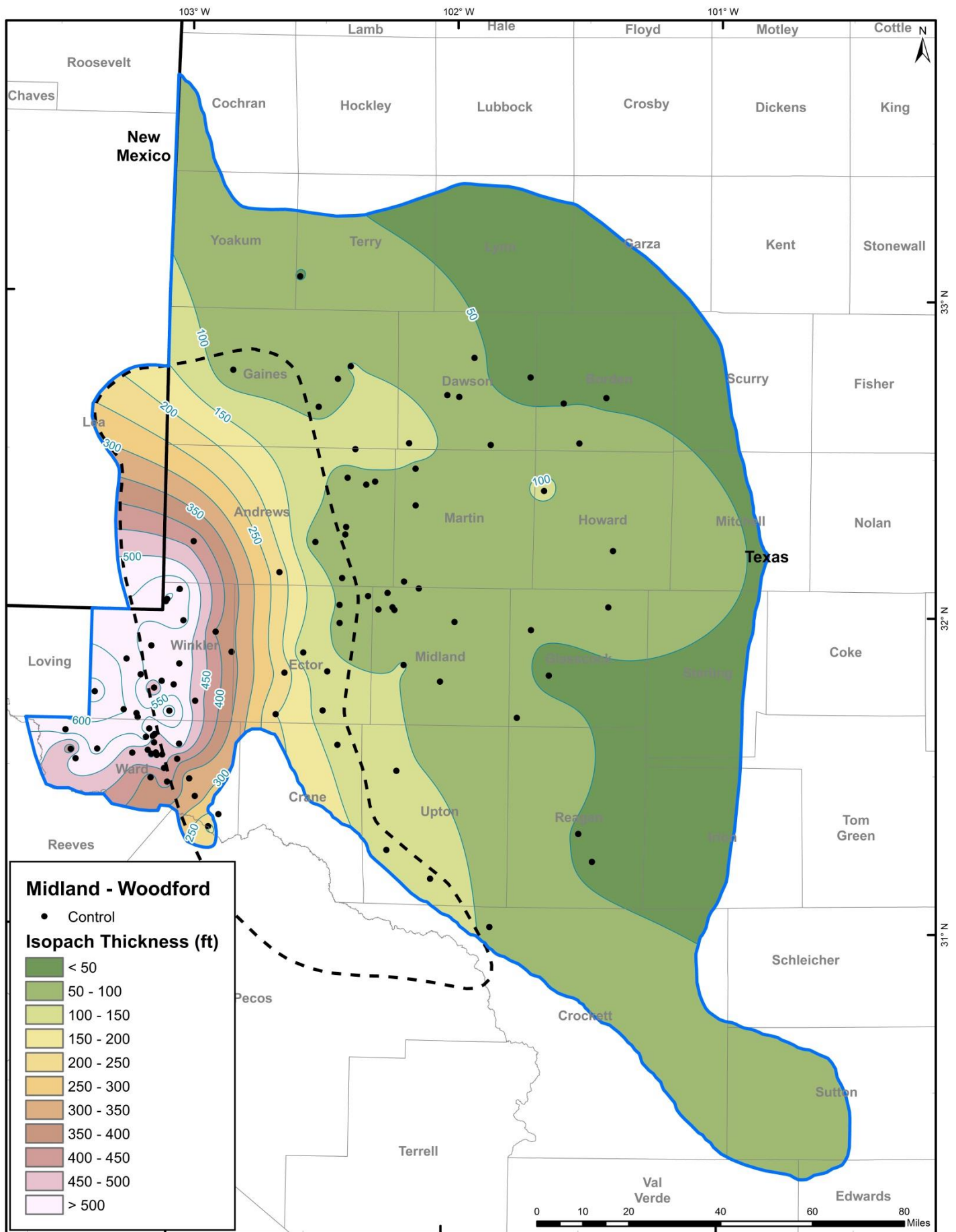
Also, as mentioned earlier, separate from this report, a map folio, in a larger physical format, allowing for maps to show more detail (e.g. control values) is also provided with this study.

Woodford Maps

Present Day Top Woodford

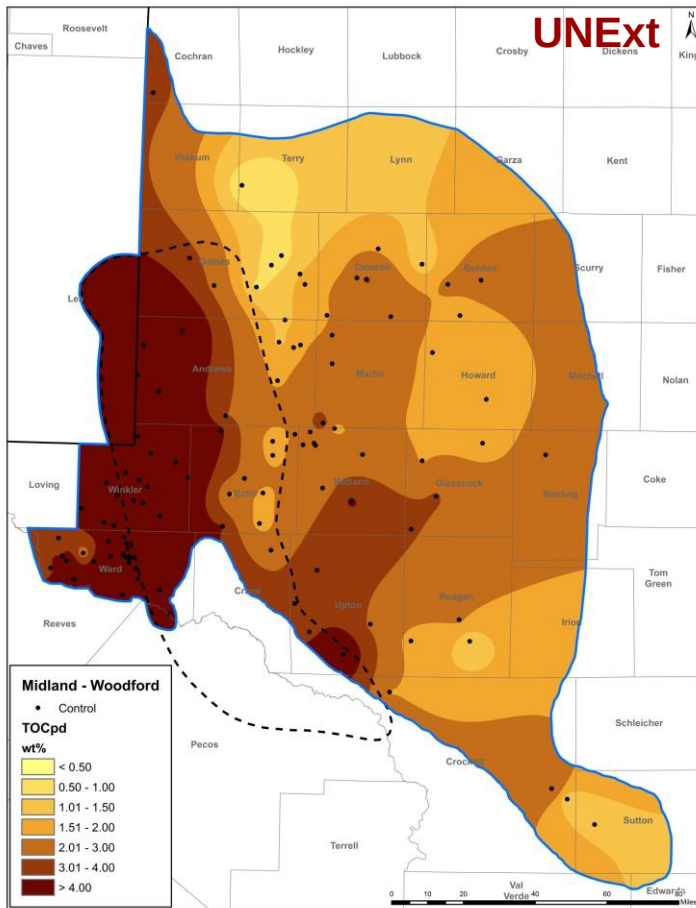


Woodford Isopach

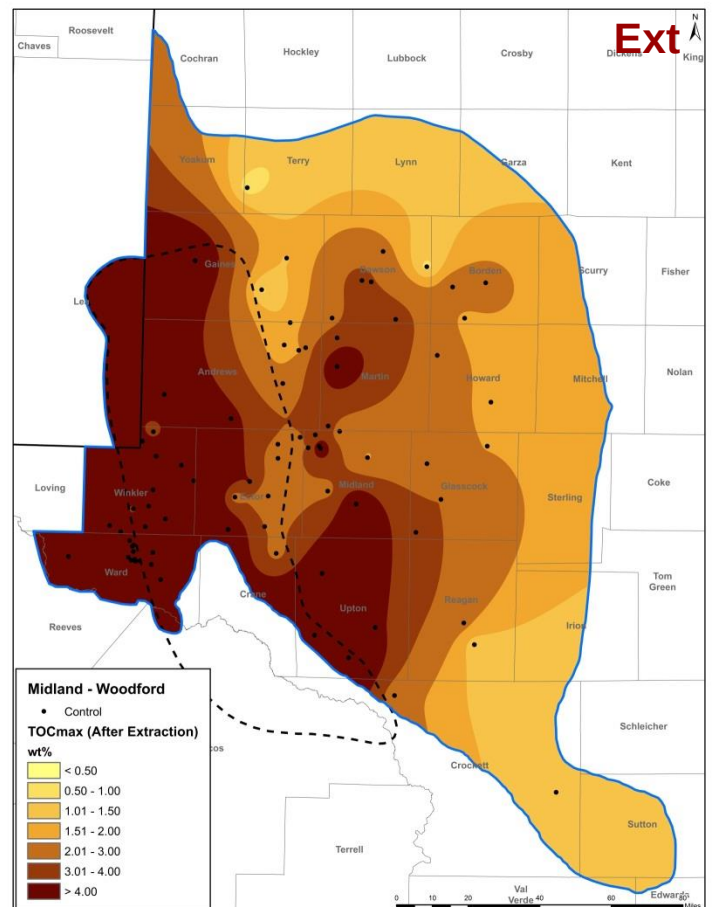
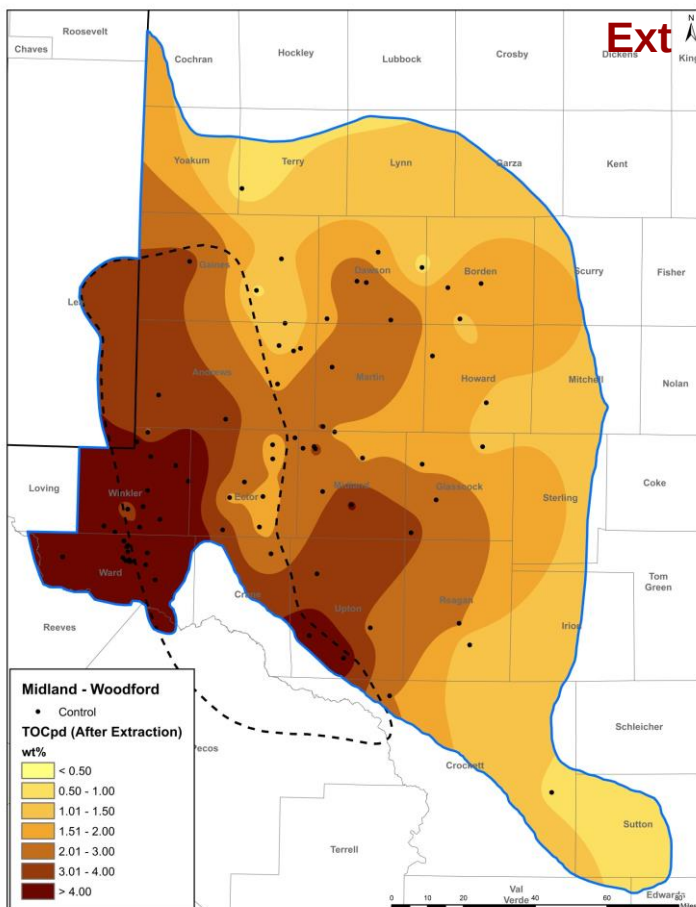
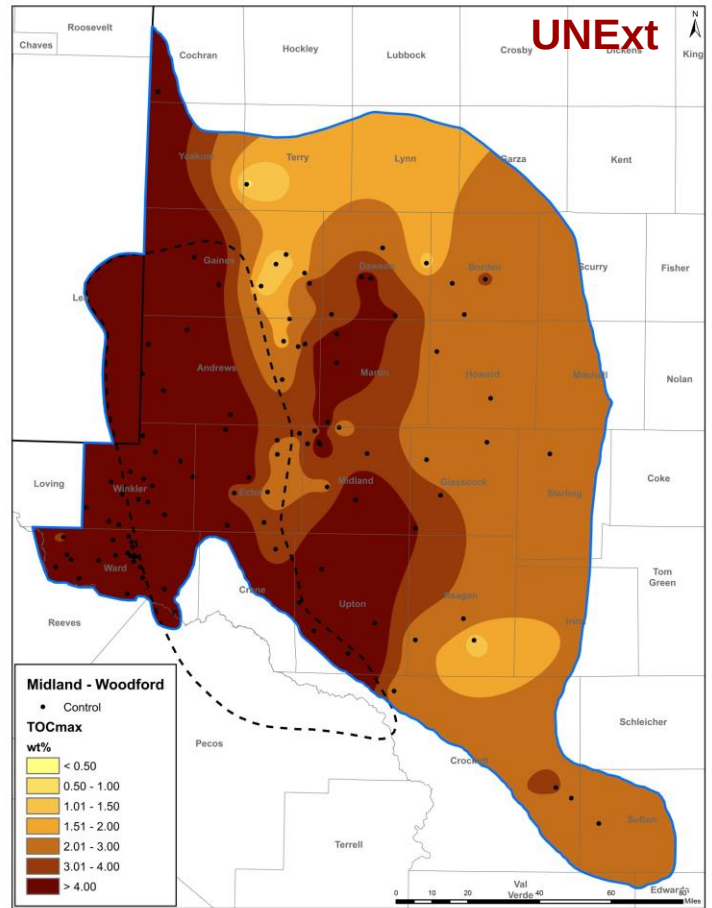


Woodford

Average TOCpd



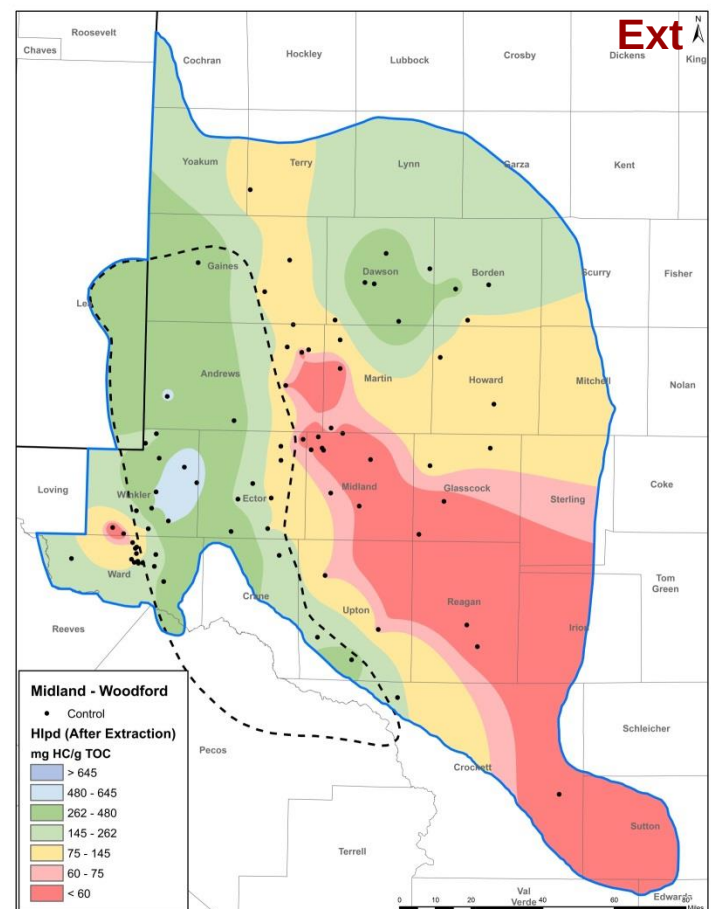
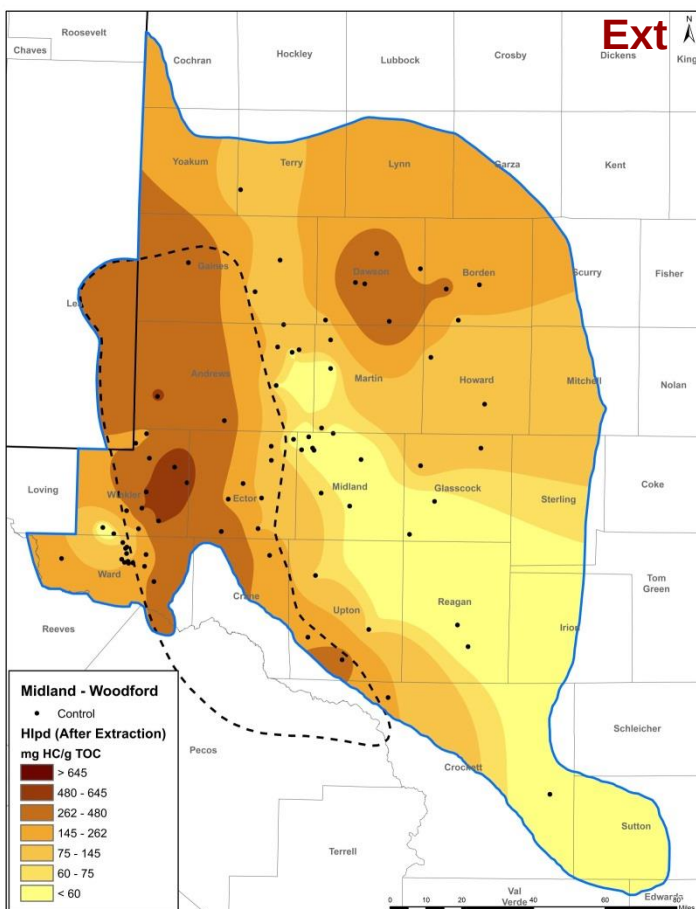
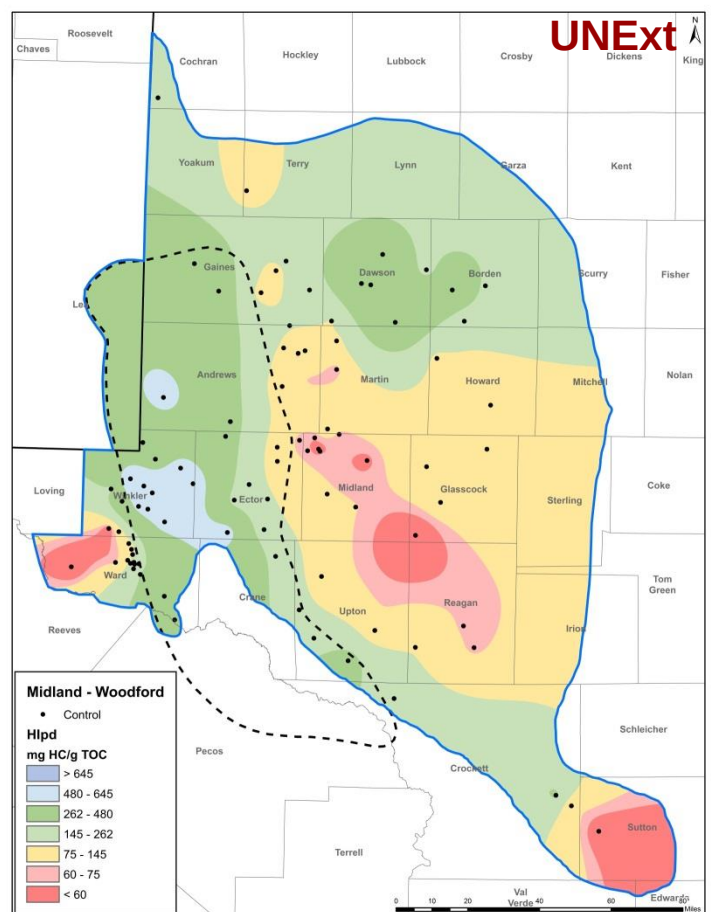
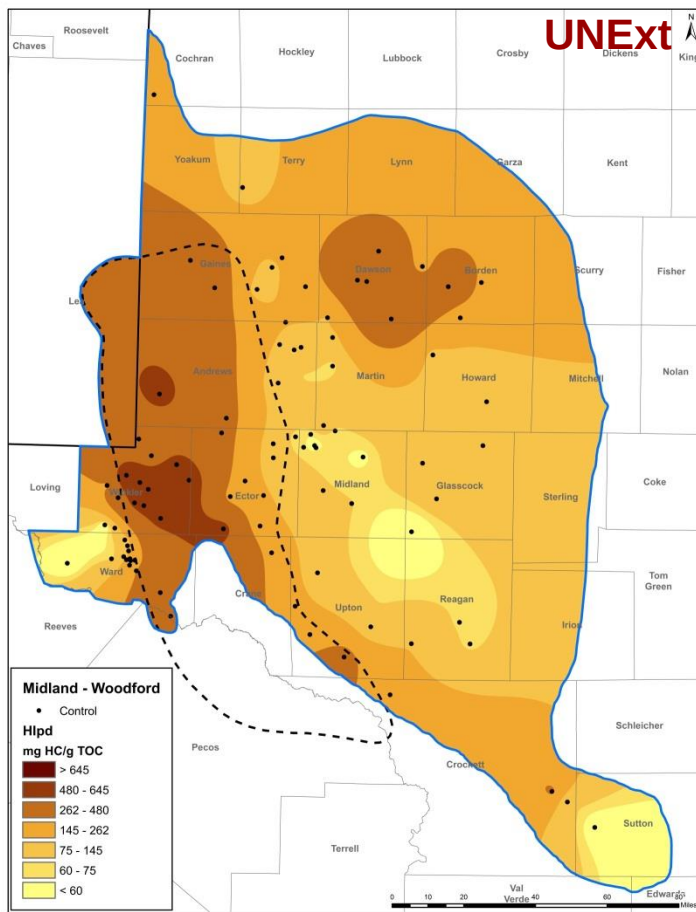
Maximum TOCpd



Woodford (H₁₀=700mgHC/gTOC)

Hl_{pd}- Source Quality

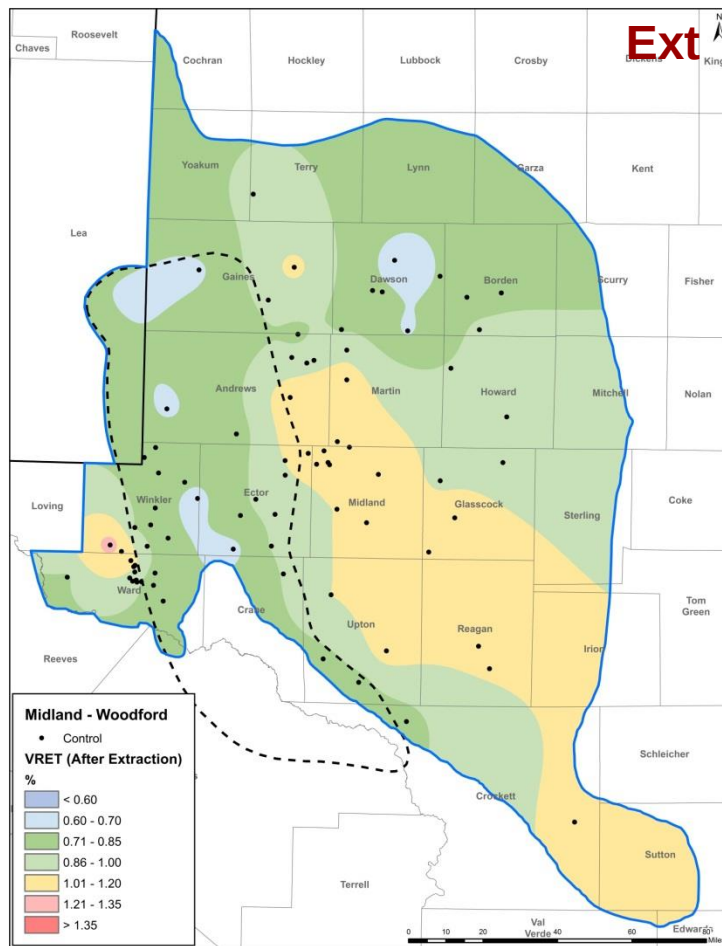
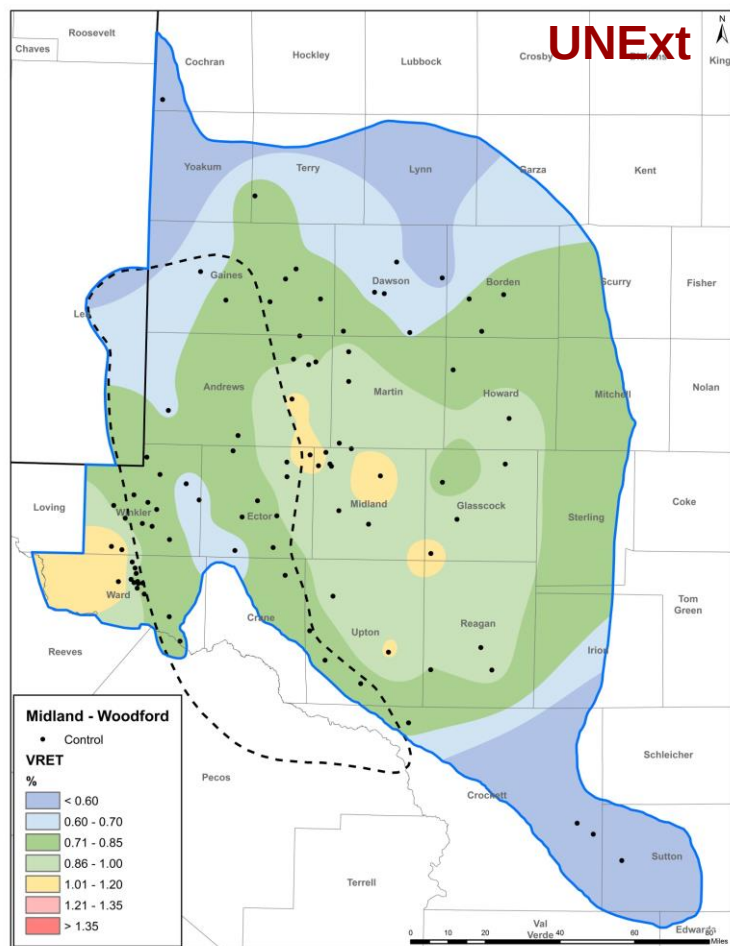
Hl_{pd}-Thermal Maturity



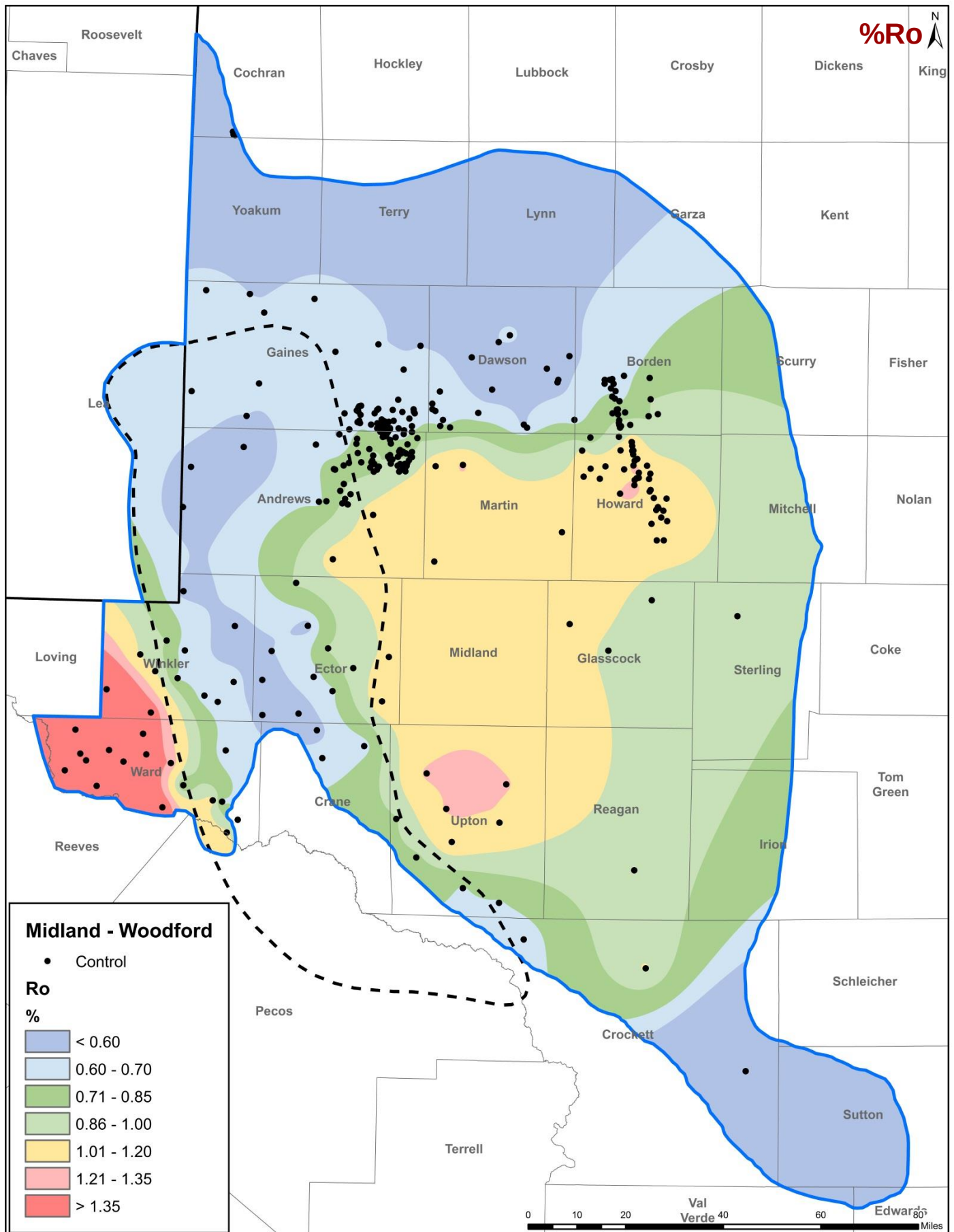
Woodford

%VRET from Tmax

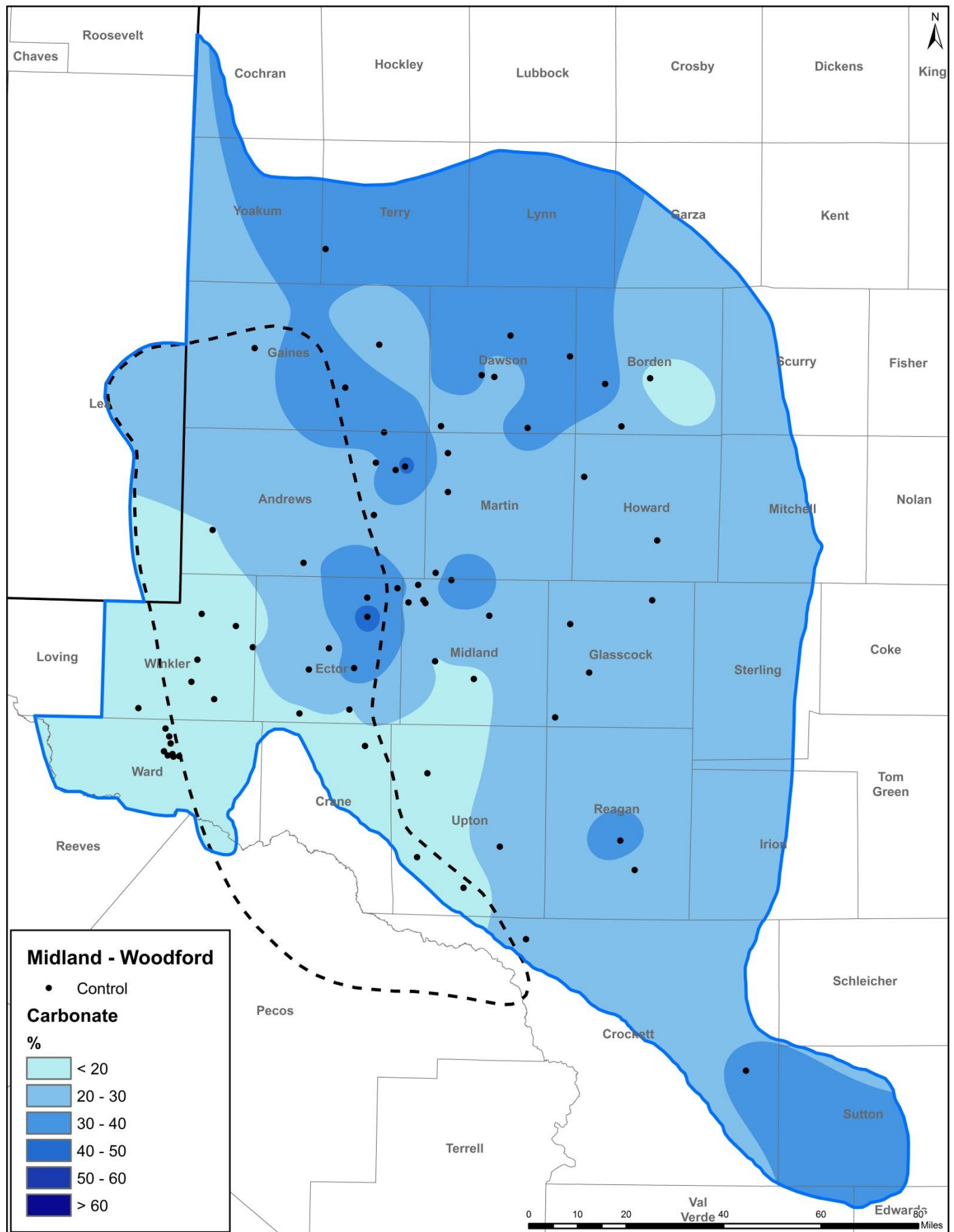
%VRET from Tmax



Woodford (Measured %Ro)

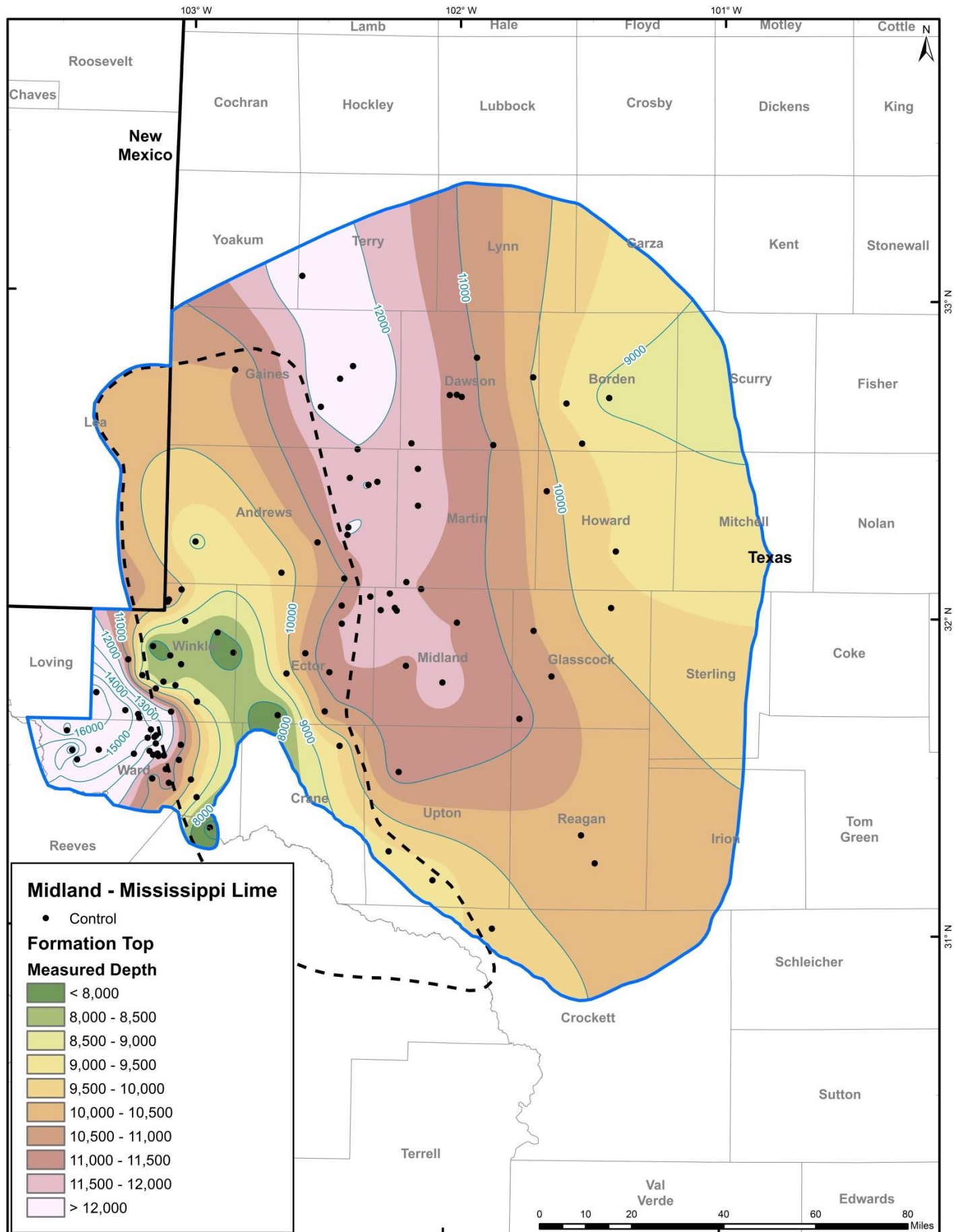


Woodford Wt % Carbonate

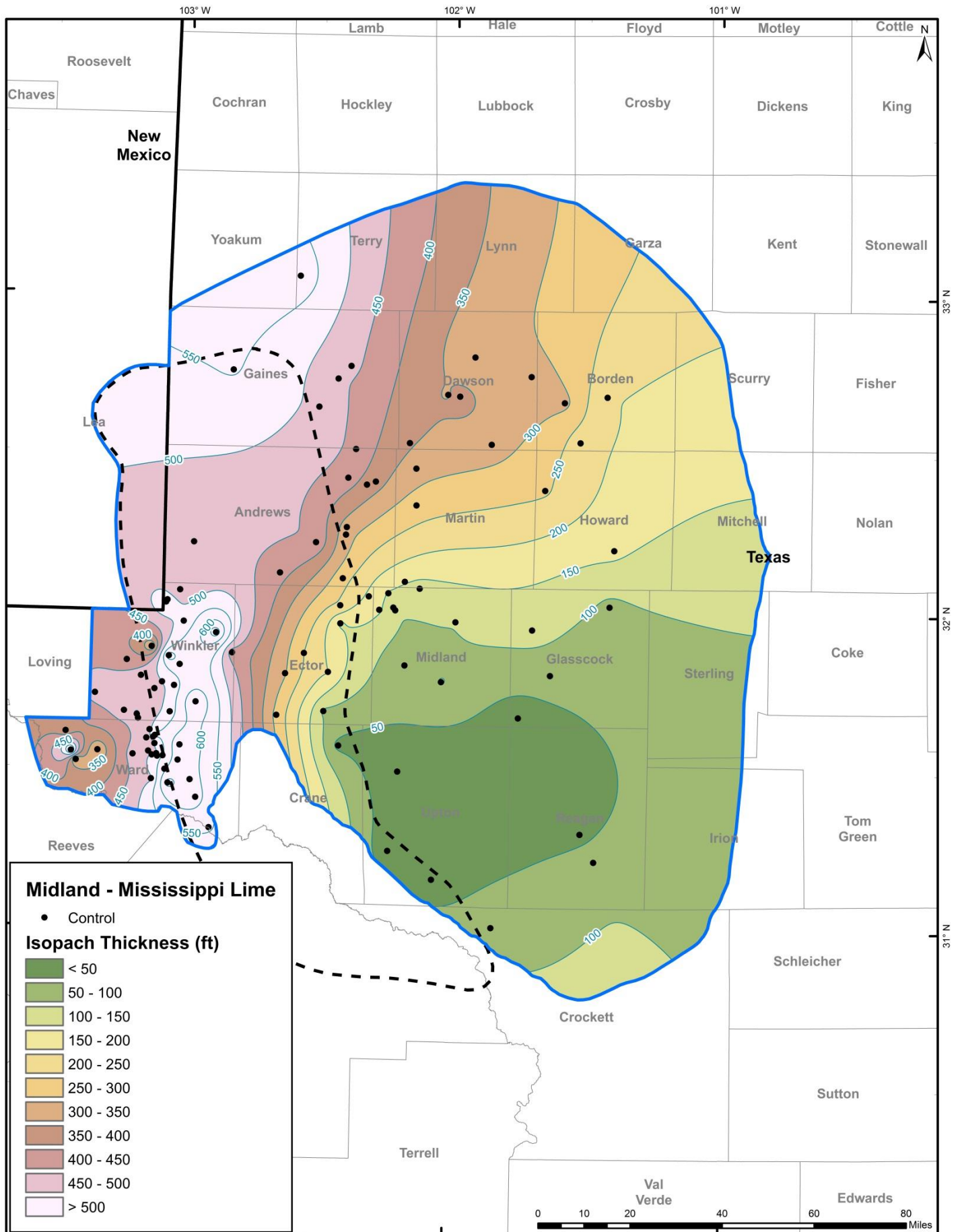


Mississippi Lime Maps

Present Day Top Mississippi Lime

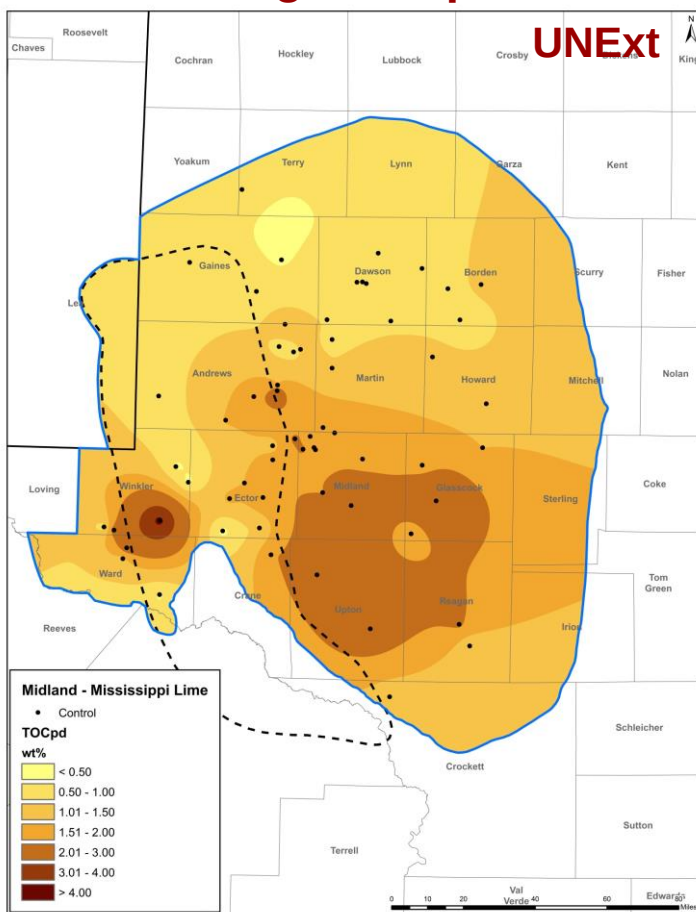


Mississippi Lime Isopach

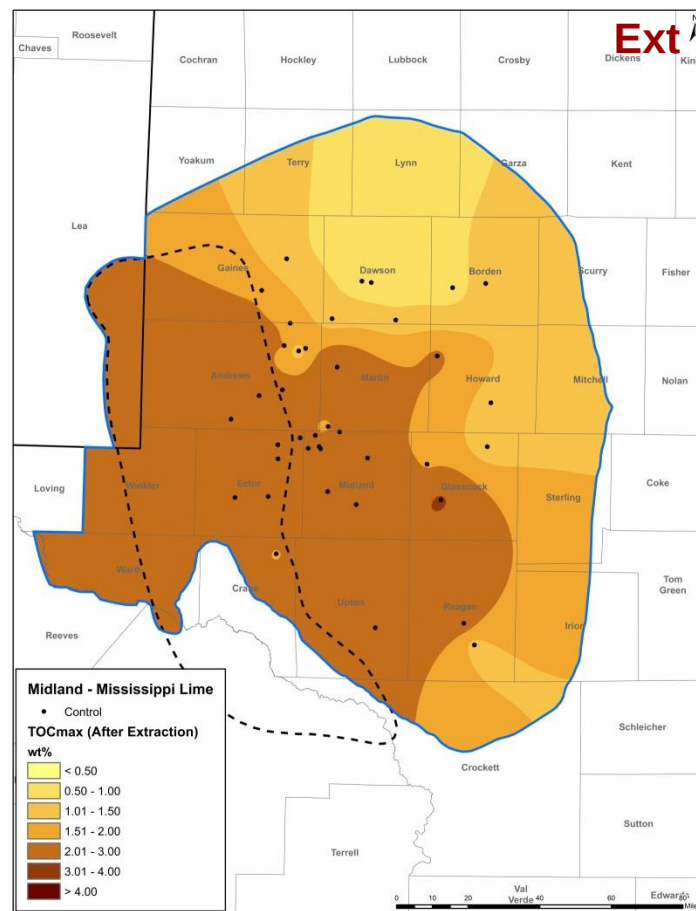
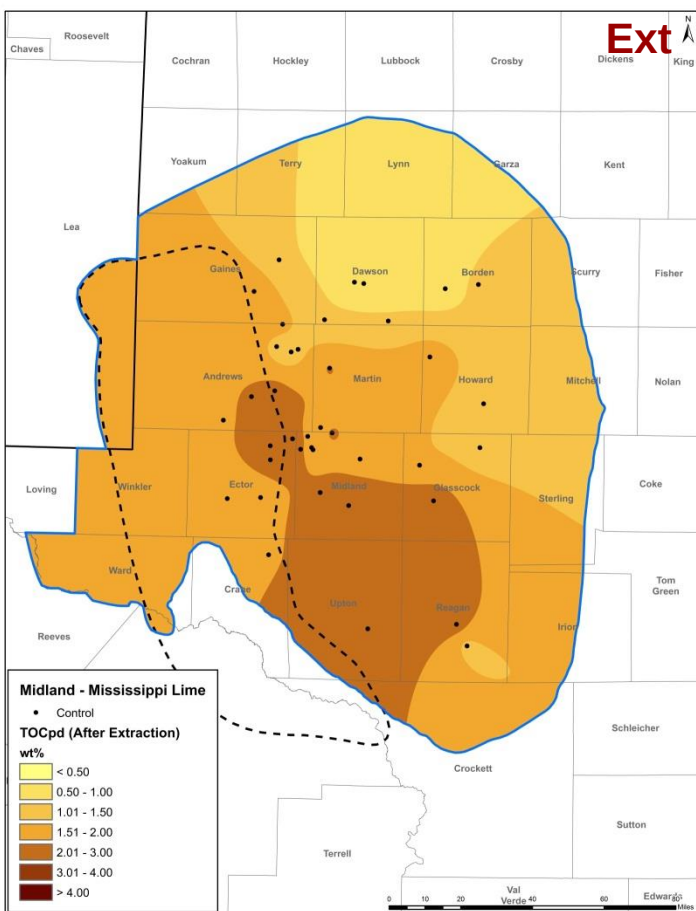
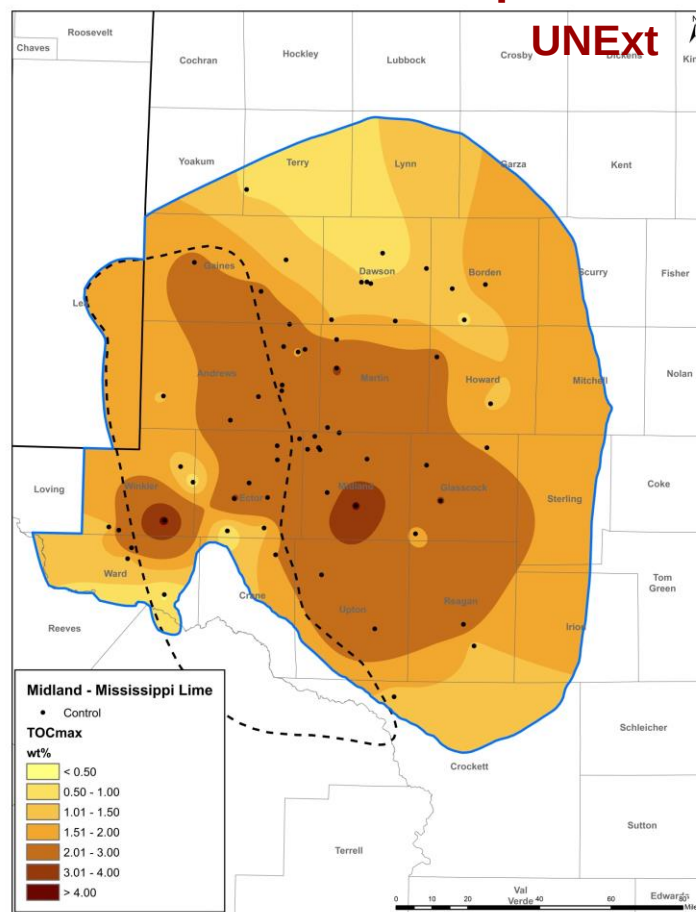


Mississippi Lime

Average TOCpd



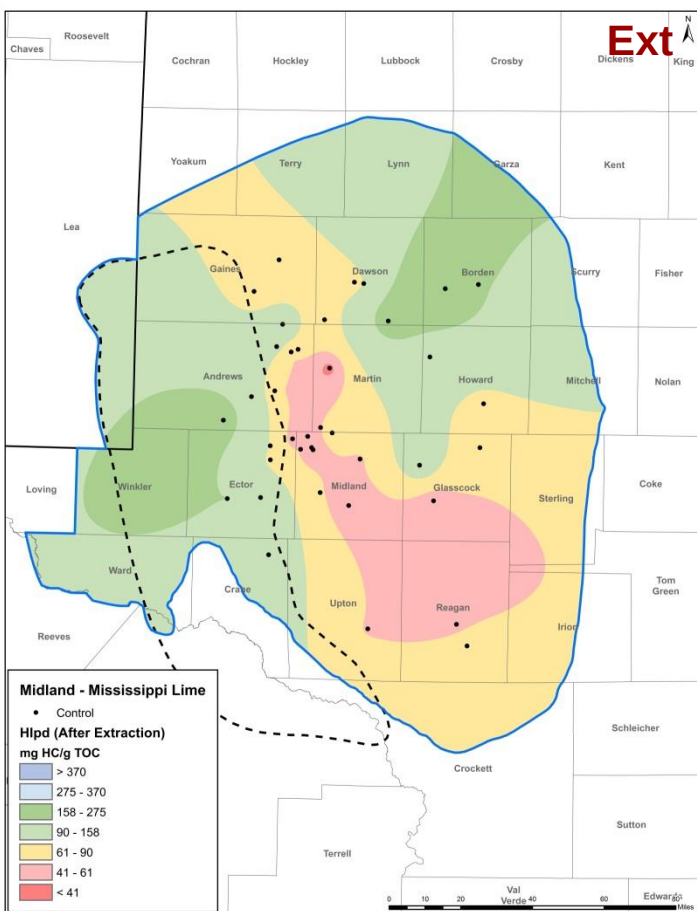
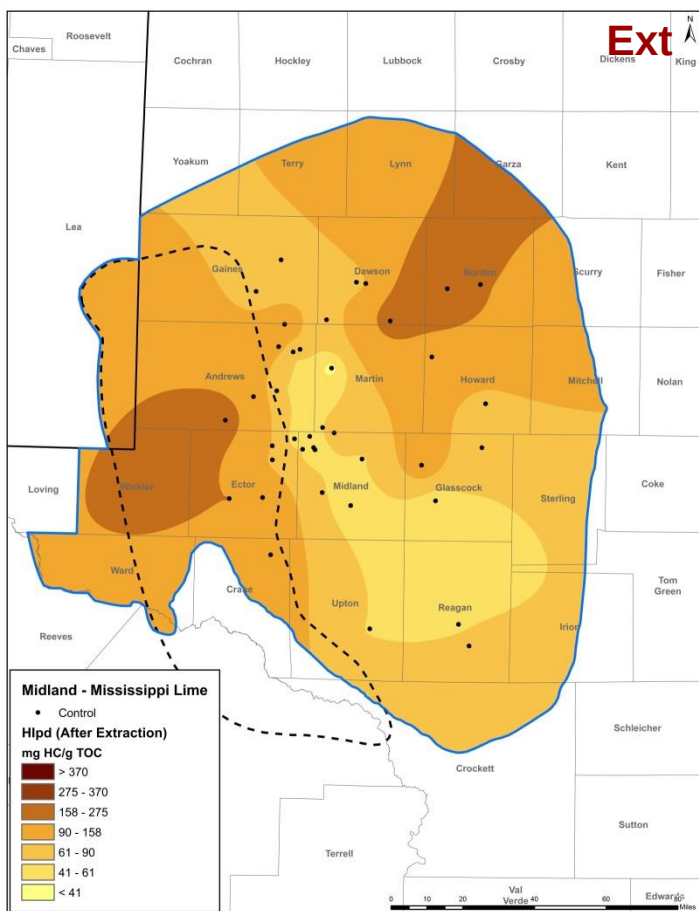
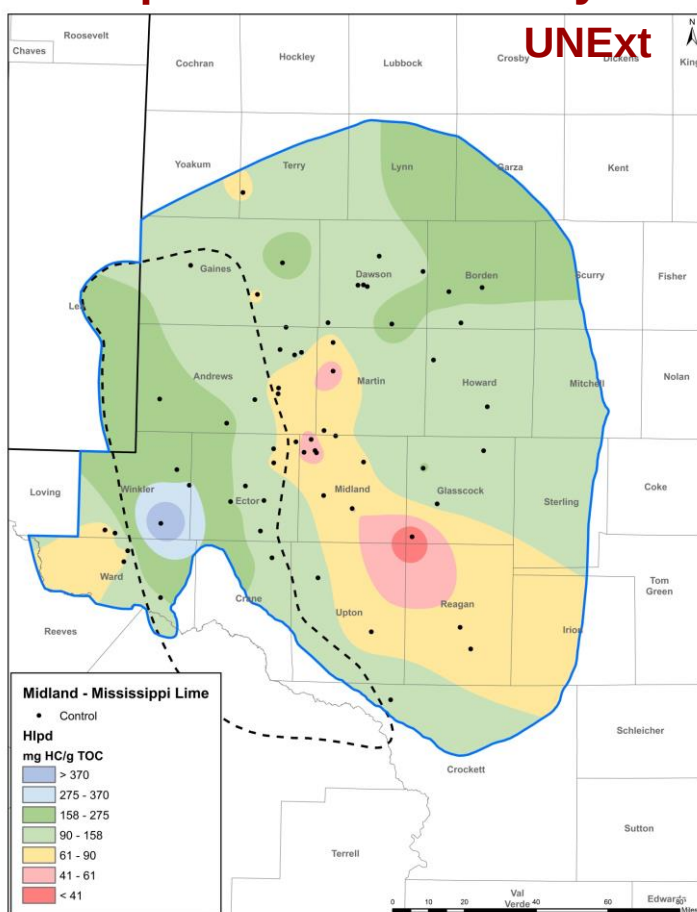
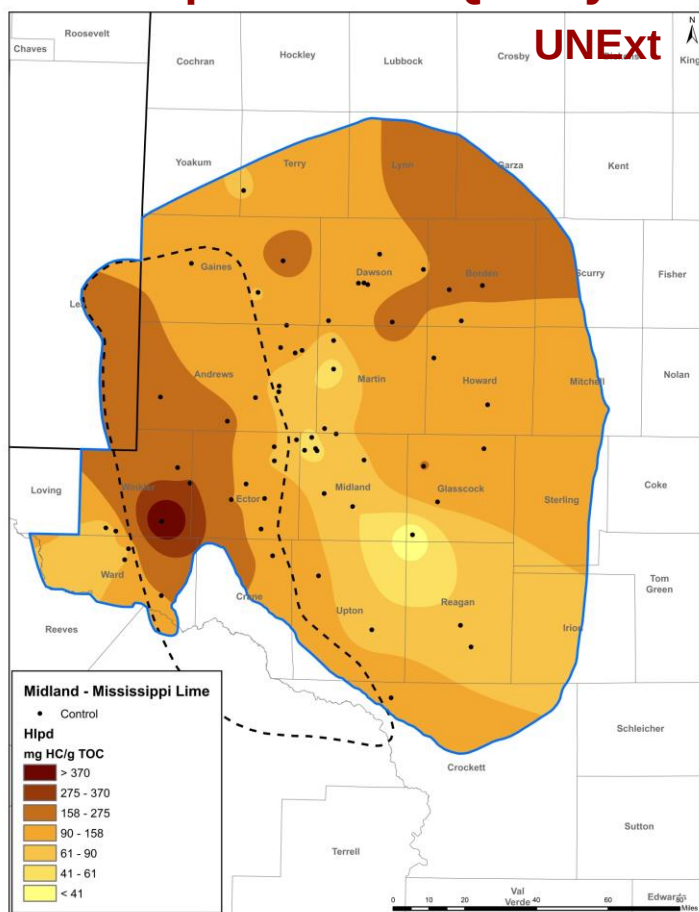
Maximum TOCpd



Mississippi Lime (Hlo=400mgHC/gTOC)

Hlpd- Source Quality

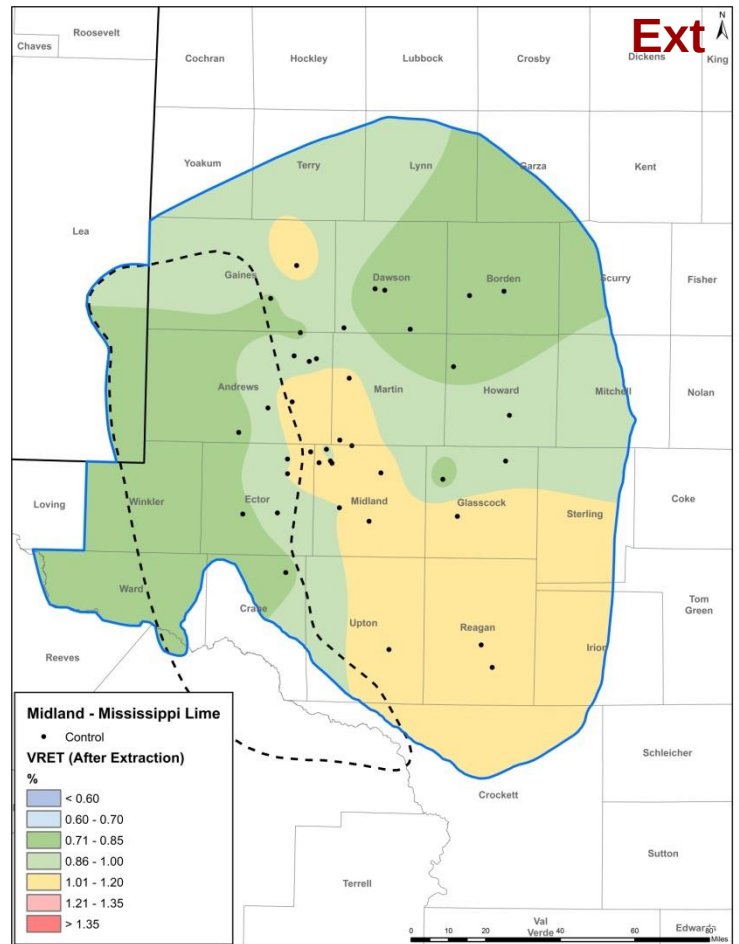
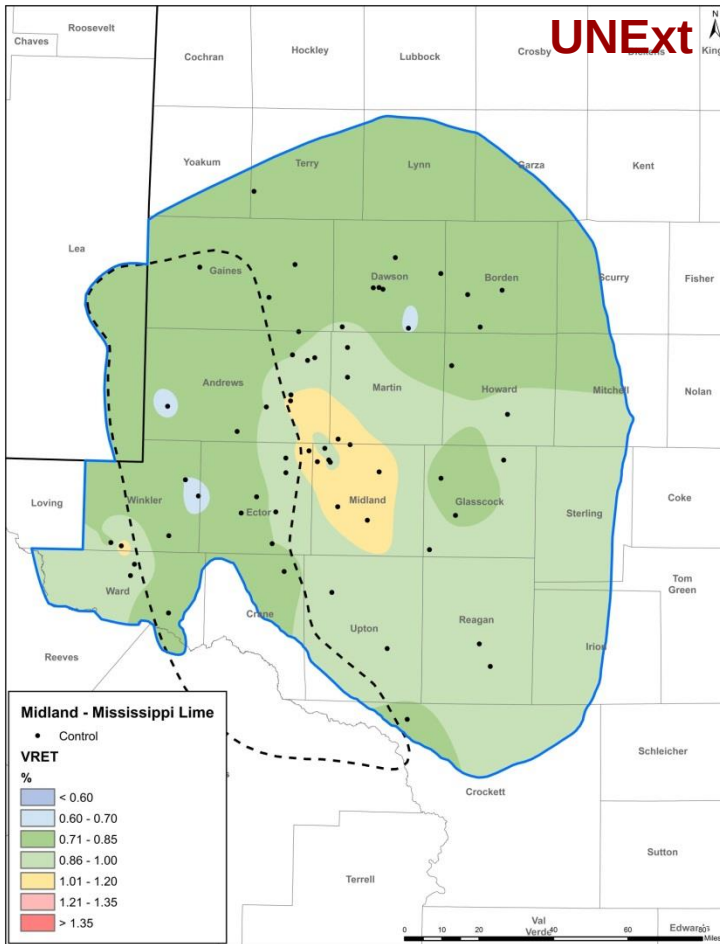
Hlpd-Thermal Maturity



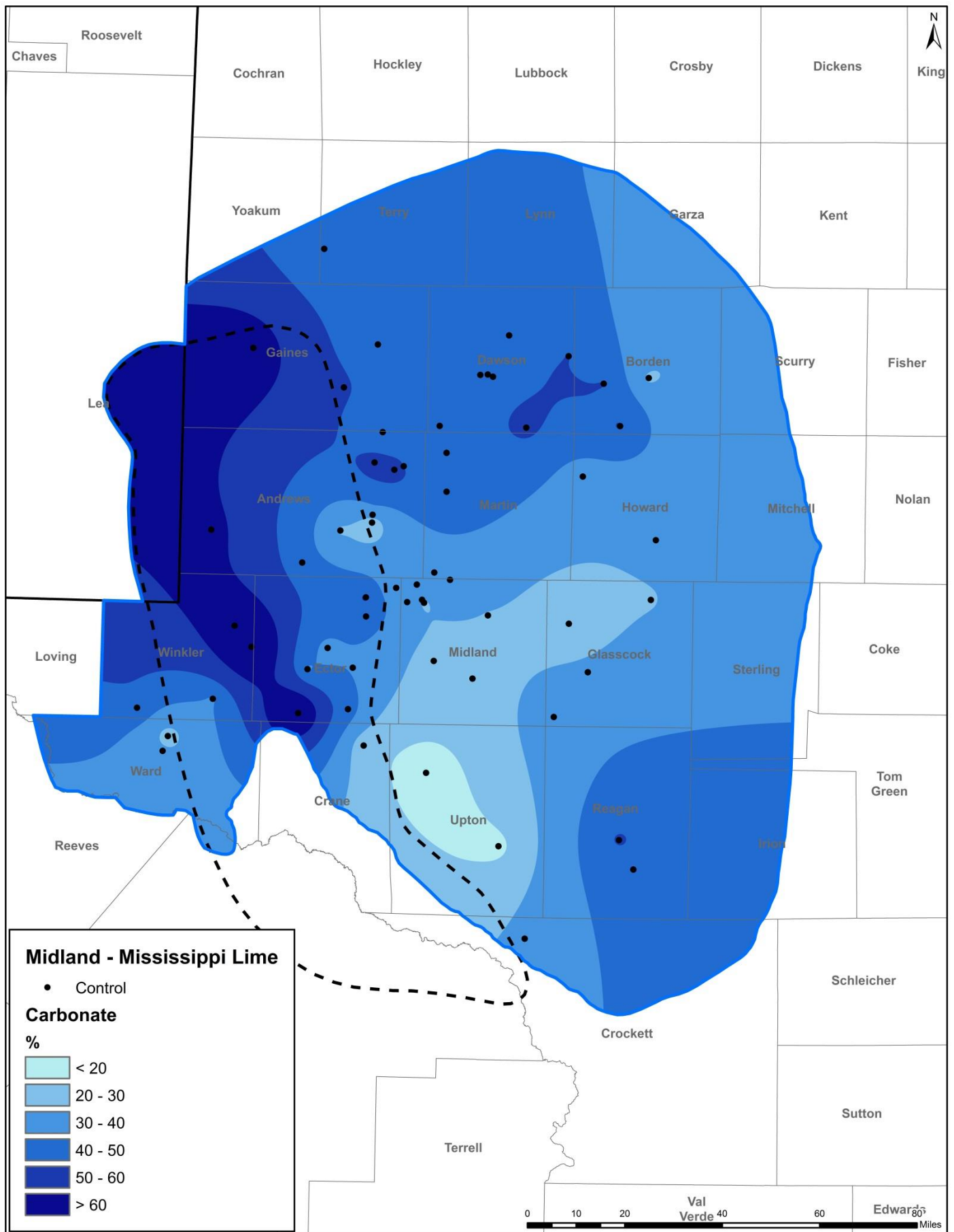
Mississippi Lime

%VRET from Tmax

%VRET from Tmax

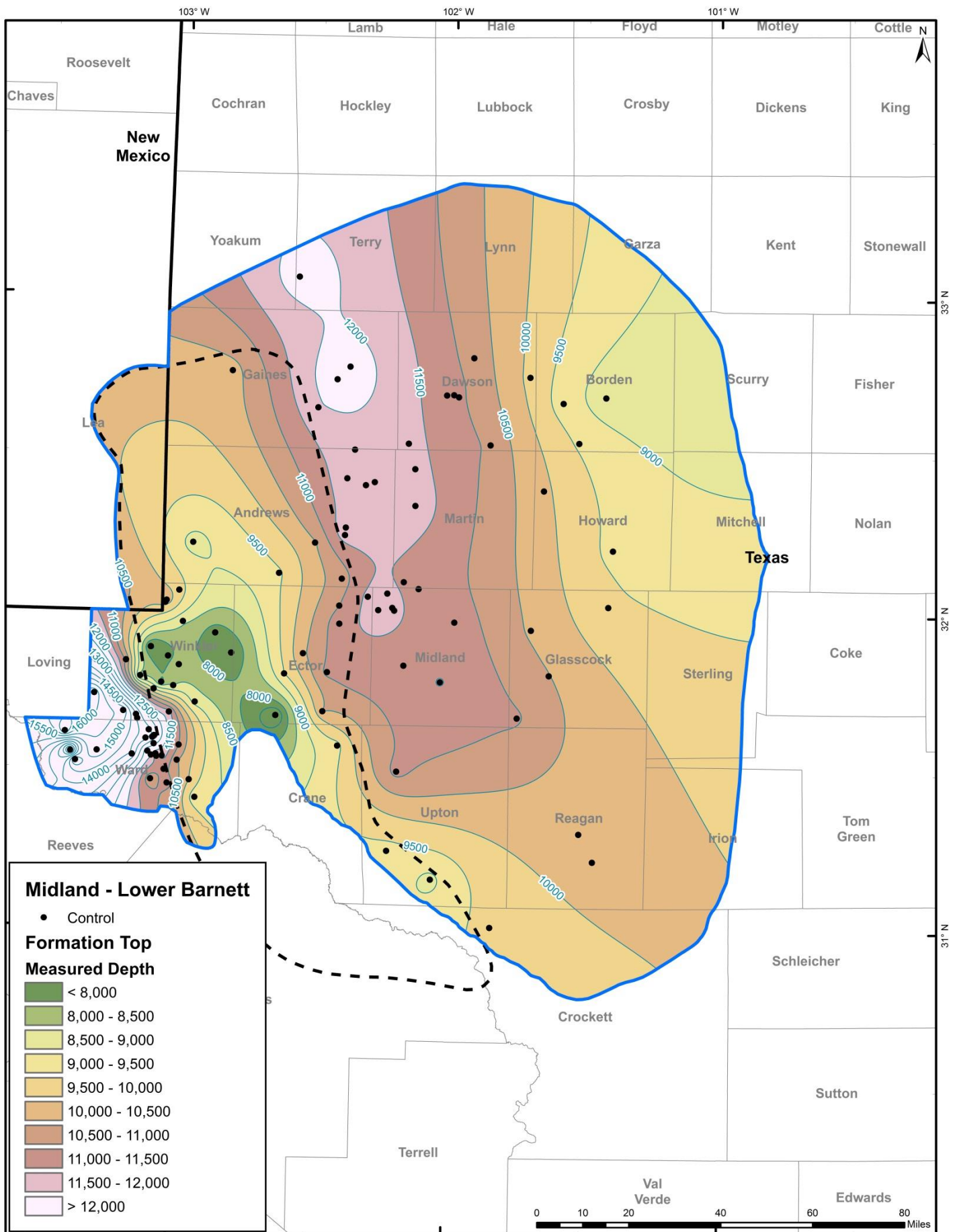


Mississippi Lime Wt% Carbonate

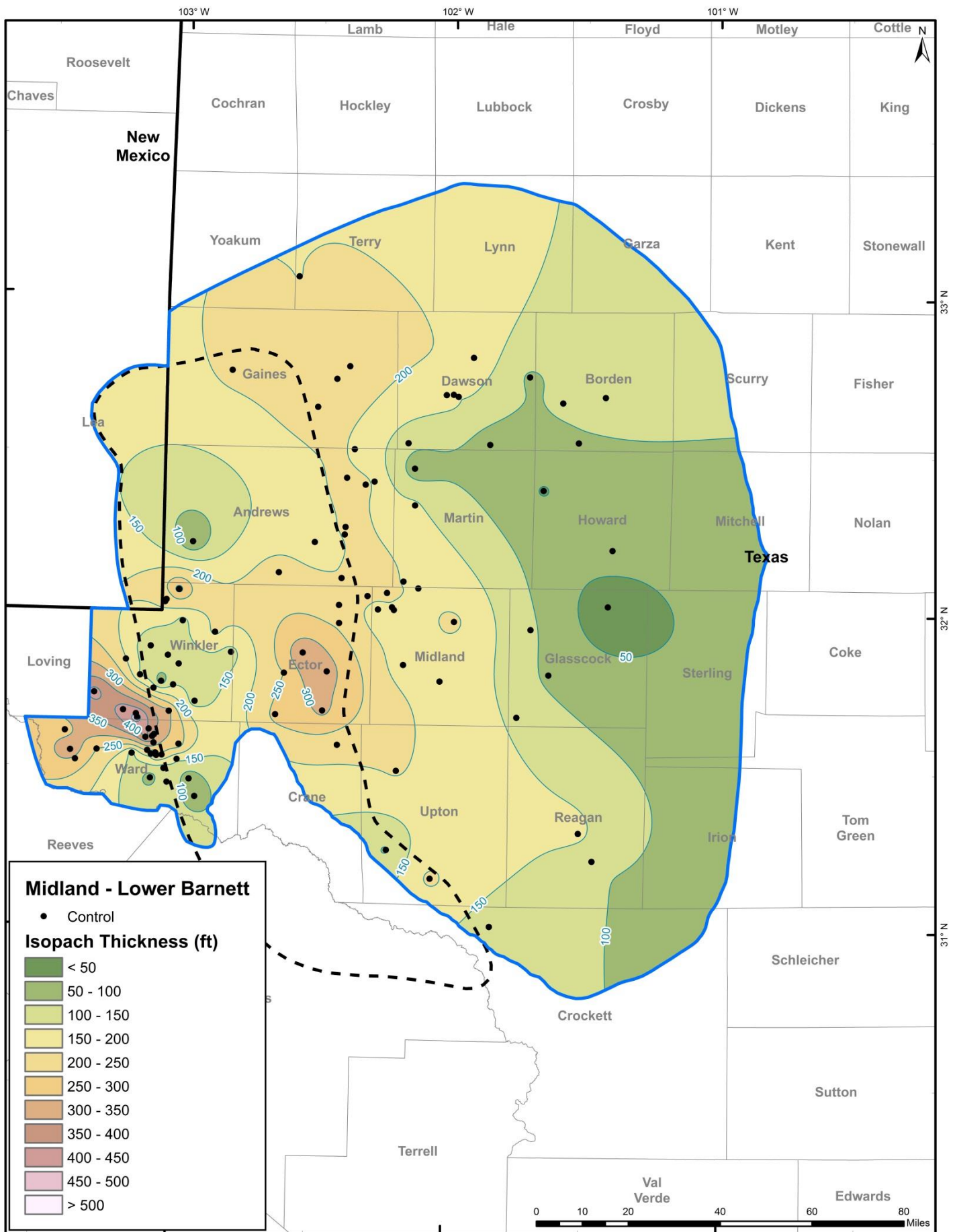


Lower Barnett Maps

Present Day Top Lower Barnett

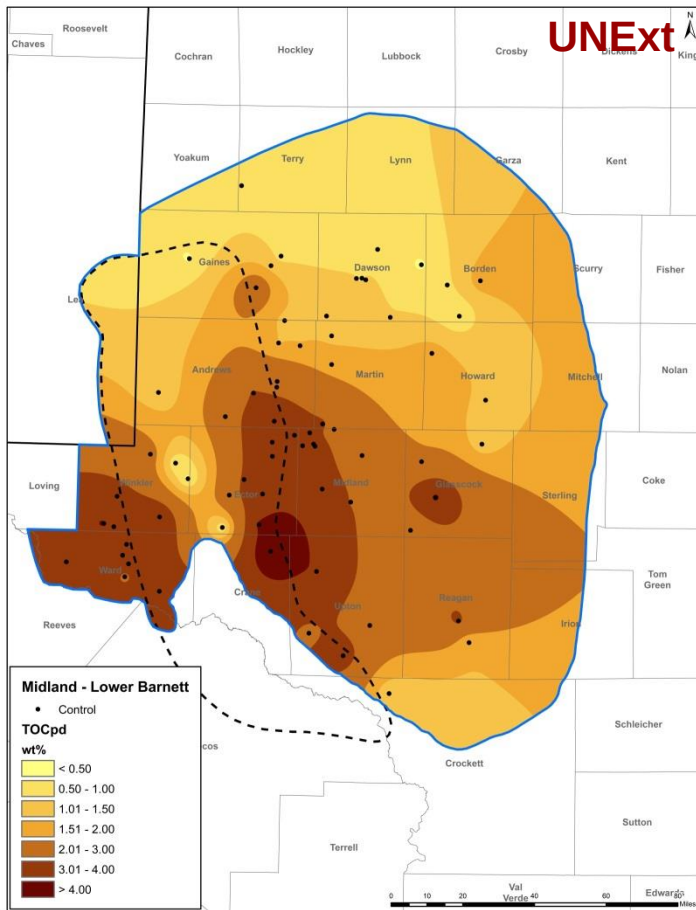


Lower Barnett Isopach

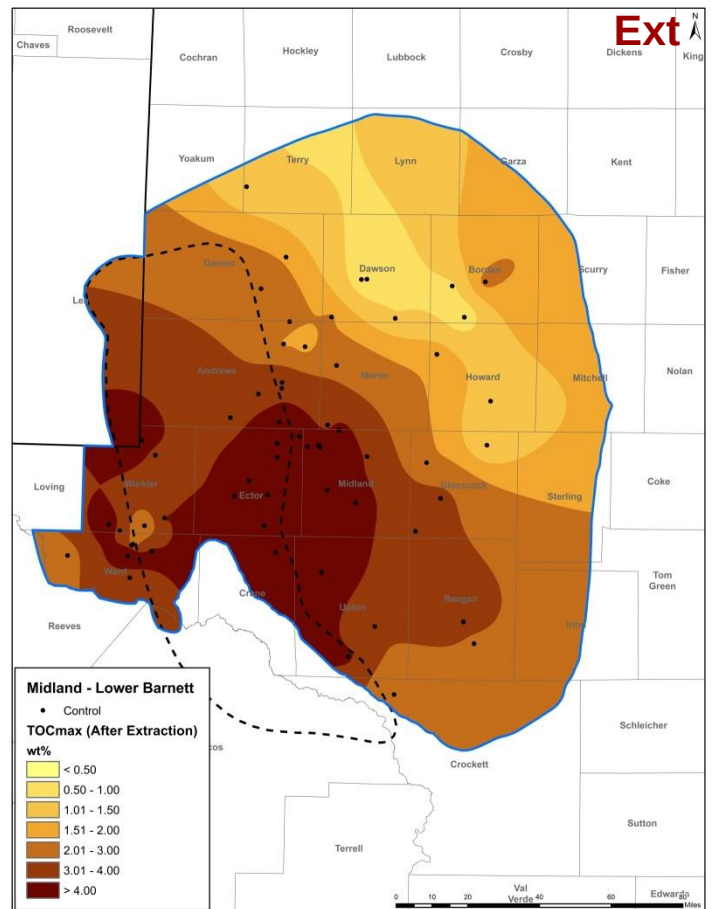
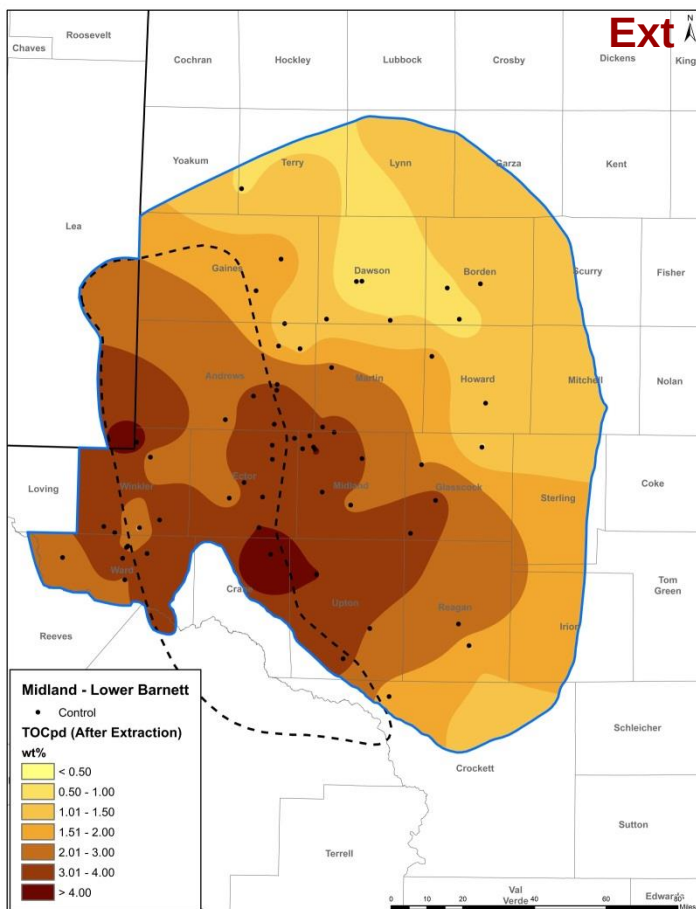
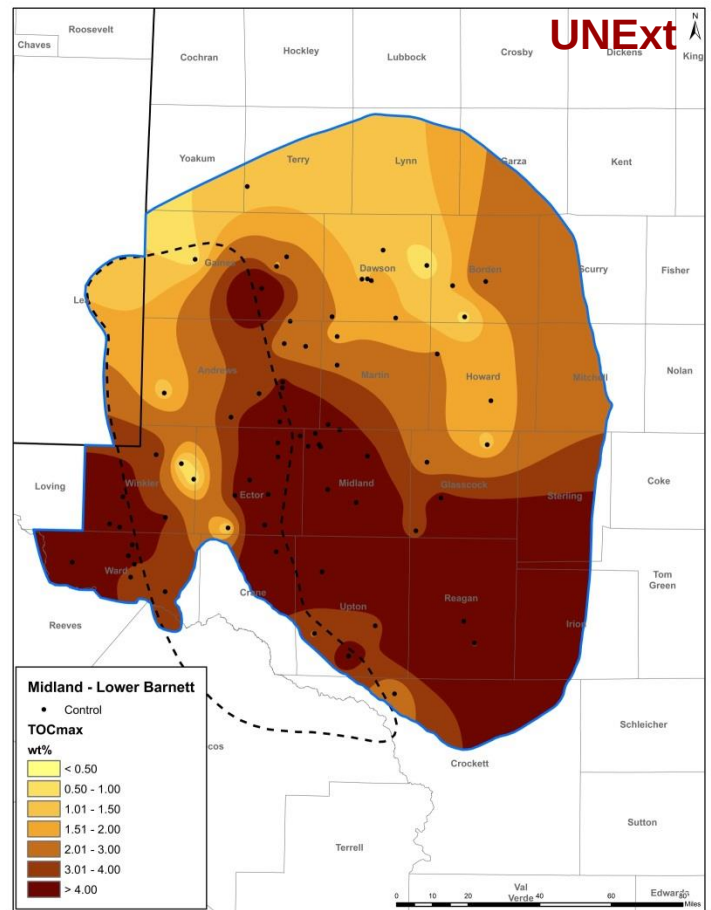


Lower Barnett

Average TOCpd



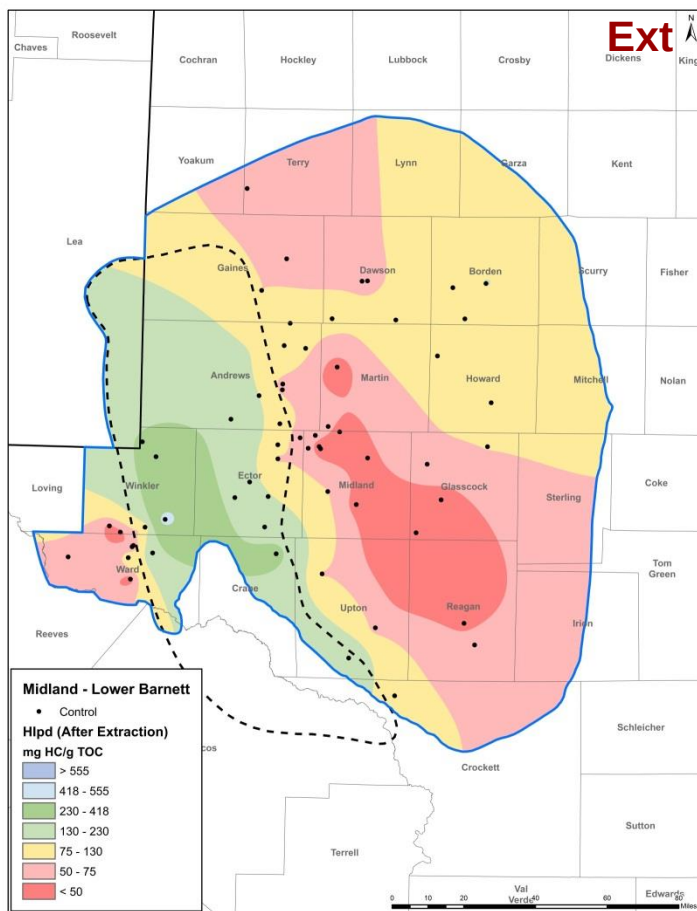
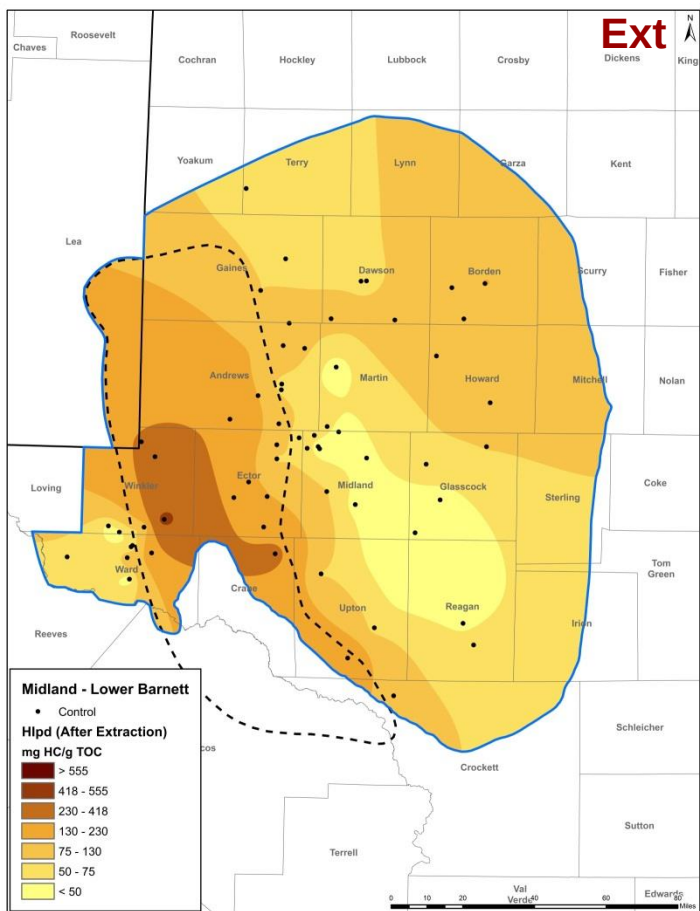
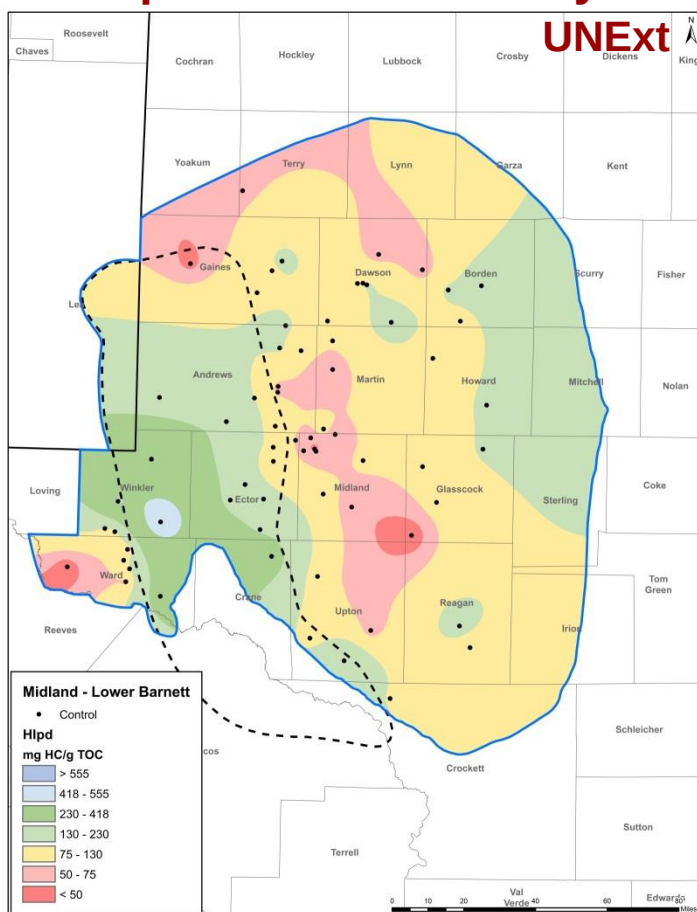
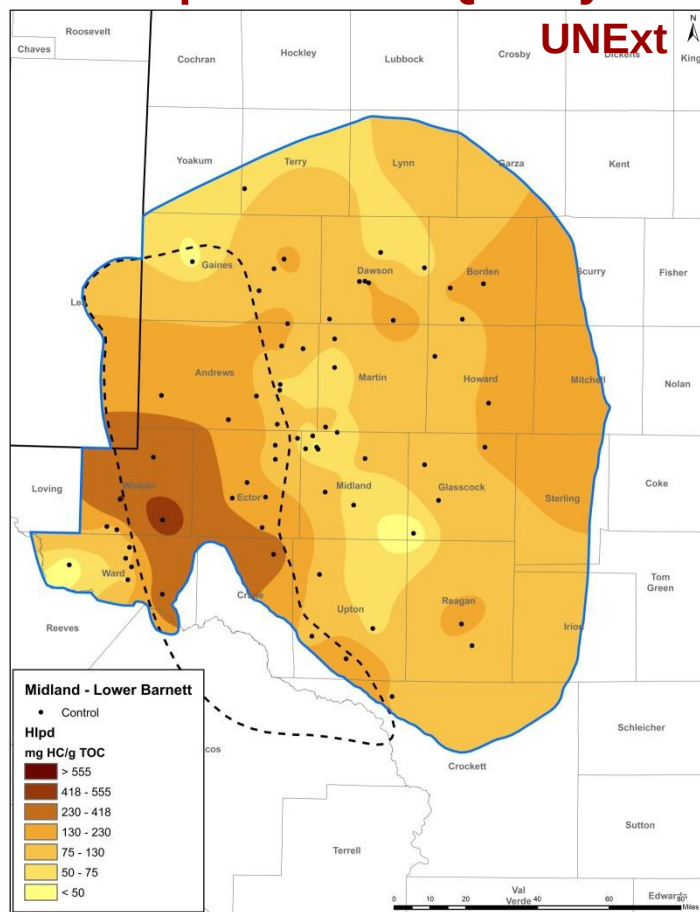
Maximum TOCpd



Lower Barnett (Hlo=600mgHC/gTOC)

Hlpd- Source Quality

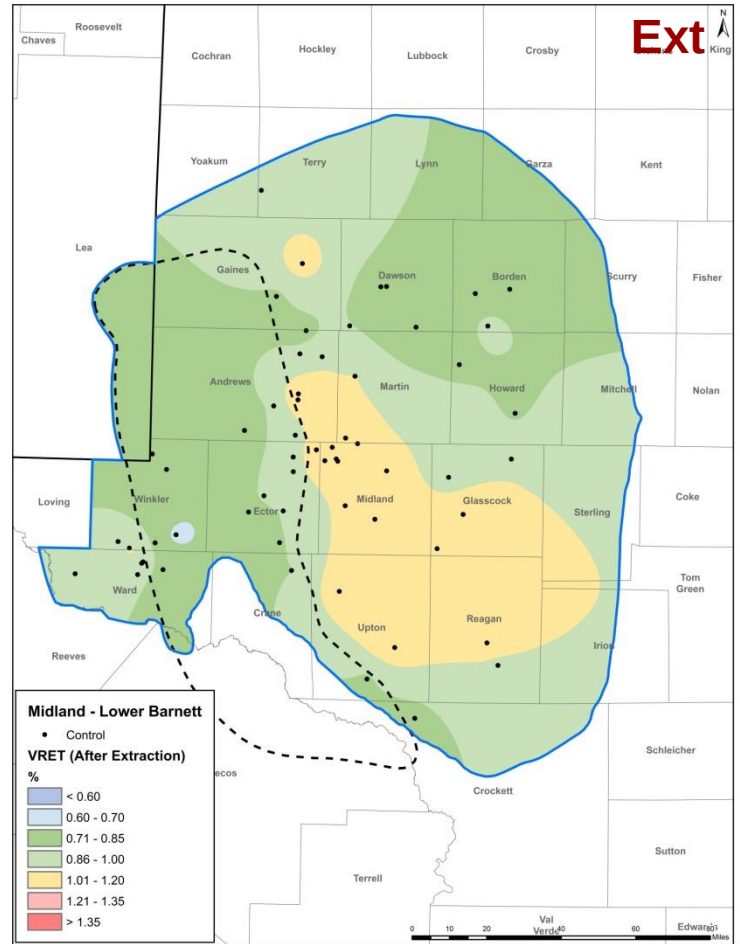
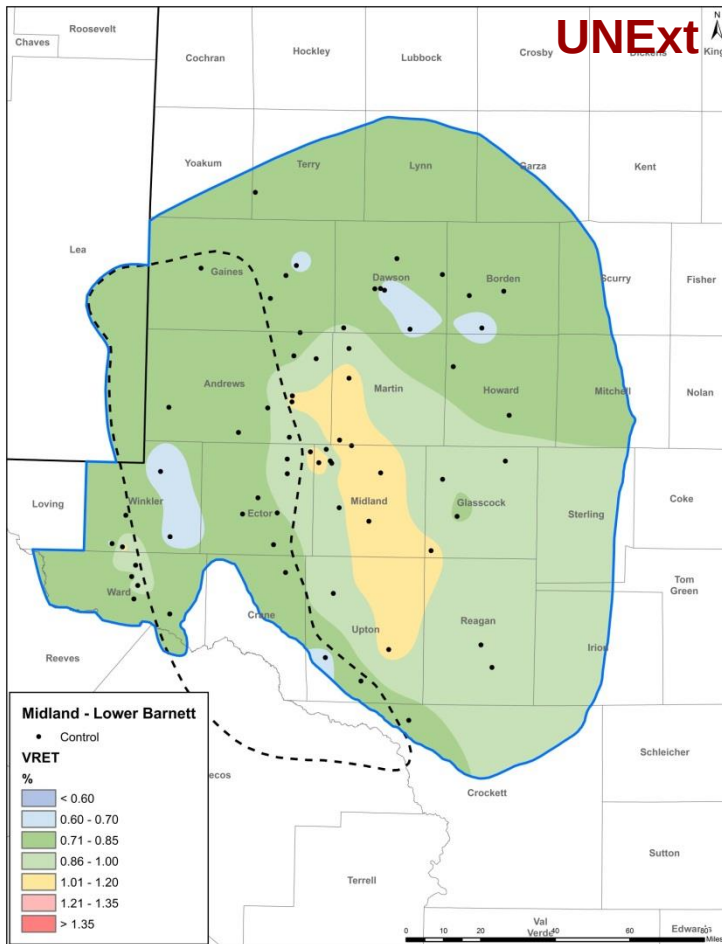
Hlpd-Thermal Maturity



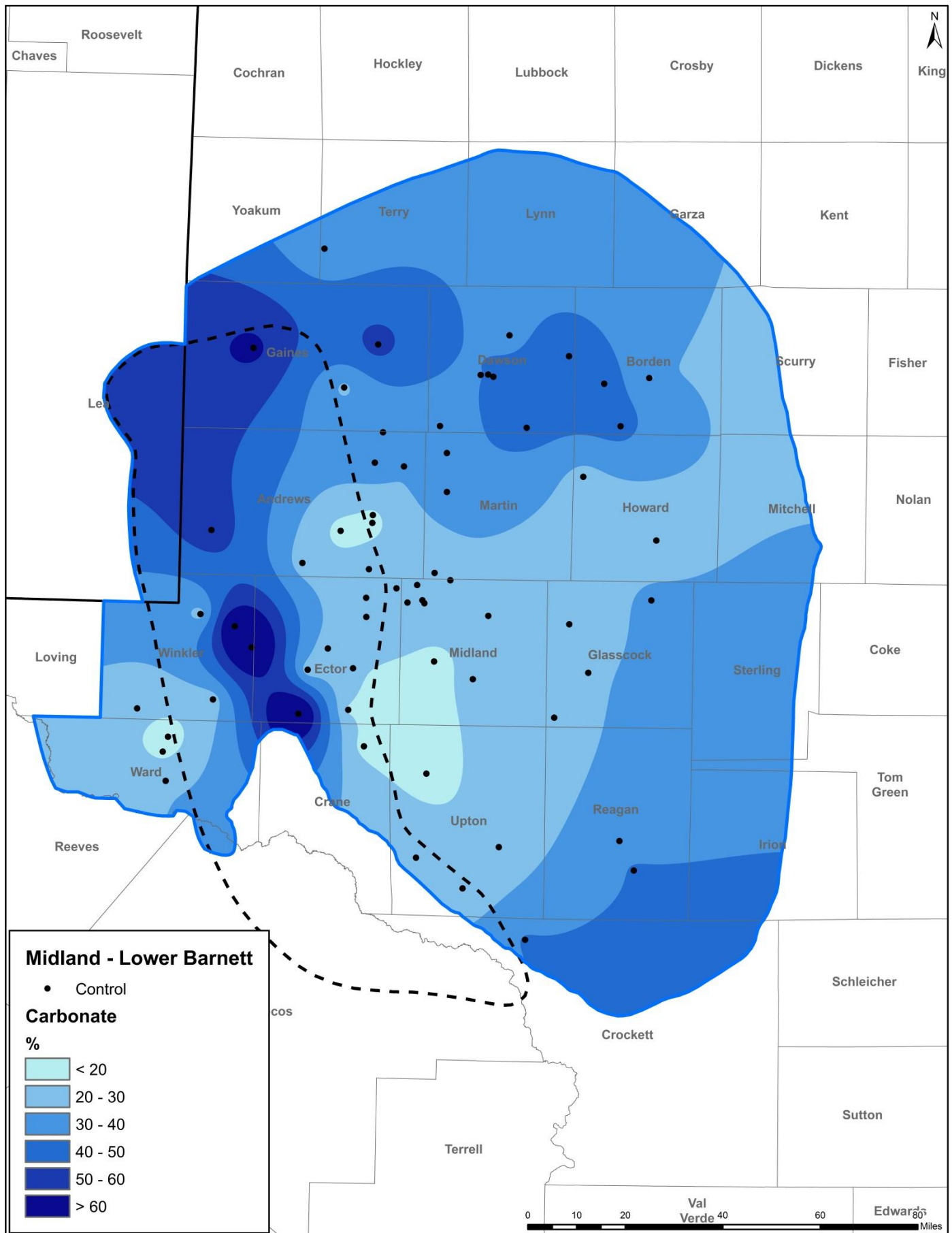
Lower Barnett

%VRET from Tmax

%VRET from Tmax

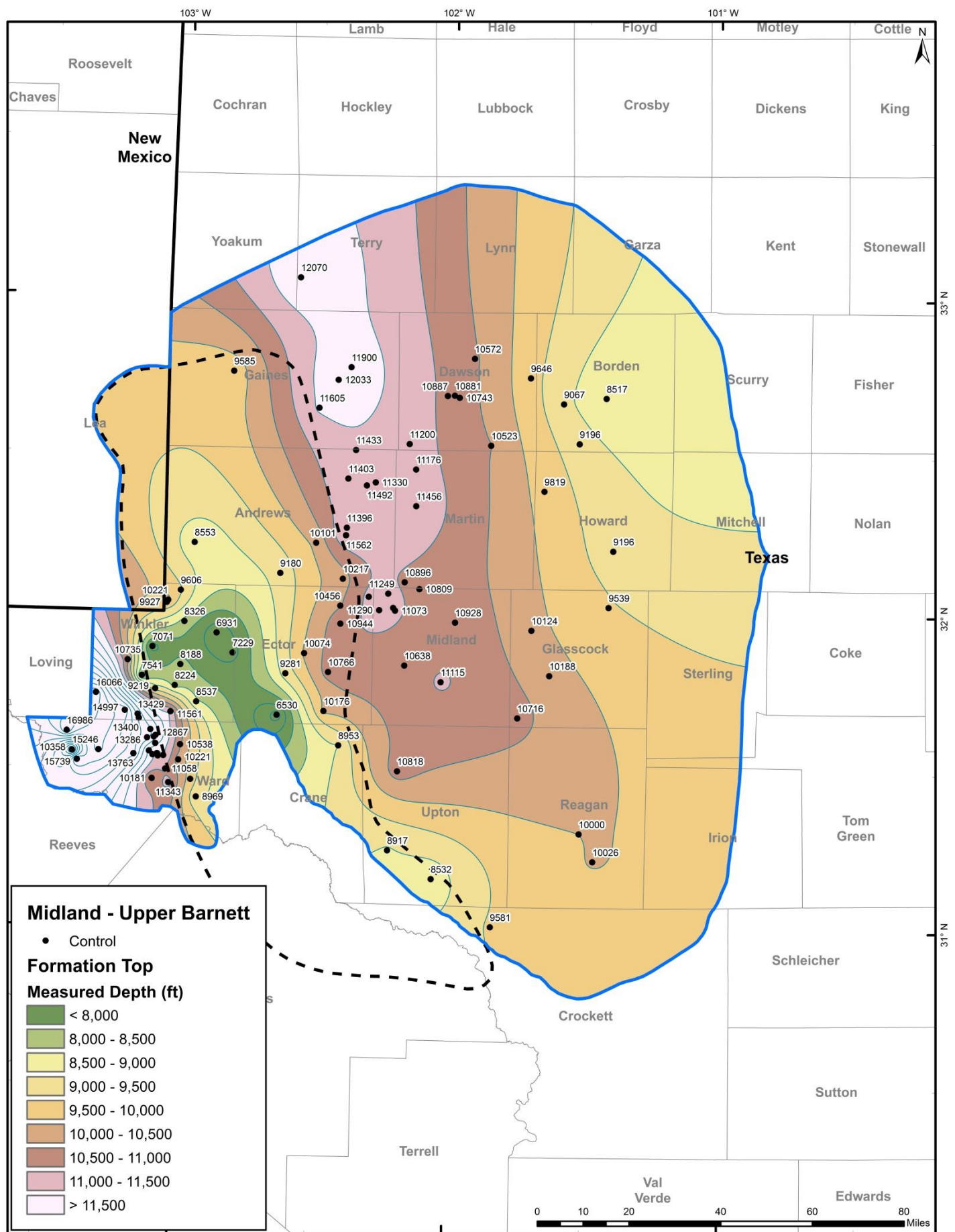


Lower Barnett Wt.% Carbonate

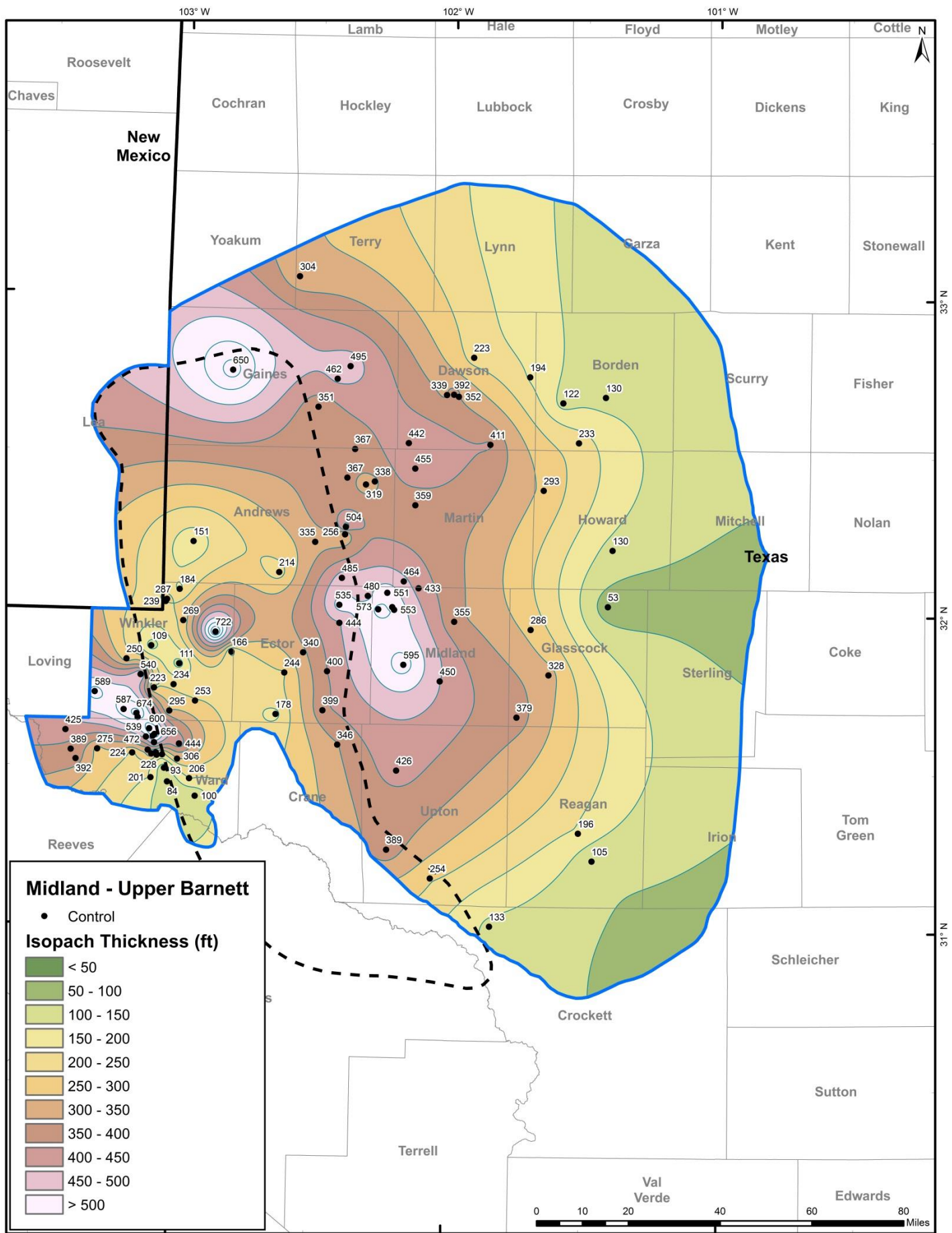


Upper Barnett Maps

Present Day Top Upper Barnett

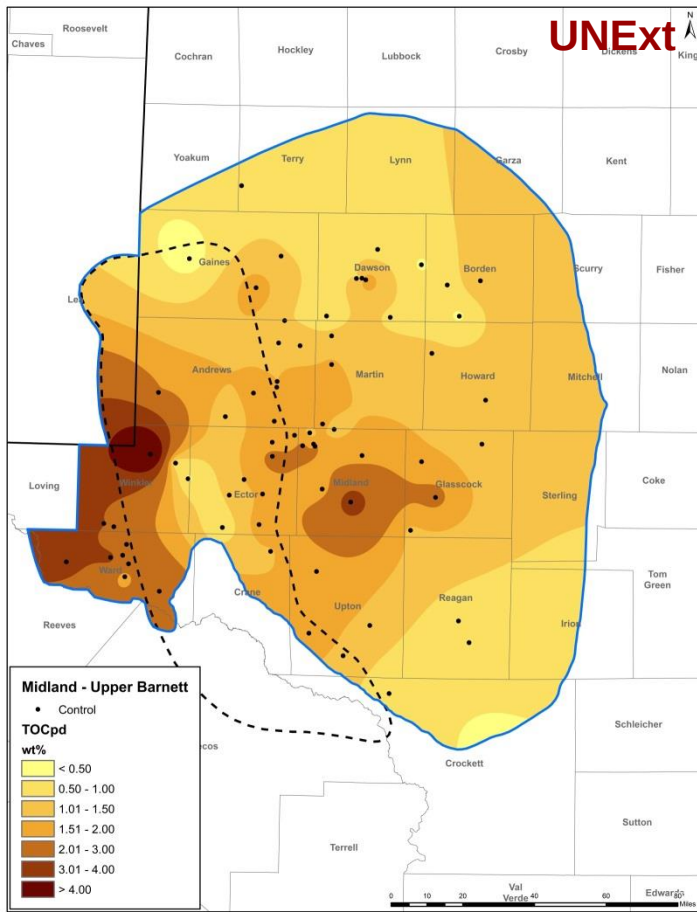


Upper Barnett Isopach

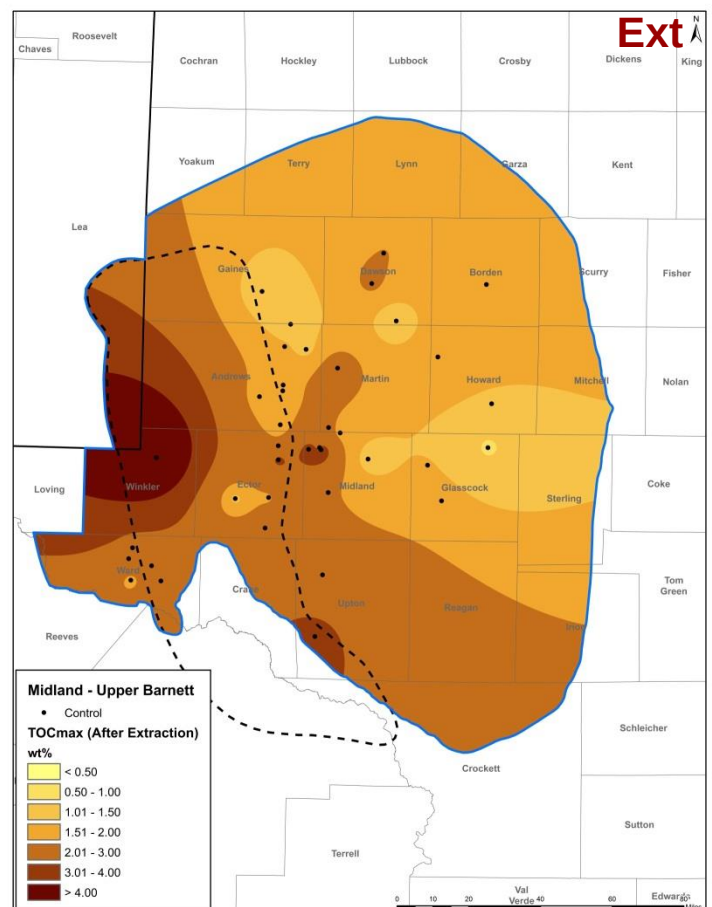
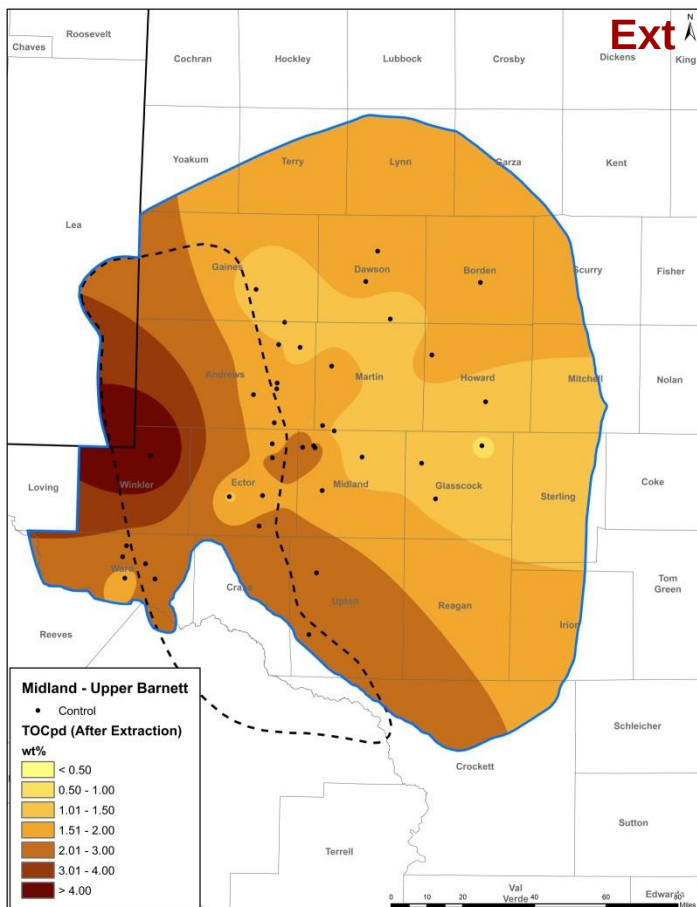
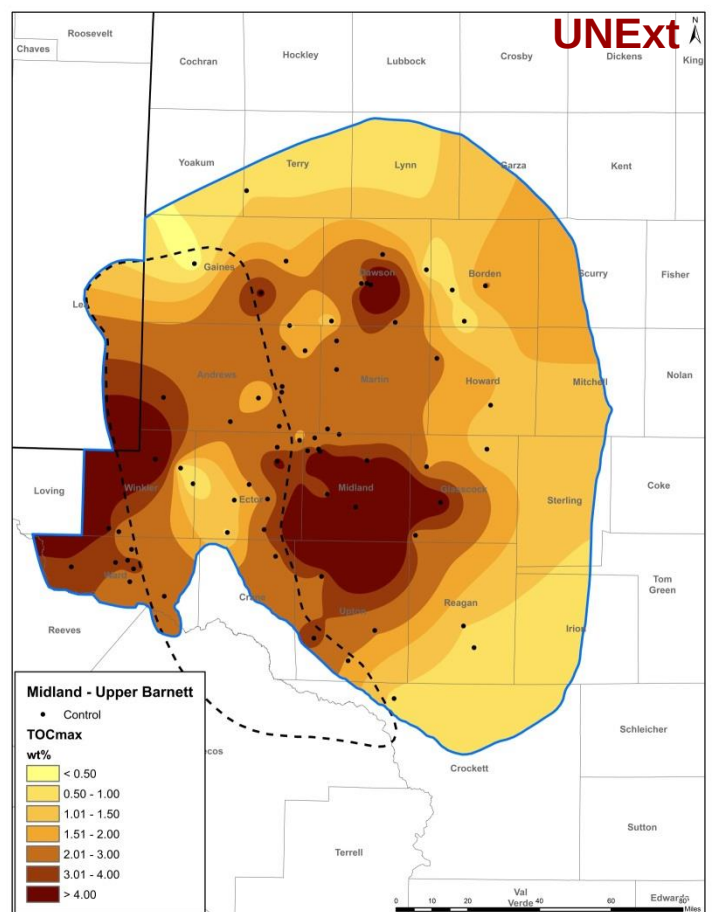


Upper Barnett

Average TOCpd



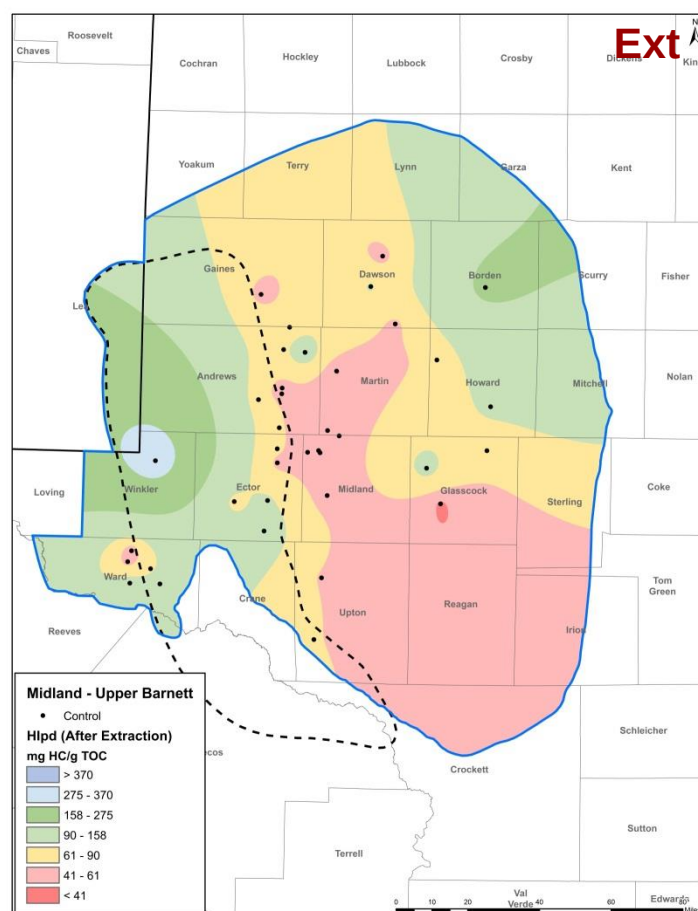
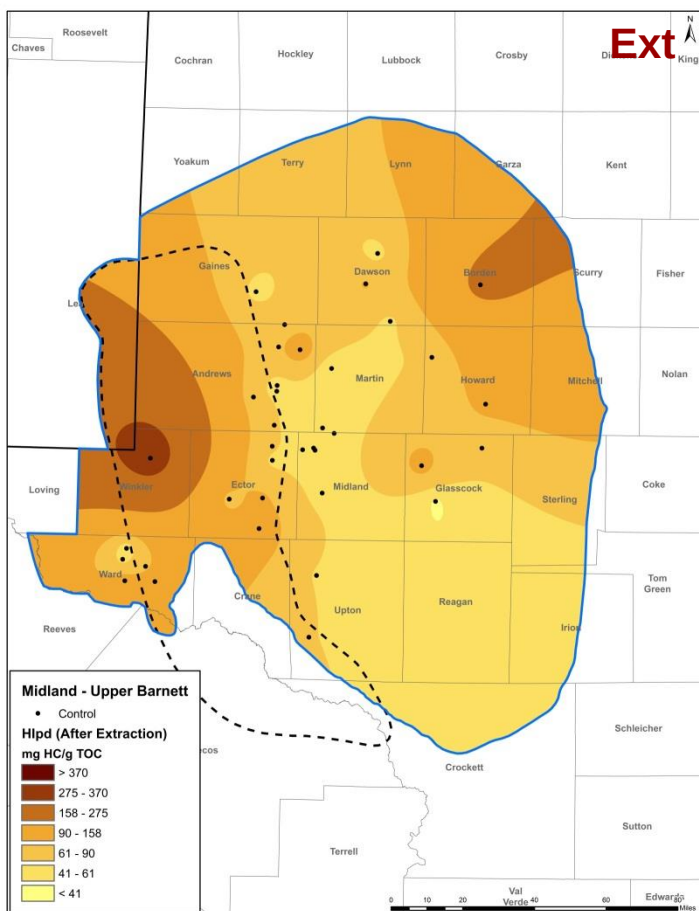
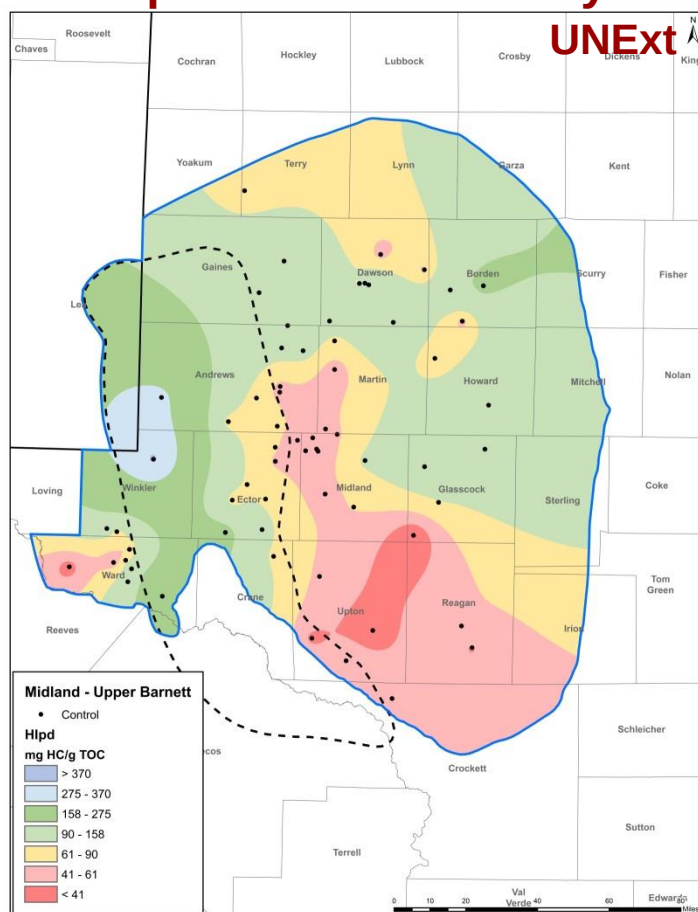
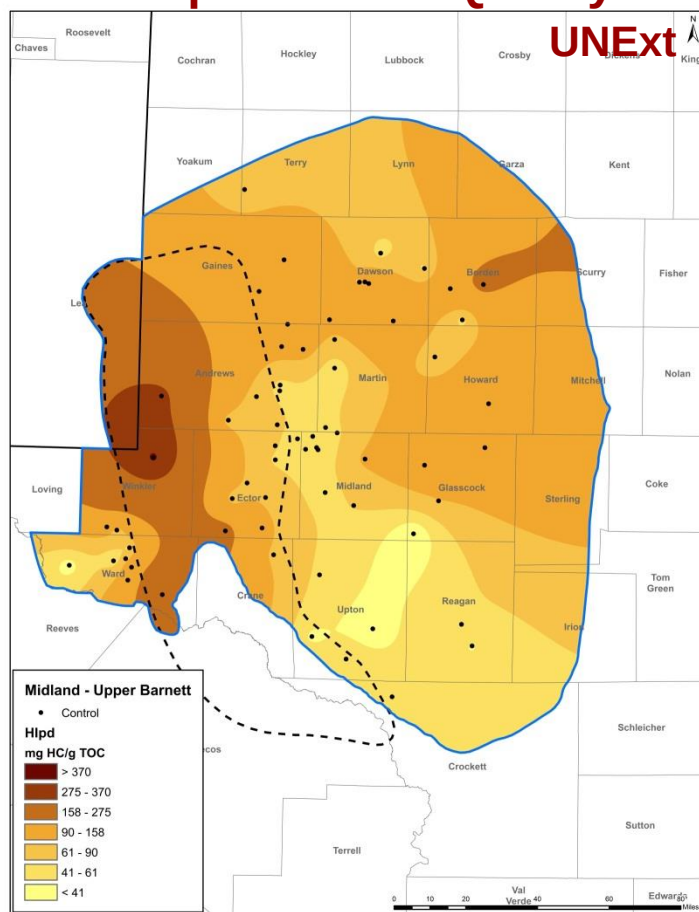
Maximum TOCpd



Upper Barnett (Hlo=400mgHC/gTOC)

Hlpd- Source Quality

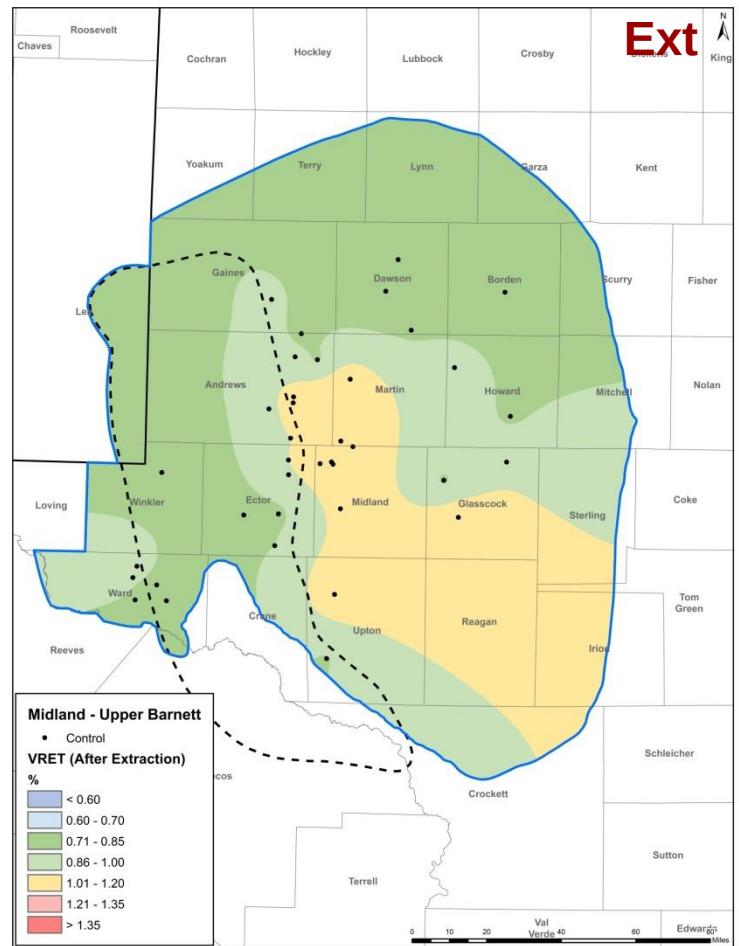
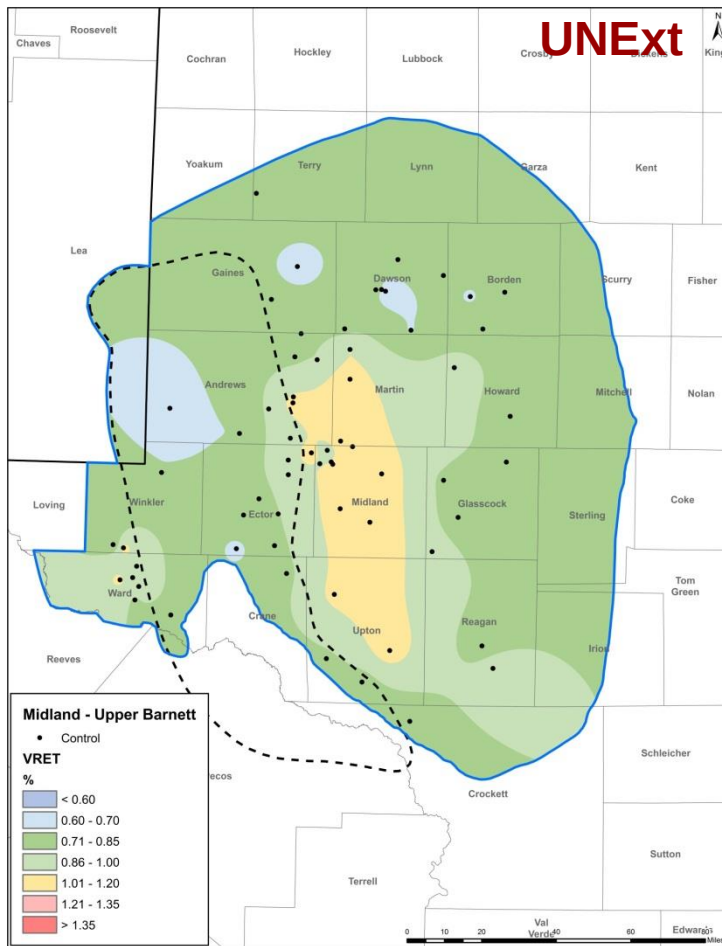
Hlpd-Thermal Maturity



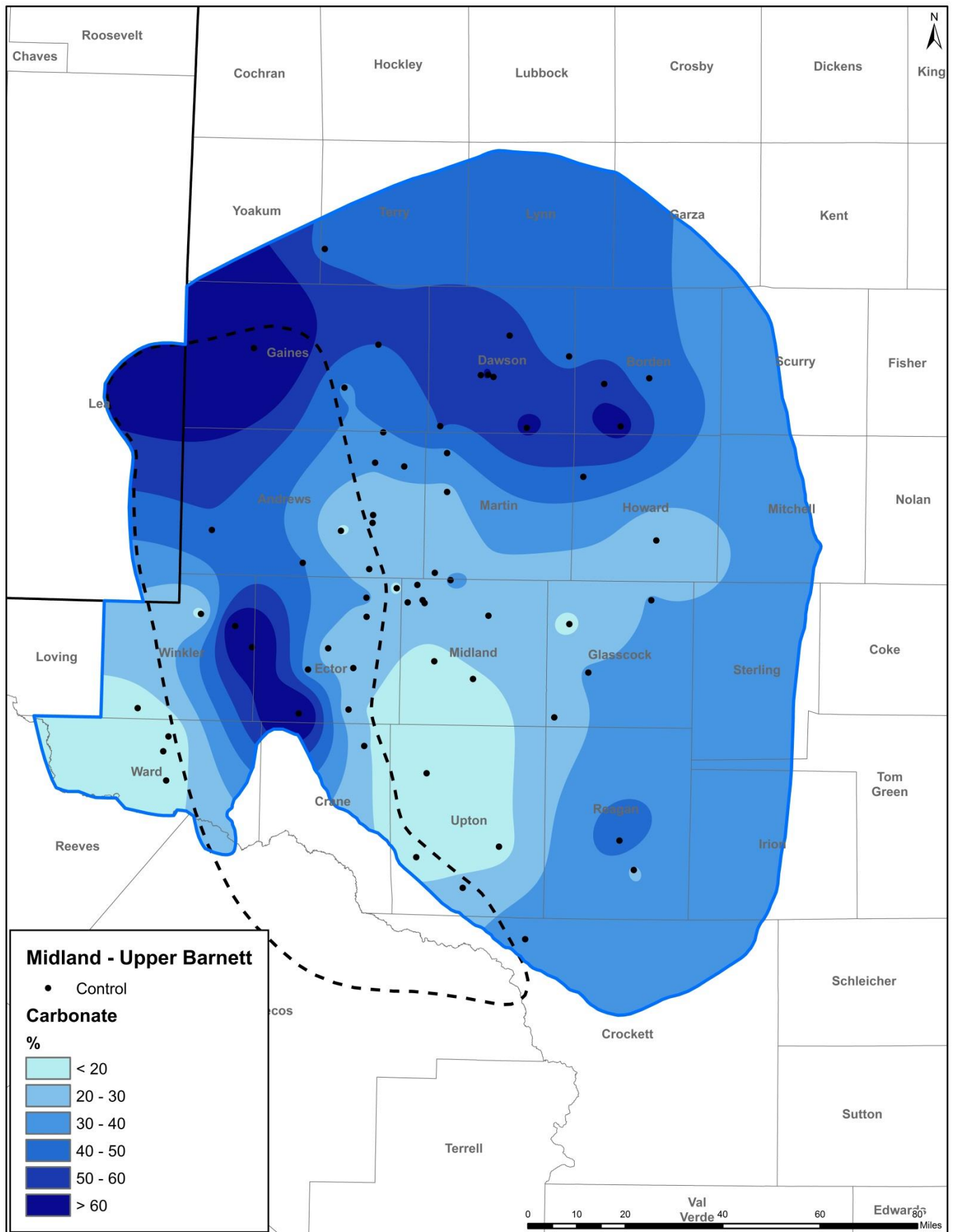
Upper Barnett

%VRET from Tmax

%VRET from Tmax



Upper Barnett Wt%.Carbonate



Sediment Interpretation

Sediment Interpretation: Woodford

The TOCpd and Hlpd maps (slides 24-25) indicate the Woodford is a very rich and oil prone source. Those data in conjunction with the other maturity parameter maps (slides 26-27) suggest that maturity, not source quality, is the major control on the Hlpd data over most of the AOI.

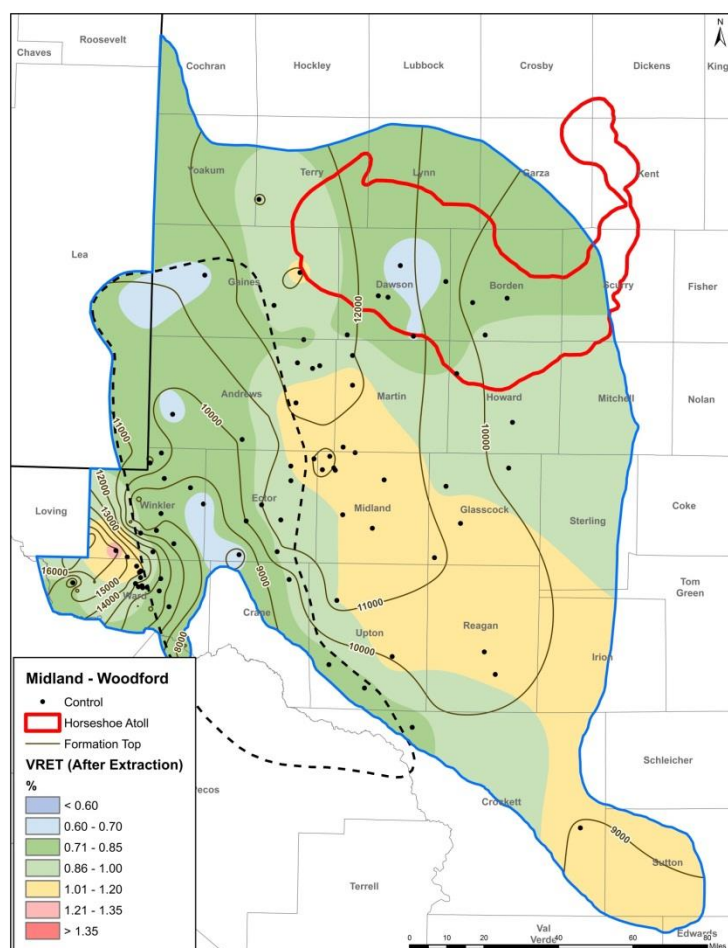
As regards maturity, however, there is a notable difference in the %VRET and measured vitrinite maps primarily in the extent of the immature/low mature area on the CBP and in the northern part of the basin in the area of Gaines/Dawson/Borden counties. The measured vitrinite reflectance map suggests that the immature area is larger than indicated by the extracted %VRET.

Measured vitrinite reflectance can be subject to suppression from the presence of insitu generated (or migrated) hydrocarbons and this issue is often most severe in the early/moderate oil window. Clearly the Woodford is very rich in these areas, and the measured vitrinite samples were not extracted prior to analysis to remove any free hydrocarbons. The somewhat higher maturity indication from the extracted %VRET and corroborated by the extracted Hlpd maturity map is the preferred interpretation in these areas. These data show the bulk of the CBP is in the zone of main oil generation.

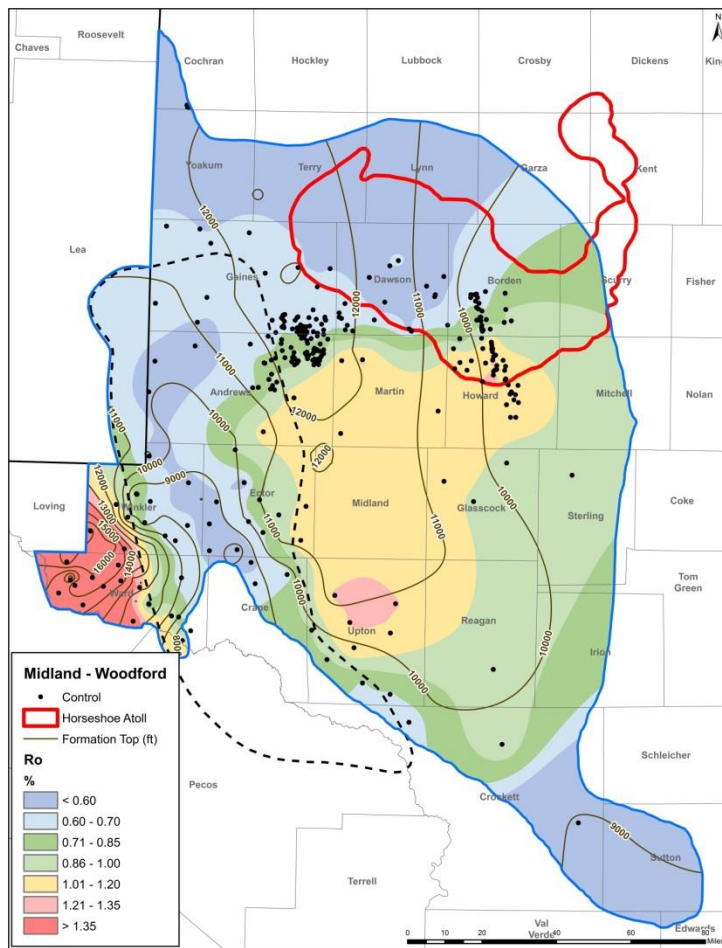
However, the present day deepest part of the basin is in eastern Gaines/Andrews over to Dawson/Martin and south to Upton/Reagan county areas (11000-12000'+). Slide 52 shows the extracted %VRET and measured %Ro maps overlaid by the present day top Woodford depth contours.

Woodford Maturity Maps with Present Day Top Woodford Contours

Extracted %VRET



Measured %Ro



Sediment Interpretation: Woodford

Interestingly, the Woodford maturity is variable in these areas from low mature/immature in the north to higher mature to the south. Additionally there is actually higher maturity to the east in Howard County (although somewhat variable between the extracted %VRET and measured %Ro maps) where depth begins to drop. These noted maturity trends are clearly inconsistent with the present day depth of burial, which suggests maturity trends should vary west to east and not north to south. Additionally the lower maturity trend (centered in Dawson County) is known to extend up into the younger sediments as seen from GeoMark's previous Wolfcamp and Spraberry studies.

Perhaps coincidentally, the area of the Horseshoe Atoll also shown in slide 52 overlays a large part of the anomalous maturity area. While it is unlikely that the atoll itself is directly related to the maturity phenomena, it may suggest a basement composition or structural change associated with a thermal anomaly across the area. A basin model approach, beyond the scope of this study, using present day corrected measured well temperatures and maturity data from this study could elucidate the extent and perhaps cause of this phenomena.

Slide 53 shows the Woodford's highest maturity in the basin in the eastern part of Andrews/western Martin counties, south to the Upton/Reagan area. At these high maturities it becomes very difficult to determine if the original kerogen quality assumptions (Hlo) are valid. Slide 17 shows the Hlpd values of several different Hlos becomes quite similar at high maturity.

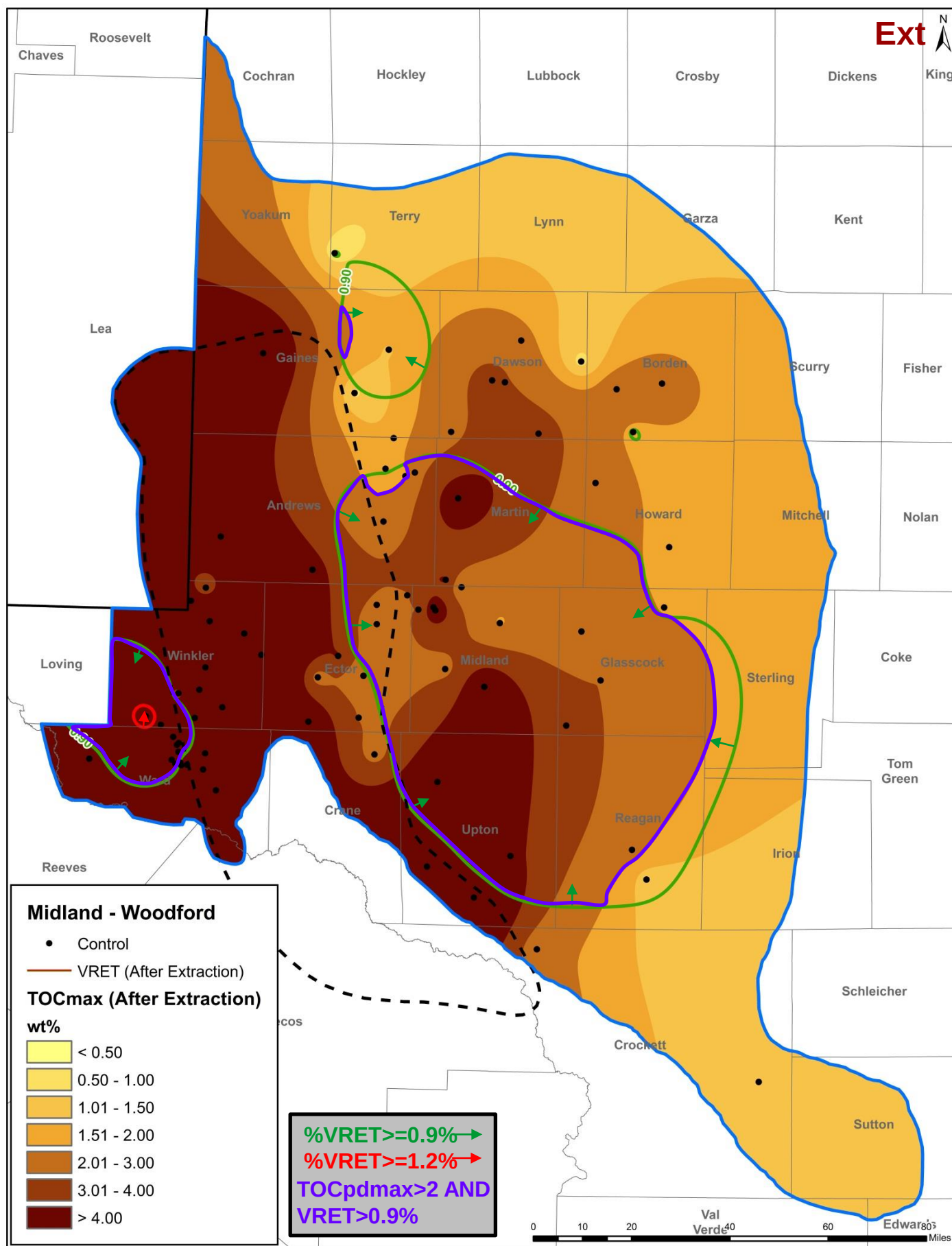
Sediment Interpretation: Woodford

However, slide 25 of extracted Hlpd (contoured for kerogen quality) and slide 24 of extracted TOCpdMax shows rich TOC values in the mature area with low Hlpd values. This suggests very good, mature Woodford source exists in this area and that the Hlo assumption of 700mgHC/gTOC is valid, or certainly a reasonable approximation.

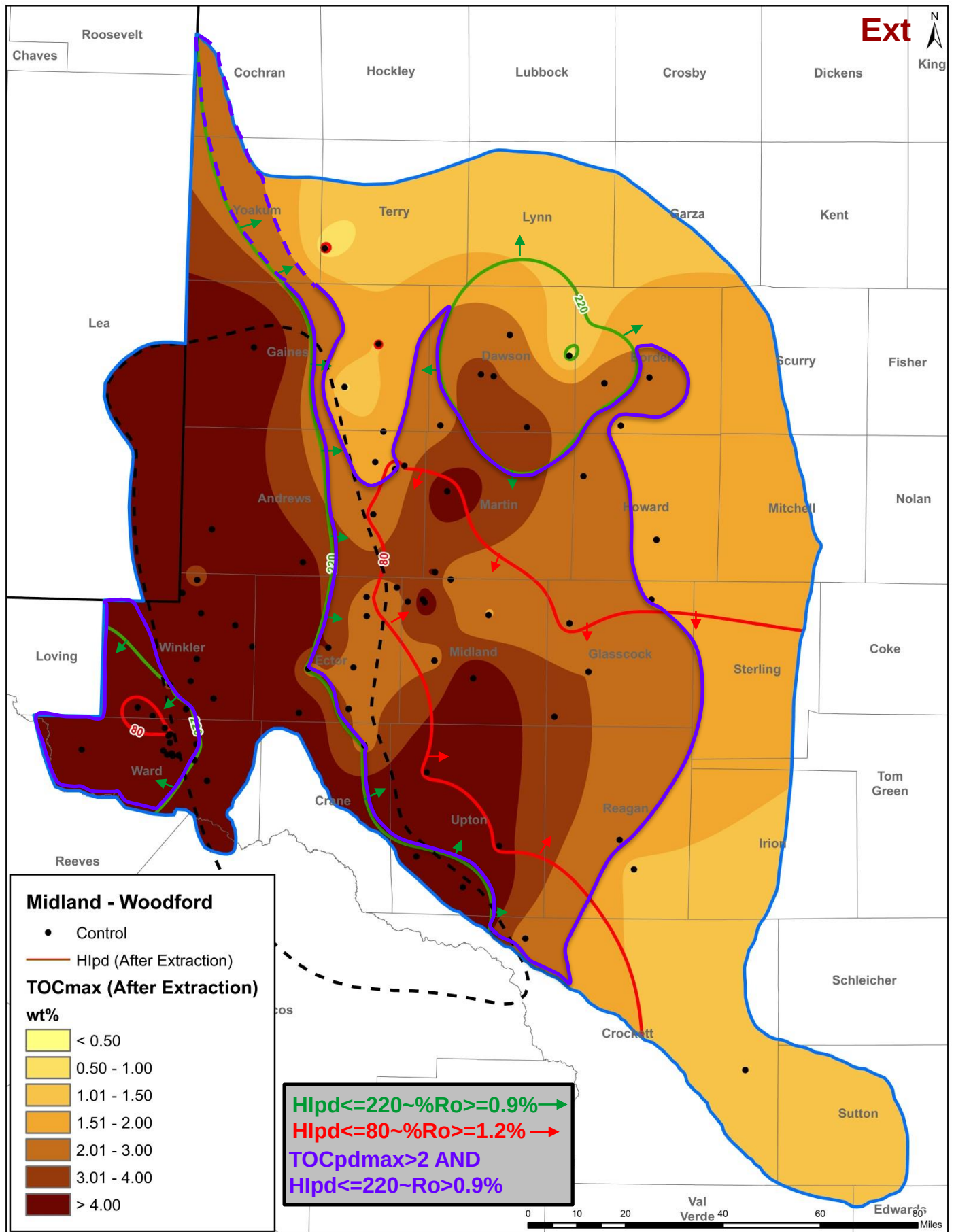
Experience has shown that generally the best unconventional wells from a geochemical point of view (certainly rock and fracture quality play an important role as well), occur where there is organic richness greater than about 2-2.5 % TOCpd and thermal maturity ≥ 0.90 -0.95%Ro/Roeq. Although hydrocarbon generation occurs at somewhat lower maturity, the higher thermal maturity is needed to lower the oil viscosity and raise the GOR to give better flow characteristics in unconventional plays.

Slide 55-57 are integrated, interpretive maps showing the extracted TOCpdMax map in conjunction with overlay contours of 0.9%, and 1.2% Ro/Roeq from the various maturity parameters. Slide 55 overlays with the extracted %VRET contours, slide 56 with extracted Hlpd (depicting %Roeq) contours, and slide 57 with the measured %Ro contours. These maps show, highlighted in purple, the rich areas ($>2\%$ extracted TOCpdmax) in the main-late oil window (Ro/Roeq $>0.9\%$). These optimal, purple highlighted areas vary somewhat with the maturity parameters, but all three figures show optimal areas in generally the same locations. The maturity variability should be considered in the risking of a play.

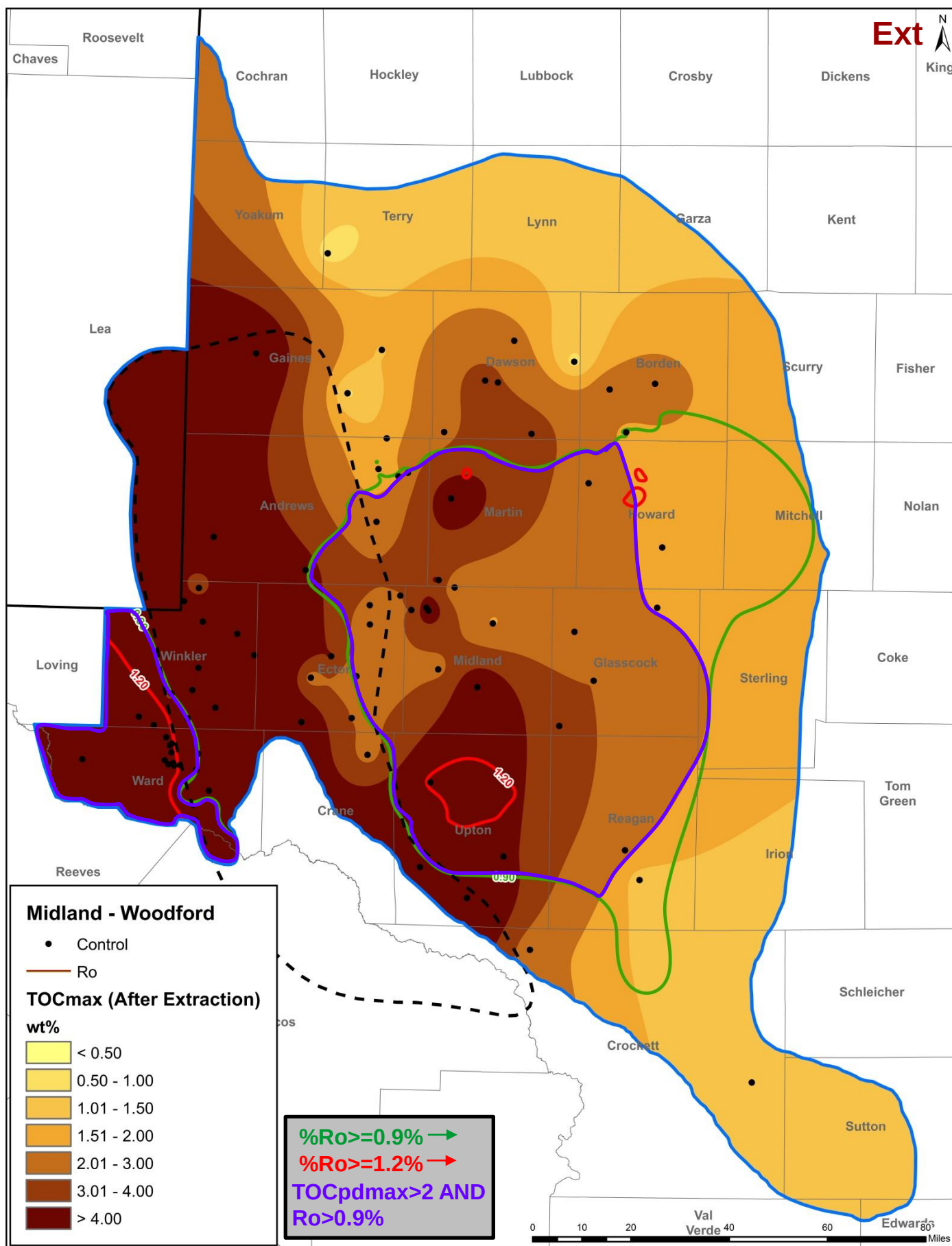
Woodford TOCpdMax and %VRET



Woodford TOCpdMax and Hlpd



Woodford TOCpdMax and Meas. %Ro



Sediment Interpretation: Woodford

To the east, on the CBP in central to eastern Andrews/Ector counties, the source and maturity conditions are optimal, with Woodford thickness about 200'. It thins notably to 100' or less farther to the east off the platform edge into Martin/Midland counties where it reaches its highest maturities. However, it is important to note that although insitu richness and maturity along with good quality fracking are positive conditions for good unconventional wells, it has also been shown from previous work by GeoMark and others that they are not always necessary conditions. We have shown that migration can and does occur within unconventional source/reservoir systems. The maturity and structure maps for the Woodford suggest that fluid migration from downdip kitchens up onto the CBP within the Woodford could also contribute to unconventional plays on the CBP.

The eastern shelf area is generally not well data constrained, although as seen from slide 22 the present-day Woodford burial depth in that area is generally less than 10000 feet, so low to modest maturity is likely. As will be seen in the oils section of this report there has been notable migration of Woodford generated and Woodford mixed conventional oils onto both the eastern shelf and CBP areas. These oils have likely migrated updip to the east and west, respectively from the deep basin.

There is also an optimal area just off the edge of the CBP to the west in Ward/Winkler counties where the Woodford averages over 500' thick.

Sediment Interpretation: Woodford

In Ward/ Winkler, as was pointed out in the original GeoMark Woodford report from 2014, the area of down dip oil-window mature Woodford is somewhat geographically limited, as the Woodford maturity increases through the main and light oil window into the wet gas to dry gas window over a fairly short geographic area. However, oil previously generated from the Woodford in this present day overmature area could have also migrated onto the CBP.

Sediment Interpretation: Miss. Lime

The dominantly calcareous Mississippi Lime (slide 35), separates the Lower Barnett from the Woodford. It is age equivalent to the Mississippi Lime in Oklahoma and Kansas where it is more a conventional reservoir that has fairly recently been drilled horizontally, with mixed success in those areas. The few attempts at drilling it as a reservoir in the northern Midland basin have been disappointing.

On the eastern CBP and to the south in Martin/Midland/Upton counties, the Mississippi Lime has moderate organic richness and is generally resistive with an enhanced gamma ray response. It is included in this study to investigate the possibility of it being self-sourcing. However, Hlpd data combined with the %VRET suggests its source quality and potential is overall notably less than the Woodford. For this reason, an Hlo of 400mgHC/gTOC is interpreted for the Mississippi Lime. It does not appear to be a very high quality target.

Sediment Interpretation: Miss. Lime

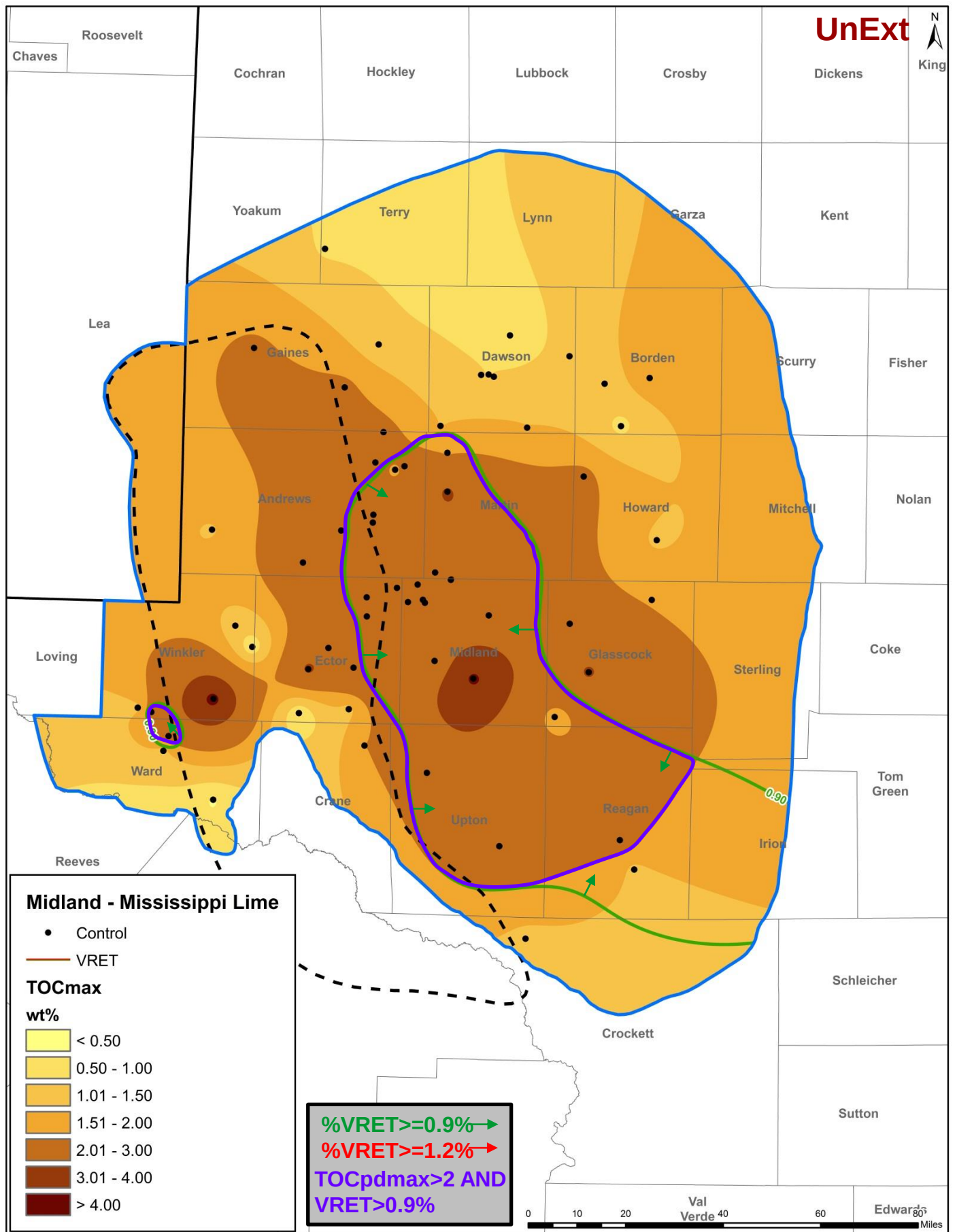
Slides 61 combines the richness and maturity parameters on one map, similar to that done for the Woodford, showing a highlighted, “optimal” area on the basis of TOCpdmax and %VRET contours. Slide 62 shows the highlighted area using the Hlpd maturity contours ($H_{lo}=400\text{mgHC/gTOC}$ source). Given the lack of data control for the extracted parameters, the TOCpdmax base and %VRET and Hlpd maturity contours are derived from unextracted samples, so these highlighted areas are very likely overly optimistic.

Slide 62 shows several separate “high maturity” (low Hlpd, $<60\text{mgHC/gTOC}$) areas in the Martin/Midland/Upton county areas off the CBP that are interpreted to indicate poor quality kerogens rather than high maturity. The concern for kerogen quality in the highlighted optimal (on the basis of richness and thermal maturity) areas in slides 61-62 needs to be strongly considered.

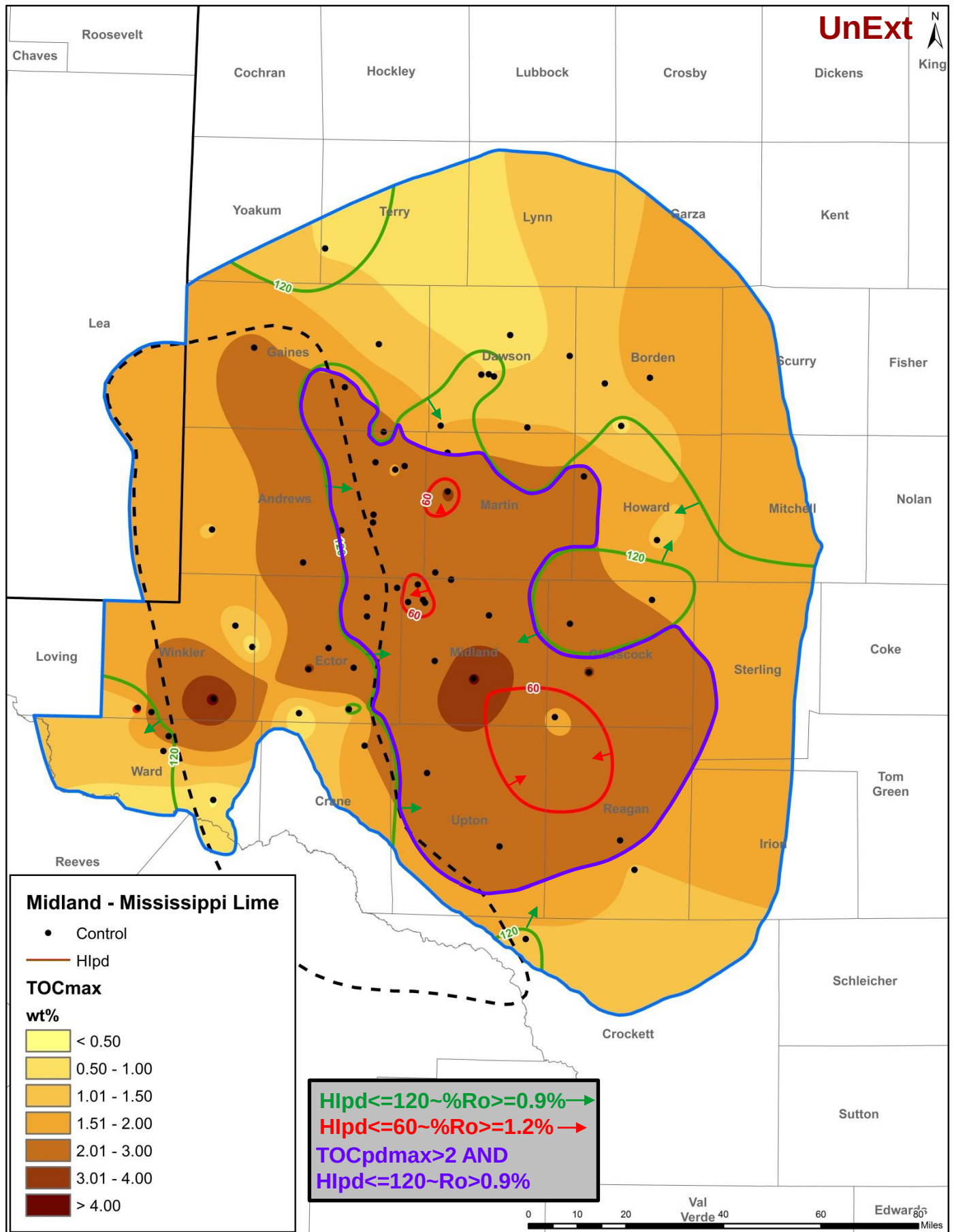
A fairly small area that could be an exception to the cautions described above is in the highlighted area on the far eastern side of Andrews/Ector counties on the CBP. Here the kerogen quality appears better and the Mississippi Lime thickness varies from ~150-350’.

Finally, from this study, the Mississippi Lime cannot be ruled out as a possible reservoir target. It may be possible to find areas of good Woodford potential where fracking the Mississippi Lime might allow it to access upward migrating Woodford charge. Reservoir quality (RQ) properties, beyond the scope of this study, need to be evaluated.

Miss.Lime TOCpdMax and %VRET



Miss.Lime TOCpdMax and Hlpd



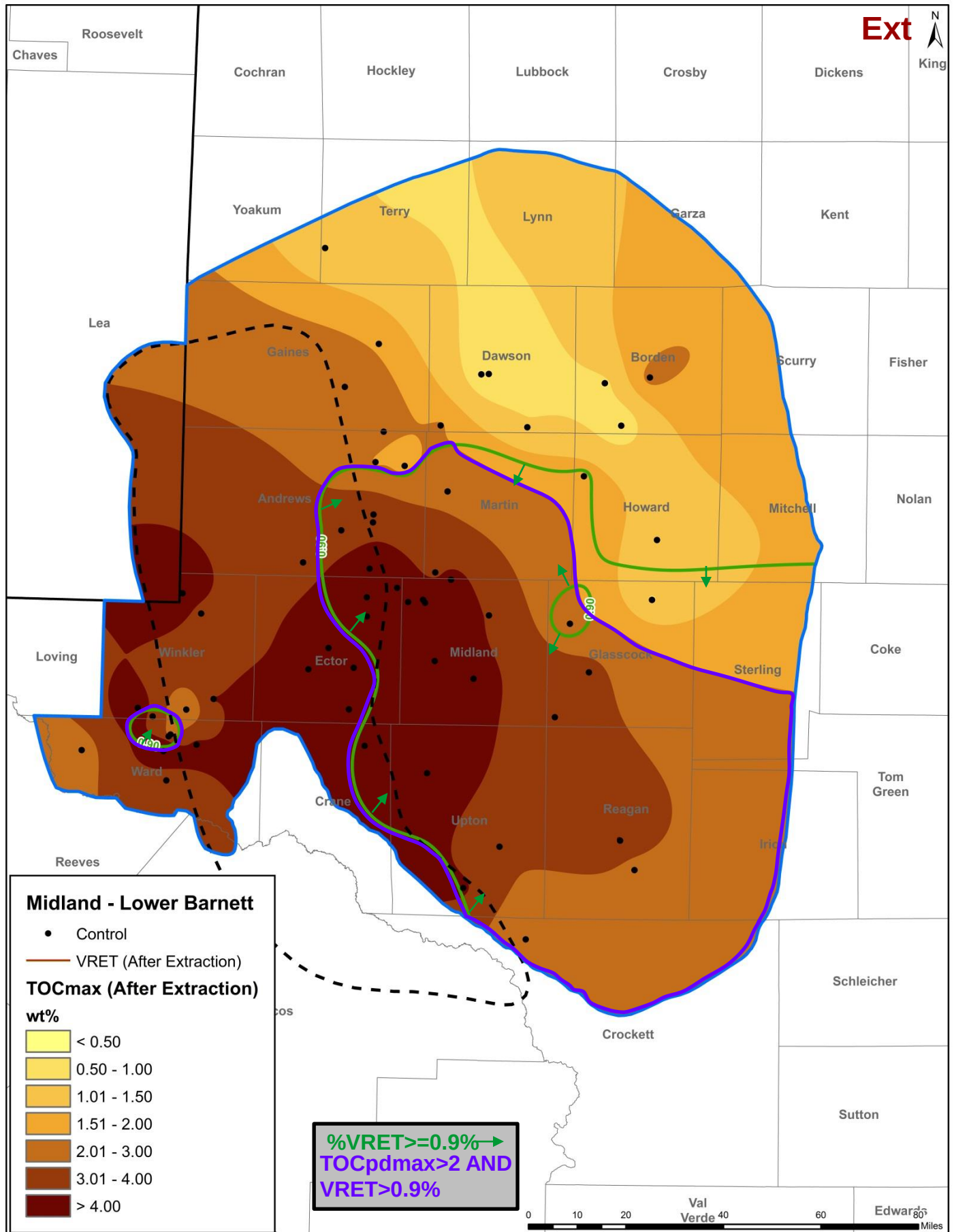
Sediment Interpretation: Lower Barnett

Similar to the Woodford, the Lower Barnett is a high quality, rich source, and its quality distribution is similar to the Woodford. It is interpreted for this study to have an Hlo of 600mgHC/gTOC.

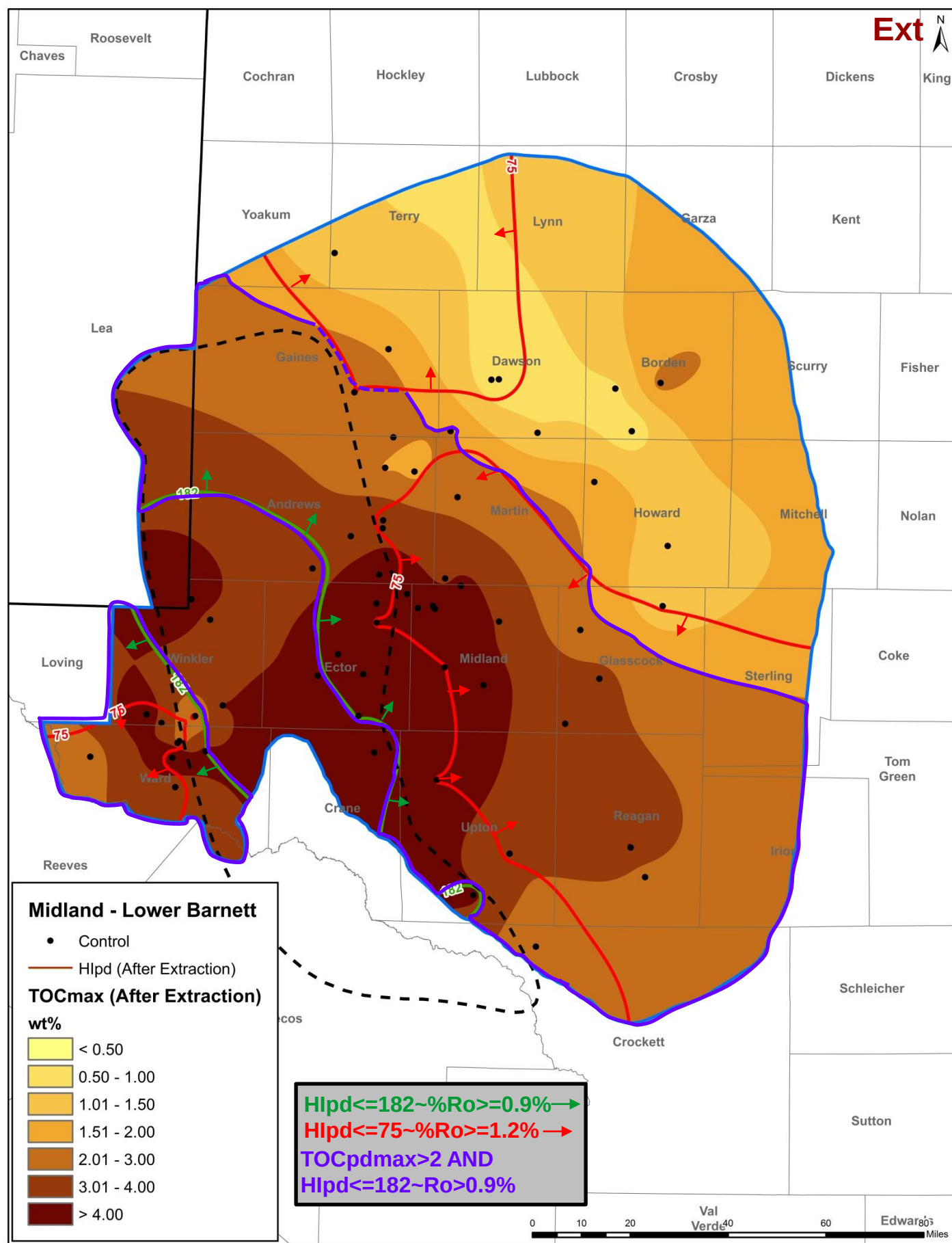
The only area where the Lower Barnett seems to be of generally poorer quality than the Woodford is in the north and northeastern part of the basin in the general area of Dawson/Borden/Howard/Mitchell counties. The source quality diminishes in those areas and Hl_{pd} is interpreted as an unreliable maturity parameter as the low values likely reflect poor kerogen quality rather than high thermal maturity.

Slides 64-65 are the integrated, interpretive maps showing the extracted TOC_{pd}Max map in conjunction with the overlay maturity contours of Ro_{eq} 0.9%, and 1.2% for extracted %VRET and extracted Hl_{pd}, respectively. These maps show the purple, highlighted optimal areas for the Lower Barnett are the central and eastern parts of Andrews/Ector counties on the CBP and extending farther east off the platform into Martin/Midland/Upton counties. Farther to the east into more central and eastern parts of Martin/Midland/Upton, kerogen quality becomes somewhat difficult to interpret due to increasing thermal maturity. However, the high extracted TOC values in that area and the extracted %VRET data suggest that not only in-situ potential, exists off the platform, but also if rock quality is adequate, migration of more mature hydrocarbons within the Barnett onto the platform can also occur.

Lower Barnett TOCpdMax and %VRET



Lower Barnett TOCpdMax and Hlpd



Sediment Interpretation: Lower Barnett

The Ward and Winkler area on the western edge of the platform is also an optimal area for the Lower Barnett. Slide 64 with the extracted %VRET contours shows the area is notably smaller than the area in slide 65 with the extracted Hlpd contours. The Hlpd maps in this area are interpreted as representing the more likely scenario. It is not clear why the %VRET data is problematic. However, this interpretation is supported by the maturity of the underlying Woodford which is roughly less than 1000' below the Lower Barnett, so only a slightly lower maturity would be expected for the Lower Barnett. The area of optimal Woodford is generally similar to that indicated in slide 65.

Sediment Interpretation: Upper Barnett

The shallowest of the formations mapped for this study is the Upper Barnett. The richness seen in the Lower Barnett does not appear to continue into the upper part of the section. In comparing the %VRET and %Hlpd maps, the overall kerogen quality for the Upper Barnett is interpreted as about 400mgHC/gTOC (this value may be somewhat optimistic). Like the Mississippi Lime, the Upper Barnett is not likely an attractive target.

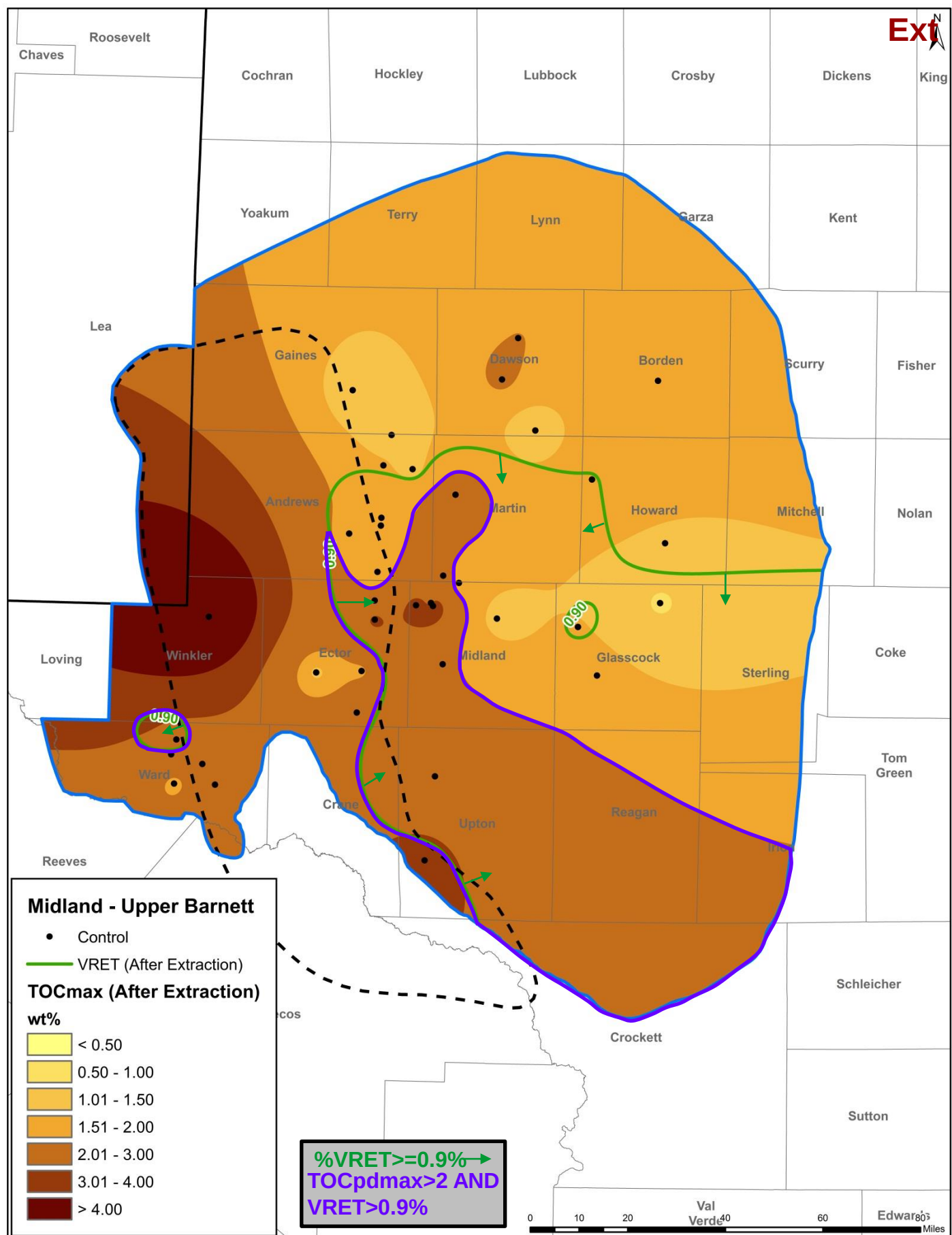
There is some richness (TOC maps) in the northern Ector/Midland county area extending south farther into Midland/Glasscock and to northern and central Upton county. However, the TOCpdmax map is very heavily influenced by some quite highly contrasting values to the TOCpd map.

Sediment Interpretation: Upper Barnett

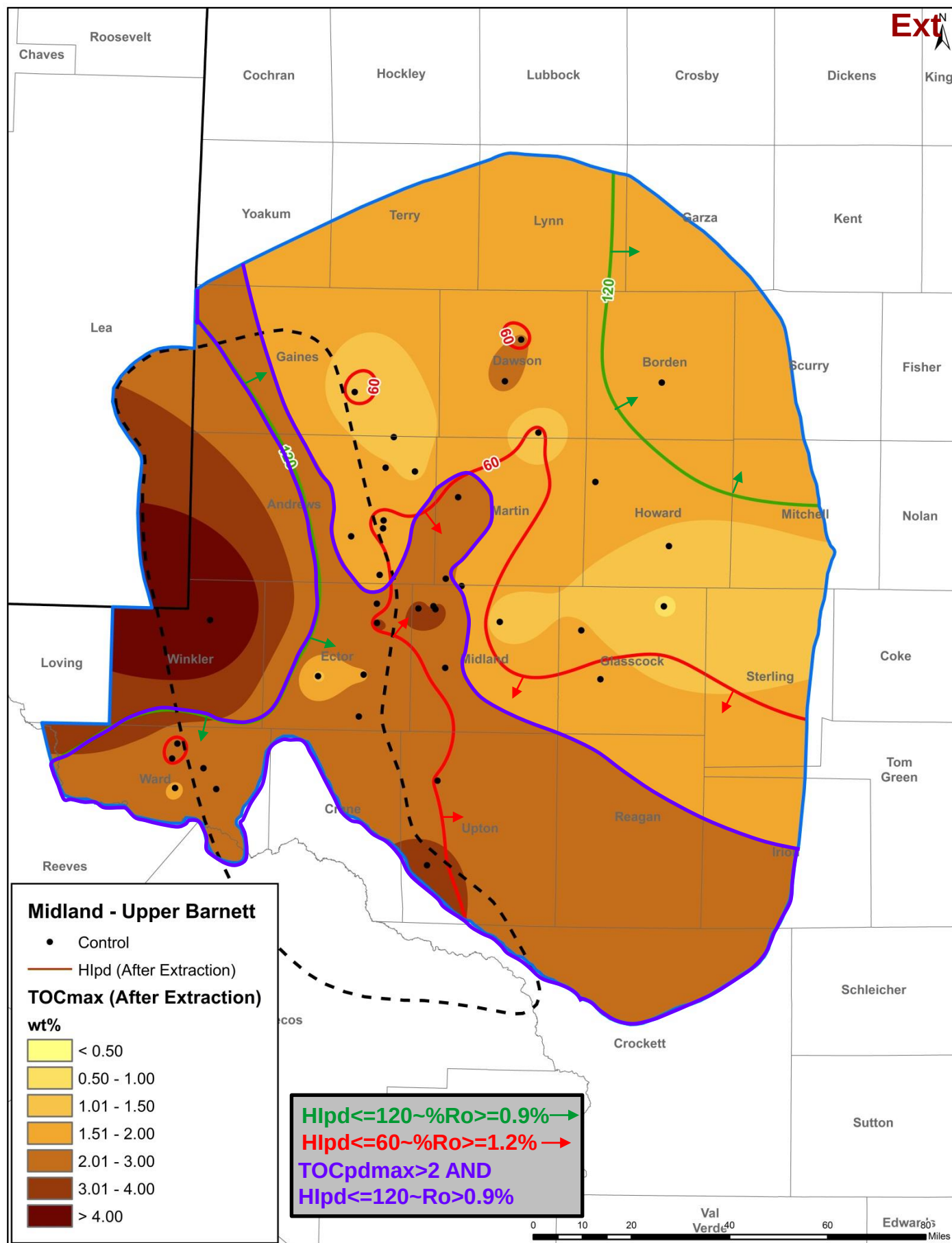
The samples with these very high values were examined for obvious contamination/coaly material (none found) and rerun, with similar results. In comparing the extracted TOCpdmax map with the unextracted map, most of the high values dropped significantly. Unfortunately, there was insufficient sample amounts to extract all the samples with high values.

The low/moderate richness values and the generally poor kerogen quality indicate the Upper Barnett is not interpreted as a good target. However maps of maturity contours overlying TOC, as made for the other formations, are shown for comparison. Slide 68 is the Upper Barnett extracted TOCpdmax map with the overlying extracted %VRET contour of 0.9% (no 1.2% contour). Slide 69 has the extracted TOCpdmax map with the extracted Hlpd contours of 120 and 60mgHC/gTOC, corresponding to 0.9% and 1.2%, respectively.

Upper Barnett TOCpdMax and %VRET



Upper Barnett TOCpdMax and Hlpd



Oils, Extracts, and Interpretation

Oils Interpretation

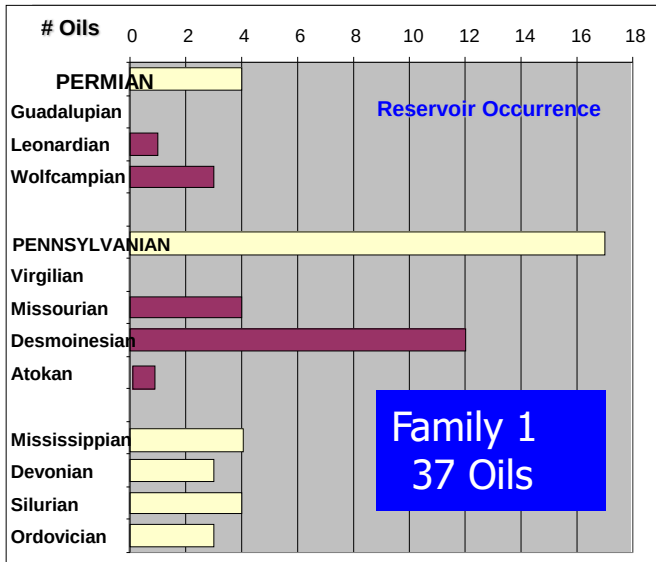
Over 100 oils from GeoMark's Regional Permian Basin Oil Study are generated from the Woodford and/or Lower Barnett and are included in this study. They are characterized and compared with sediment results. None of the oils in this study are interpreted as generated from either the Upper Barnett or the Mississippi Lime.

The oils were analyzed using multivariate statistical analysis of biomarker and carbon isotopic compositions. From that analysis, three definitive families generated from the Woodford were identified as well as one family generated from the Lower Barnett. Oils generated from the Woodford in the Anadarko and Arkoma Basins correlate well with the three families identified in this study. The Barnett family 4 oils correlate very strongly with oils from the Fort Worth Basin generated by the Barnett and are similar, but distinguishable from the Woodford oils. Slide 72 shows histograms of the reservoir occurrences for the various families.

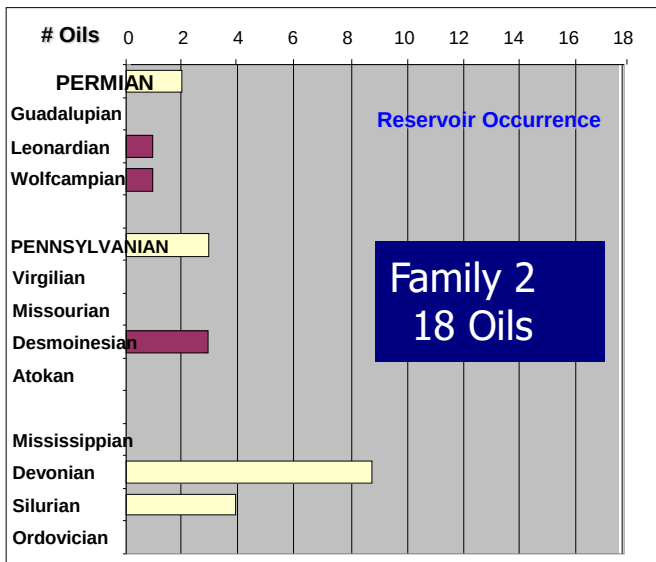
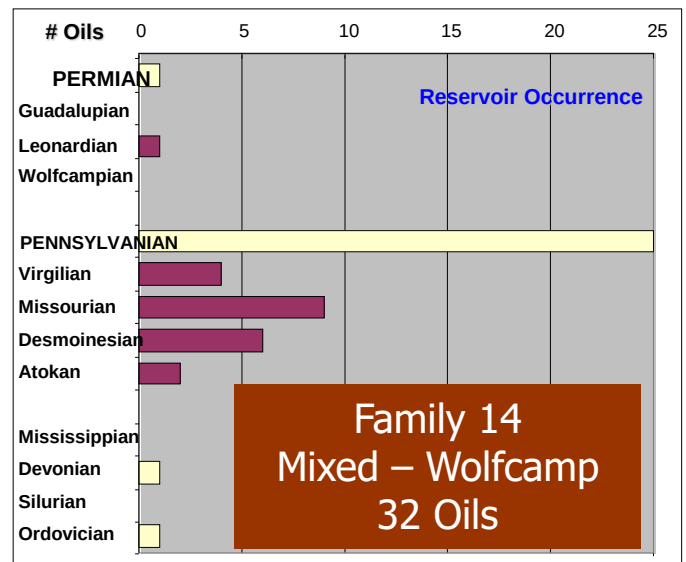
Although there are three geochemically distinguishable families generated from the Woodford, they are very similar. There is an interpreted mixed family that has both Woodford and Wolfcamp affinities (Family 14). Slides 73-74 show maps with the geographic distribution of the four Woodford influenced families (families 1-3, 14) along with Woodford sediment maturity contours of 0.7%, 0.9%, and 1.2% from two different maturity parameters, measured %Ro and extracted %VRET, respectively.

Reservoir Occurrence of Study Oils

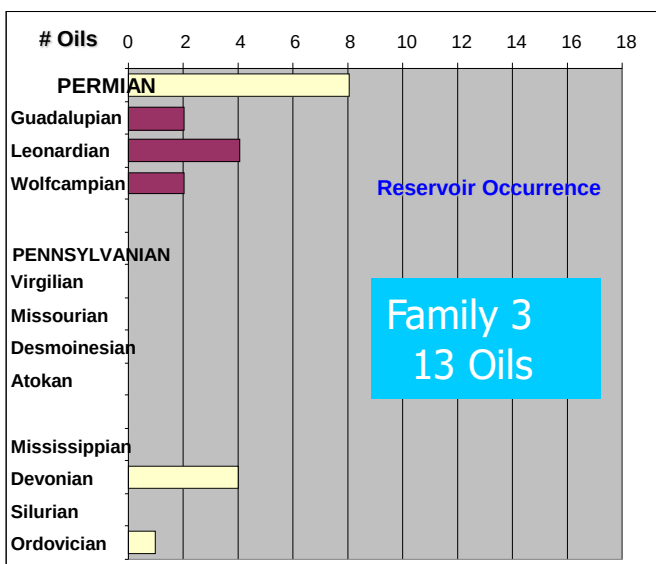
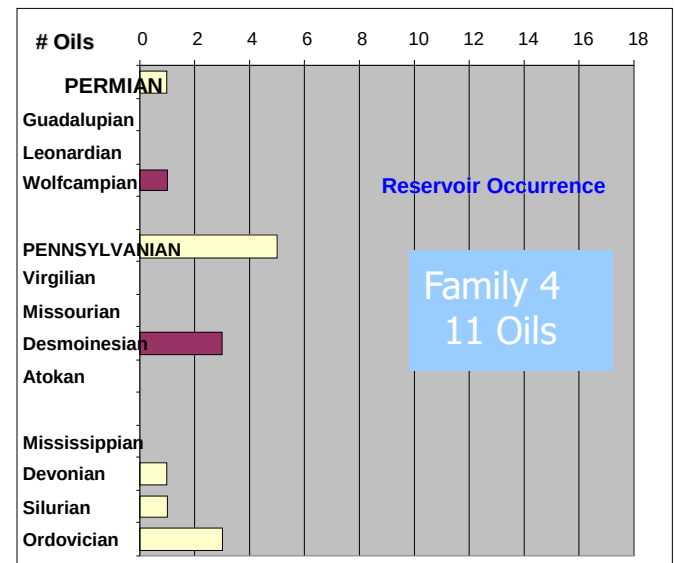
Woodford Source



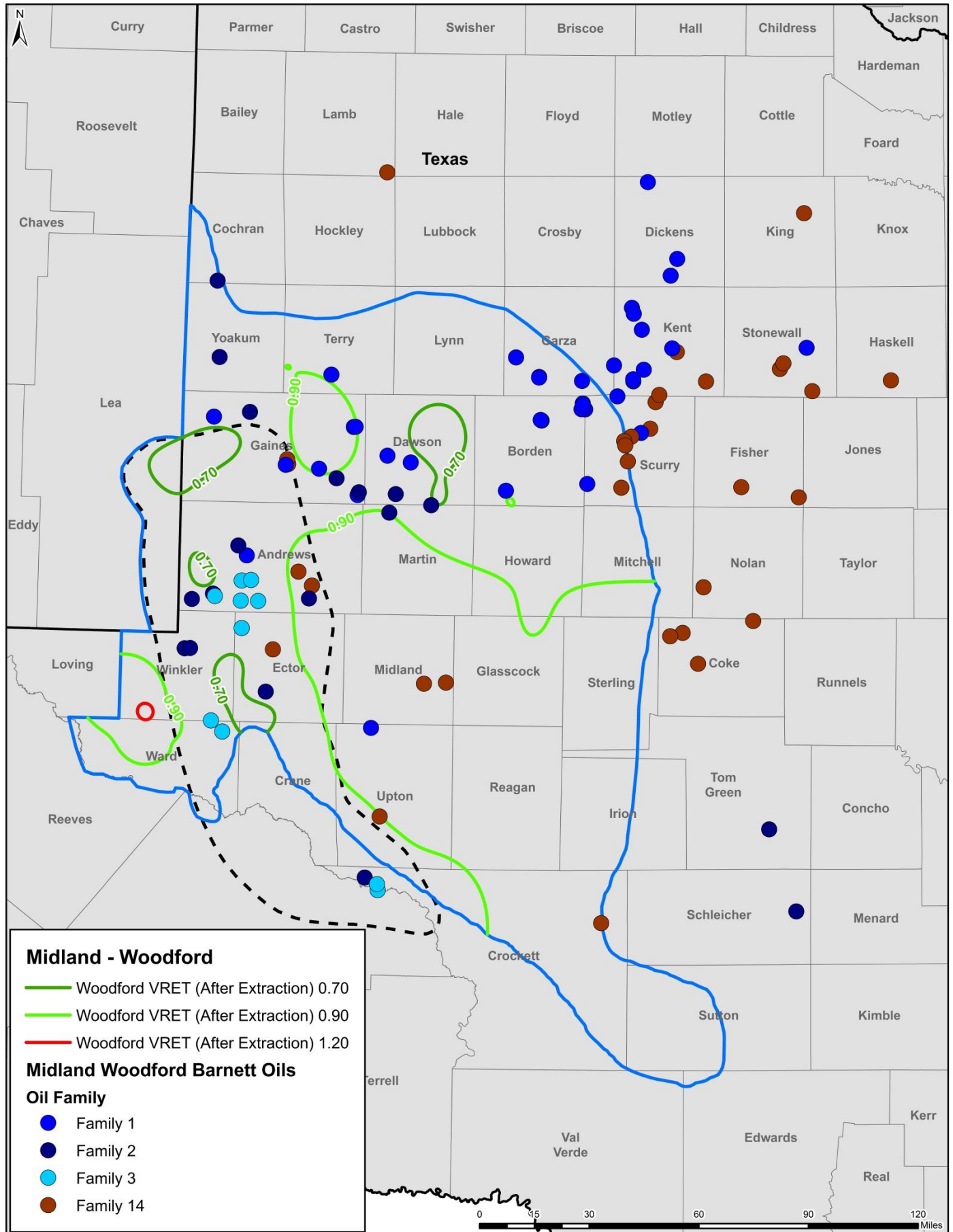
Mixed Woodford



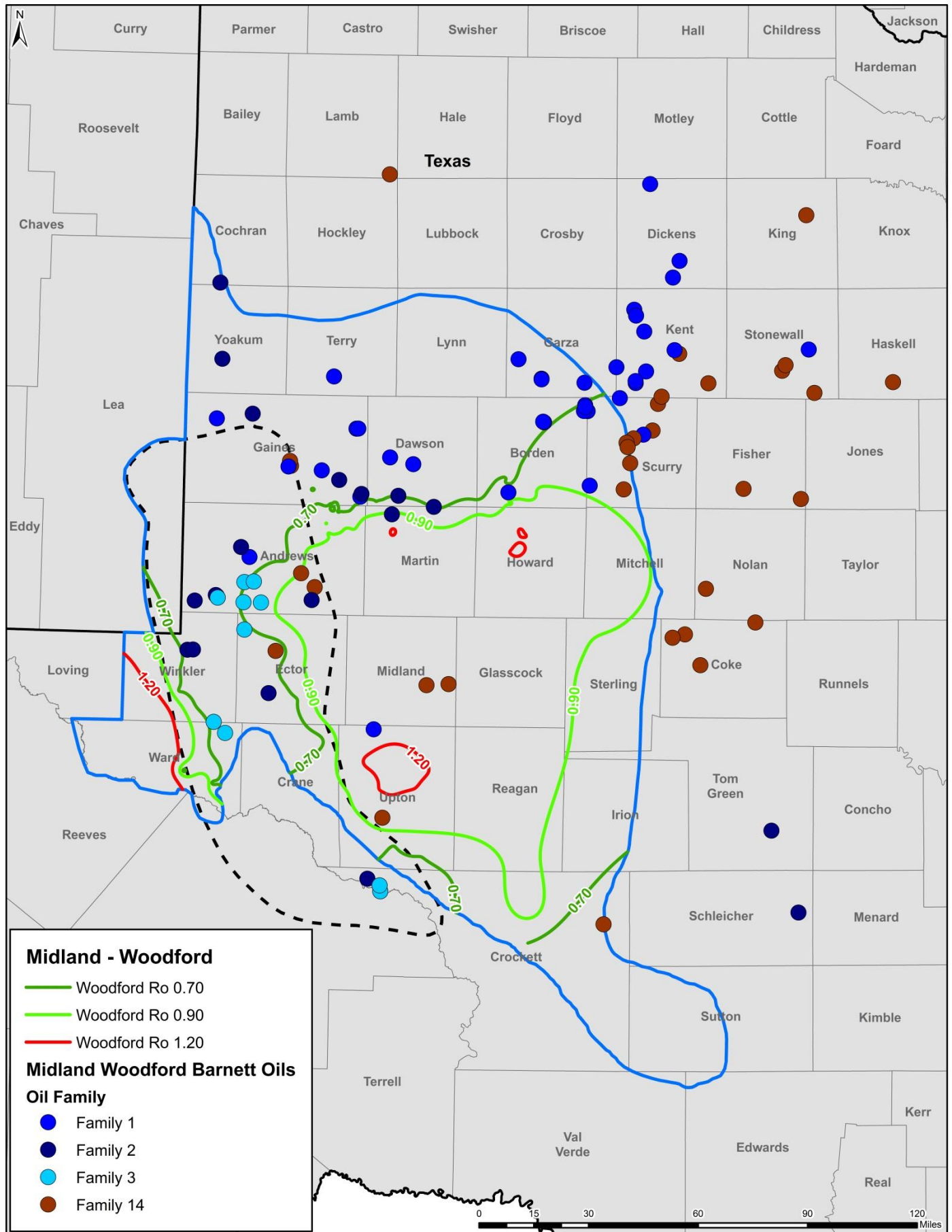
Lower Barnett Source



Woodford Oil Families 1-3,14 with Source Maturity, Extracted %VRET



Woodford Oil Families 1-3,14 with Source Maturity, Measured %Ro



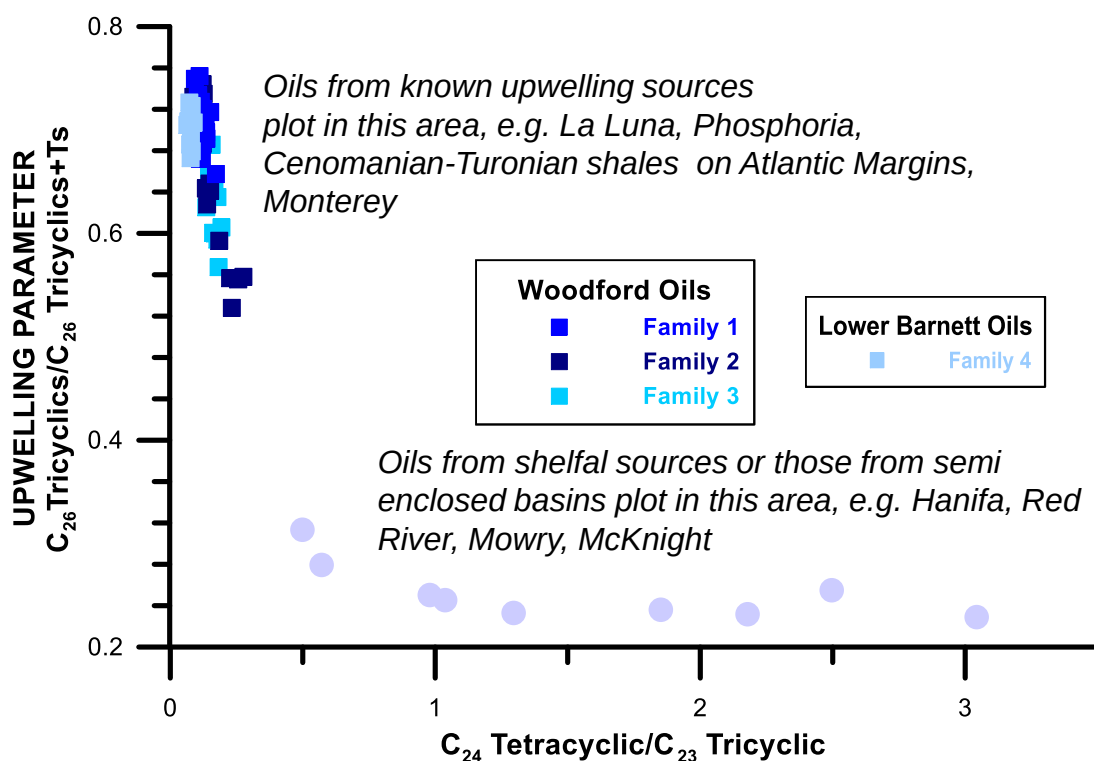
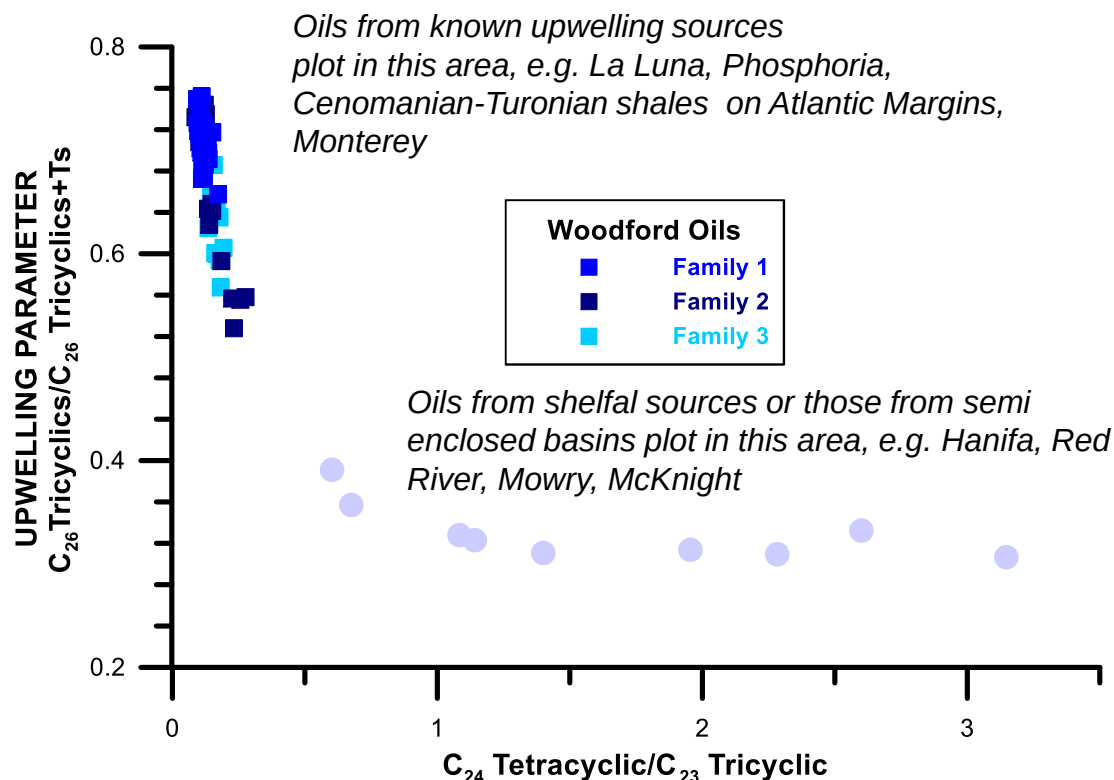
Oils Interpretation

Since the oils in the study occur in conventional reservoirs, the lower maturity contour of 0.7% is included on these maps since generation and expulsion of low/early mature oils into higher quality (relative to unconventional) conventional reservoirs certainly occurs. The maps show there is some geographic distinction in the families.

The Family 1 oils are dominantly in the northern Midland Basin trending eastward toward the basin edge. Many clearly occur where the Woodford sediment maturity is immature to low mature. Family 2 has a scattered distribution, with many on the CBP where the Woodford sediment is generally immature except near its edges where maturity increases rapidly into the basin. Family 3 is only on and near the CBP. Family 14, the mixed family, has a number of occurrences on the eastern shelf area and a few near the western edge of the CBP. The occurrences of both the Family 1 and Family 14 oils suggests they have migrated north and east likely out of the deeper parts of the basin.

Woodford deposition occurred during a time of upwelling and high productivity conditions in the Permian (Tobosa) Basin. Its oils are all characterized by a notable abundance of a homologous series of extended tricyclic biomarkers that are associated with upwelling environments. Slide 76 shows a plot (upper plot) of the Family 1-3 oils showing two sets of biomarkers that distinguish upwelling source input from those of shelfal sources.

Upwelling Biomarkers



Oils Interpretation

All the oils have fairly strong upwelling characteristics, but as with their geographic distribution, the three families have some variability. Family 1 has the strongest upwelling characteristics but it overlaps with Families 2 and 3, with Family 2 having the greatest spread.

The principal component analysis strongly indicate that the variation in the three main Woodford oil families is due to depositional environment variability with upwelling conditions the dominant influence. The oils may all be mixtures generated from various Woodford depositional environment end member facies with the upwelling facies being likely the most organic rich and, therefore, dominantly represented in the oils.

The upwelling conditions associated with elevated productivity during Woodford deposition contributed significantly to the present day Woodford organic richness and its excellent initial source potential. However, preservation of the accumulated organic matter was also extremely important.

The oils also contain a series of aromatic biomarkers (aryl isoprenoids) indicating enhanced preservation of Woodford organic matter which occurred during conditions of photic zone euxinia (PZE). PZE occurs when sulfate reducing bacteria living in the water column make H_2S , creating water column anoxia. The anoxia results in enhanced preservation of organic matter in the underlying sediments.

Oils Interpretation

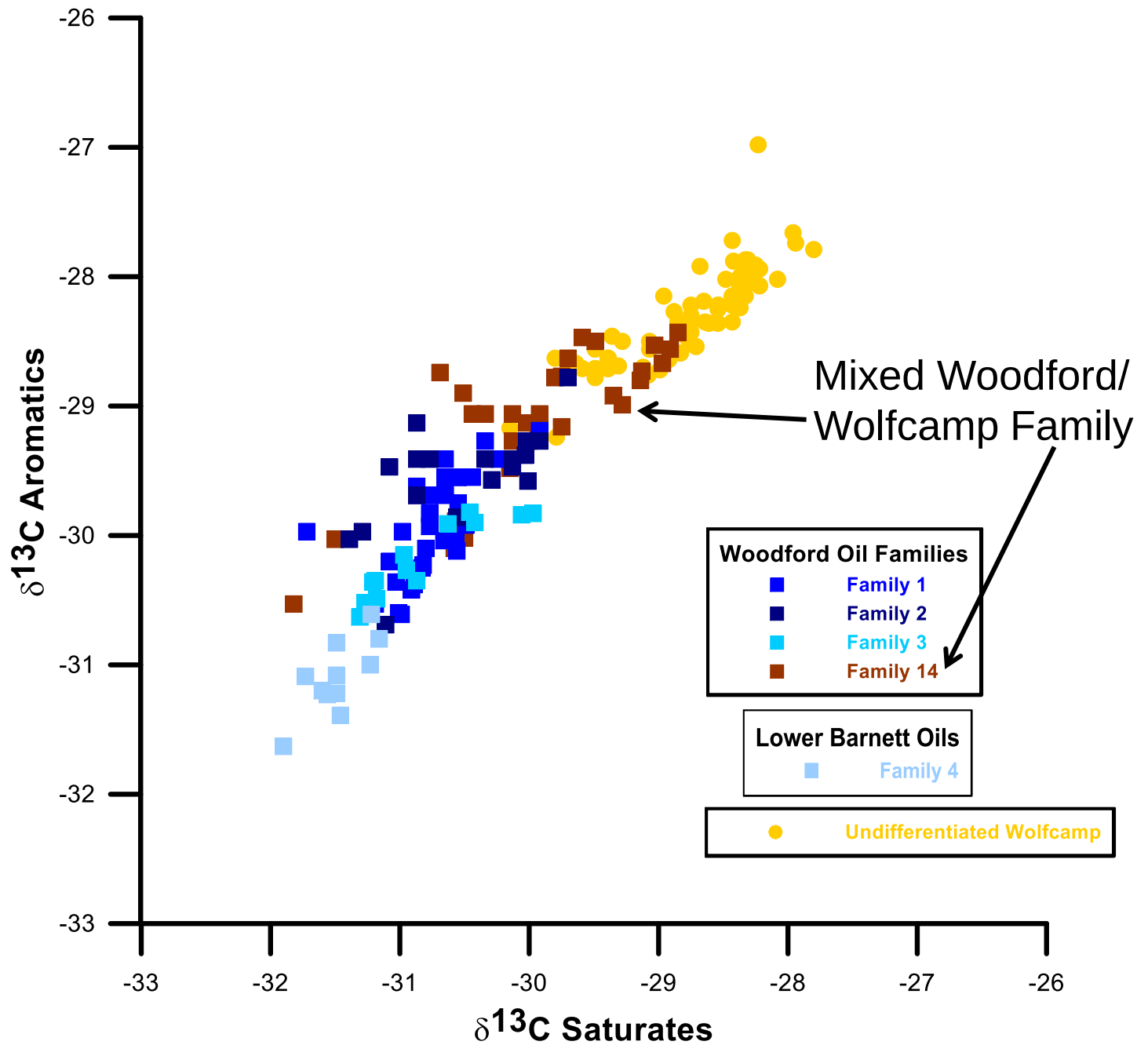
Aryl isoprenoids are a metabolic byproduct of a species of green photosynthetic sulfur bacteria that live in the photic (shallow water) zone and consume H_2S . Therefore, the presence of the aryl isoprenoid biomarkers in oils associates them with conditions of enhanced organic matter preservation.

Aryl isoprenoids are somewhat thermally labile and their concentration diminishes significantly in oils in the upper oil maturity window, around 0.95%-1.0% Ro. The chemistry of the oils is a strong indicator that the excellent Woodford source quality is the result of both enhanced productivity and preservation.

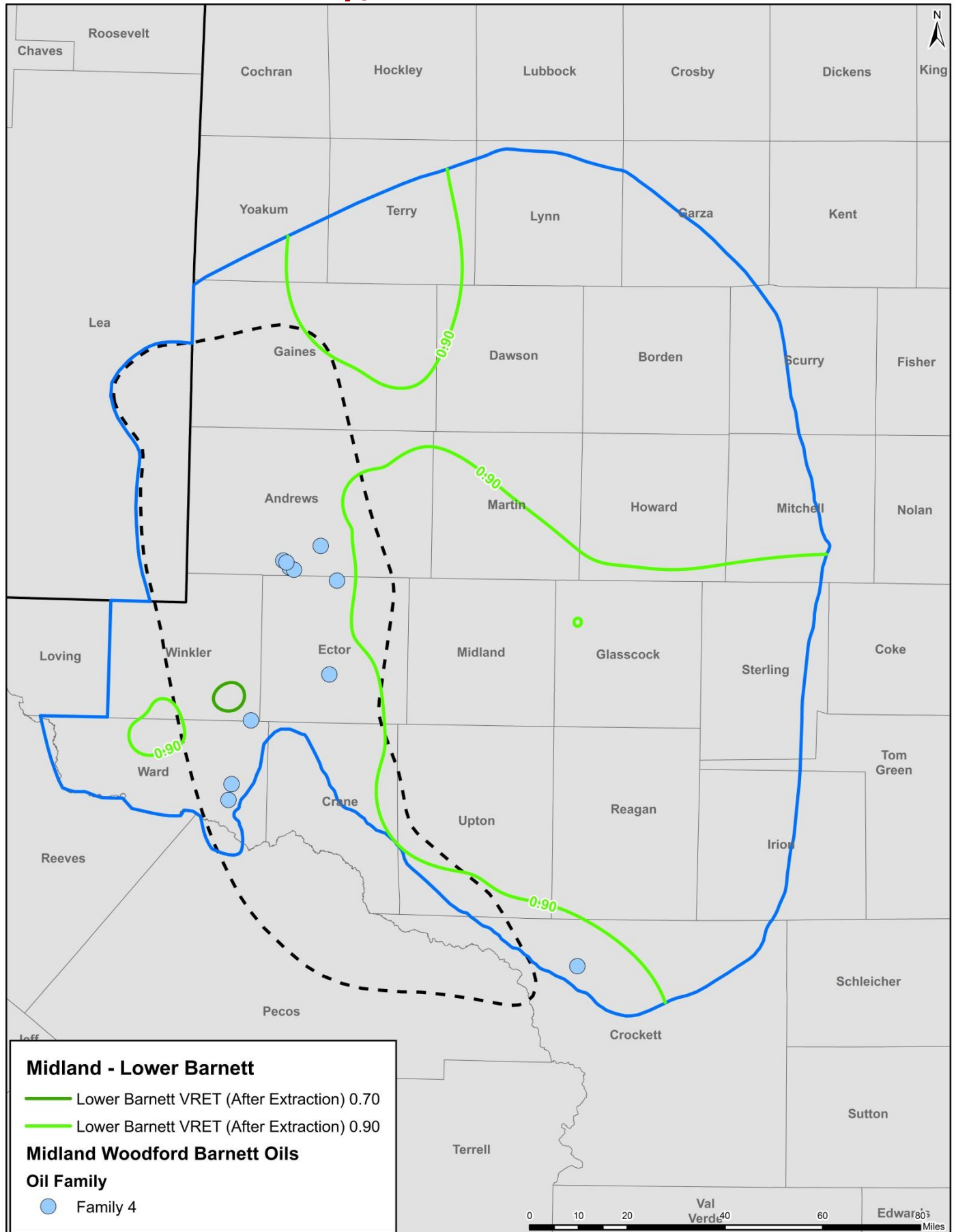
Family 14 oils have an intermediate composition between the end member Woodford and Wolfcamp oils. Slide 79 shows Family 14 occurs compositionally, on the basis of isotopes, between the Wolfcamp and Woodford family oils. All the Family 14 oils contain the Permian specific tetracyclic biomarker. As mentioned previously, they are generally located on the edges of the Midland Basin (slides 73-74), many to the east where the Woodford sediment is absent or immature, suggesting they have migrated outward from the deeper parts of the basin.

The Family 4 oils are very similar to the main Woodford families 1-3 oils but they correlate with the Lower Barnett shale. They occur in Permian-Ordovician aged reservoirs and are generally located on the CBP as shown in slide 80 which also shows the Lower Barnett extracted %VRET maturity contours of 0.7% and 0.9% Ro.

Isotopic Compositions



Lower Barnett Oil Family 4 with Source Maturity, Extracted %VRET



Oils Interpretation

Like Woodford oils, the Lower Barnett oils also have upwelling biomarkers (slide 76, lower plot) and are isotopically fairly light, generally lighter than most Woodford oils (slide 79).

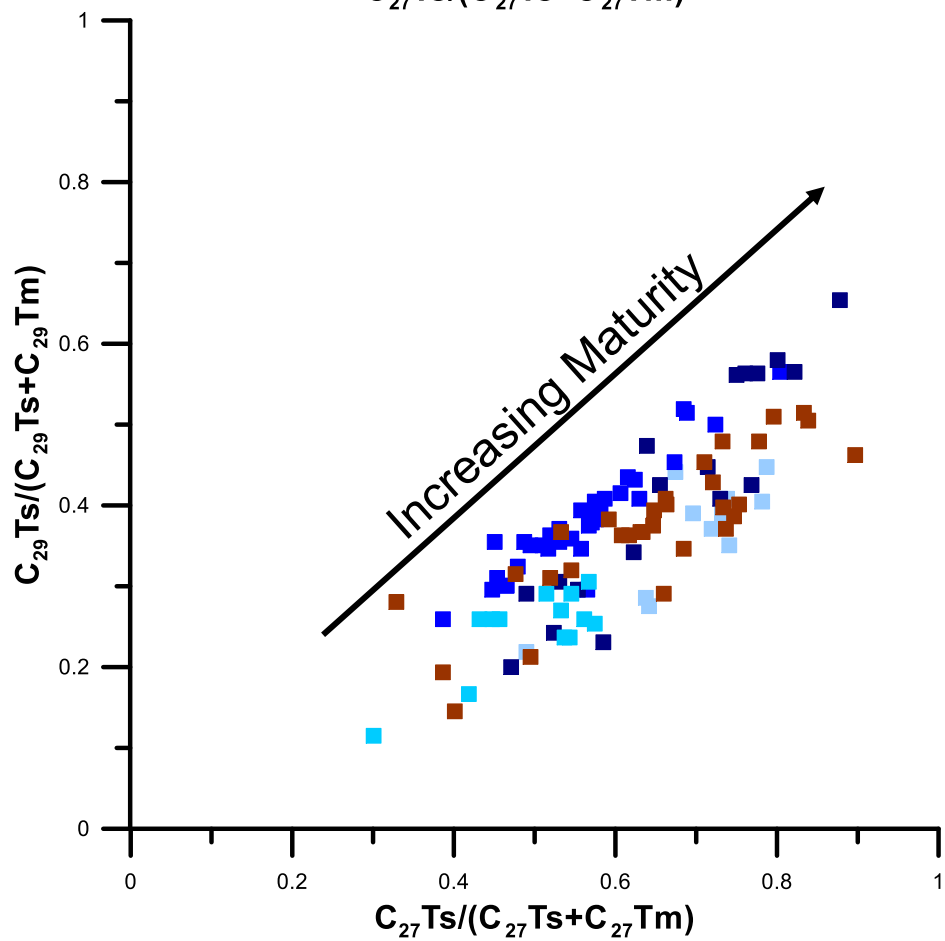
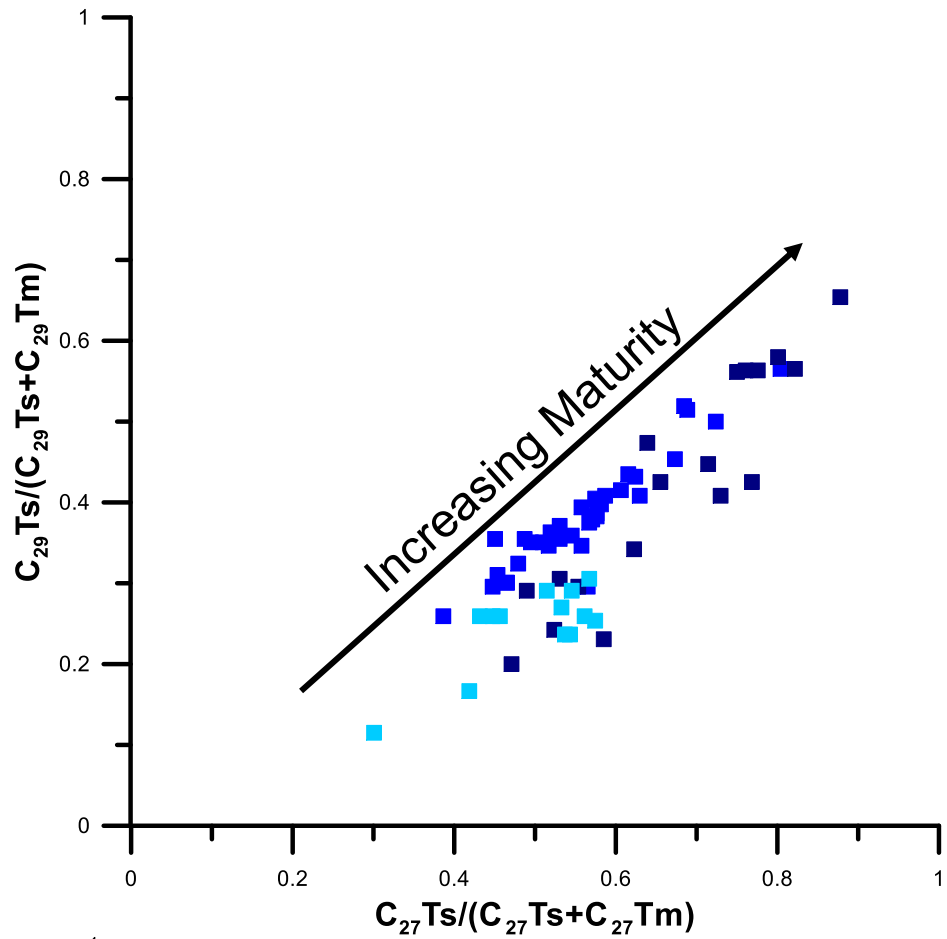
Family 4 oil chemistry also indicates the Lower Barnett source is high quality, characterized by good organic productivity. However, the Lower Barnett oils do not generally show strong indications of PZE conditions as do the Woodford oils. This indicates that while still a very good source, the Lower Barnett overall quality is not quite as strong as the Woodford, likely due to slightly poorer organic matter preservation.

Oil maturity can be assessed by various biomarker relationships. Two sets of maturity biomarkers are plotted on slide 82, upper plot for only the Woodford oils, Families 1-3, and on the bottom for all families. Maturity increases with increasing values of both parameters.

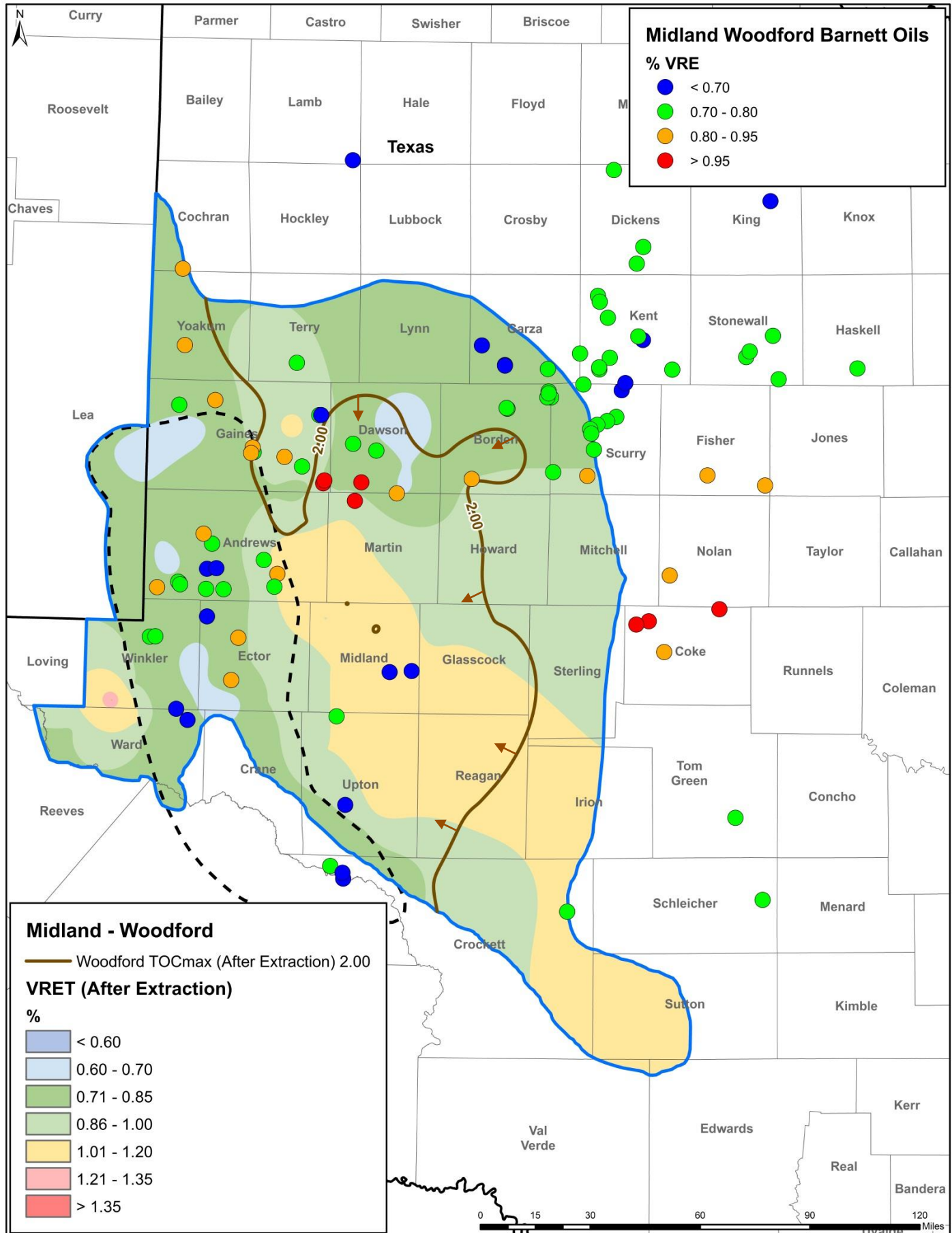
Family 1 and 2 oils are the most mature although as noted previously Family 2 is the most scattered. Each oil in the study has an interpreted %Roeq value based on a combination of those in slide 82 and several additional maturity parameters.

The maps on slides 83-84, 86-87 integrate the oil and sediment maturity data. On slide 83 the maturities (%Roeq) for oil Families 1-3 and 14 are plotted on the Woodford extracted %VRET map. Additionally, areas of Woodford greater than 2% TOCpd are also shown.

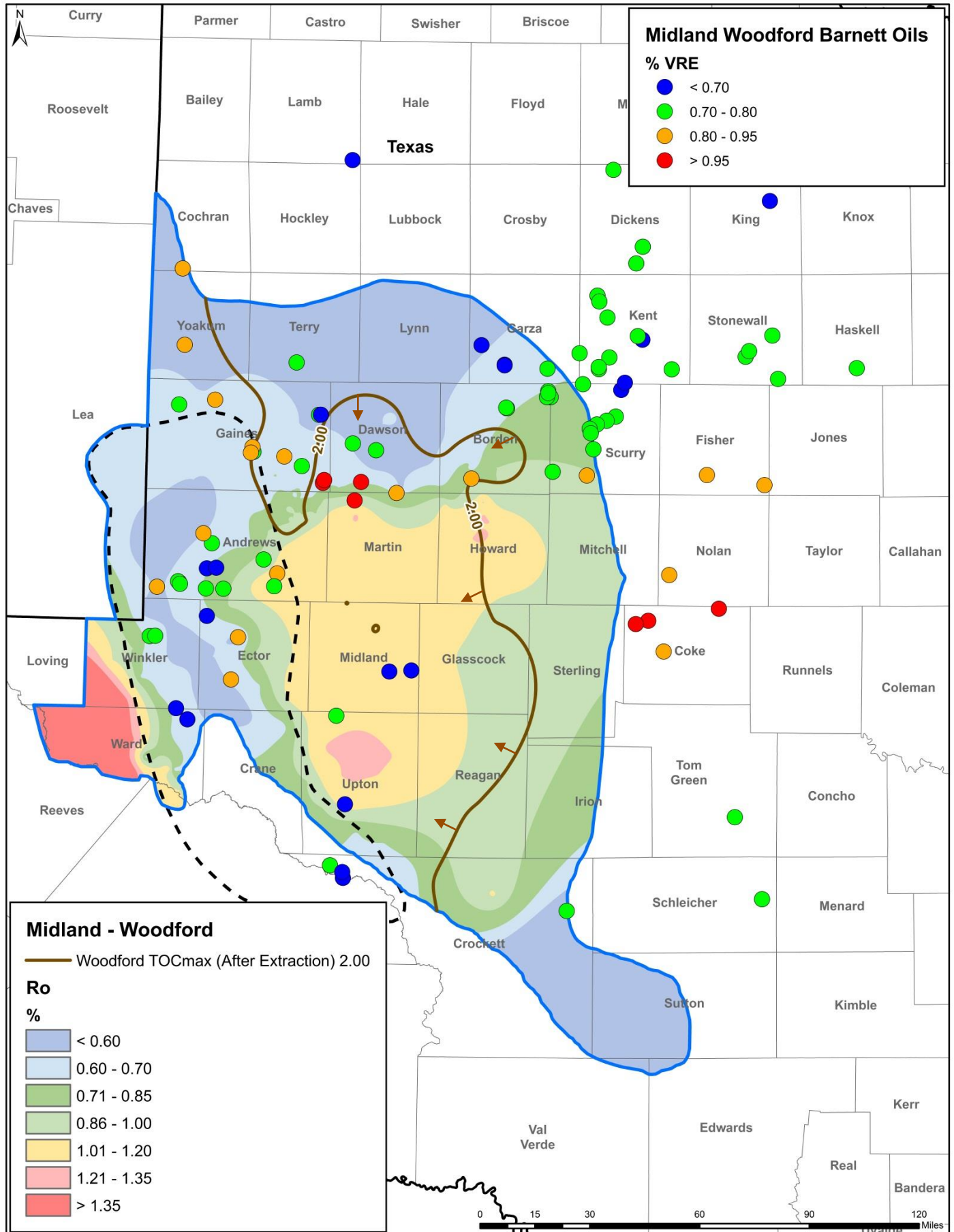
Biomarker Maturity Parameters



Woodford Sediment Maturity (%VRET), Richness, and Oil Families 1-3,14 Maturity



Woodford Sediment Maturity (Measured %Ro), Richness, and Oil Families 1-3,14 Maturity



Oils Interpretation

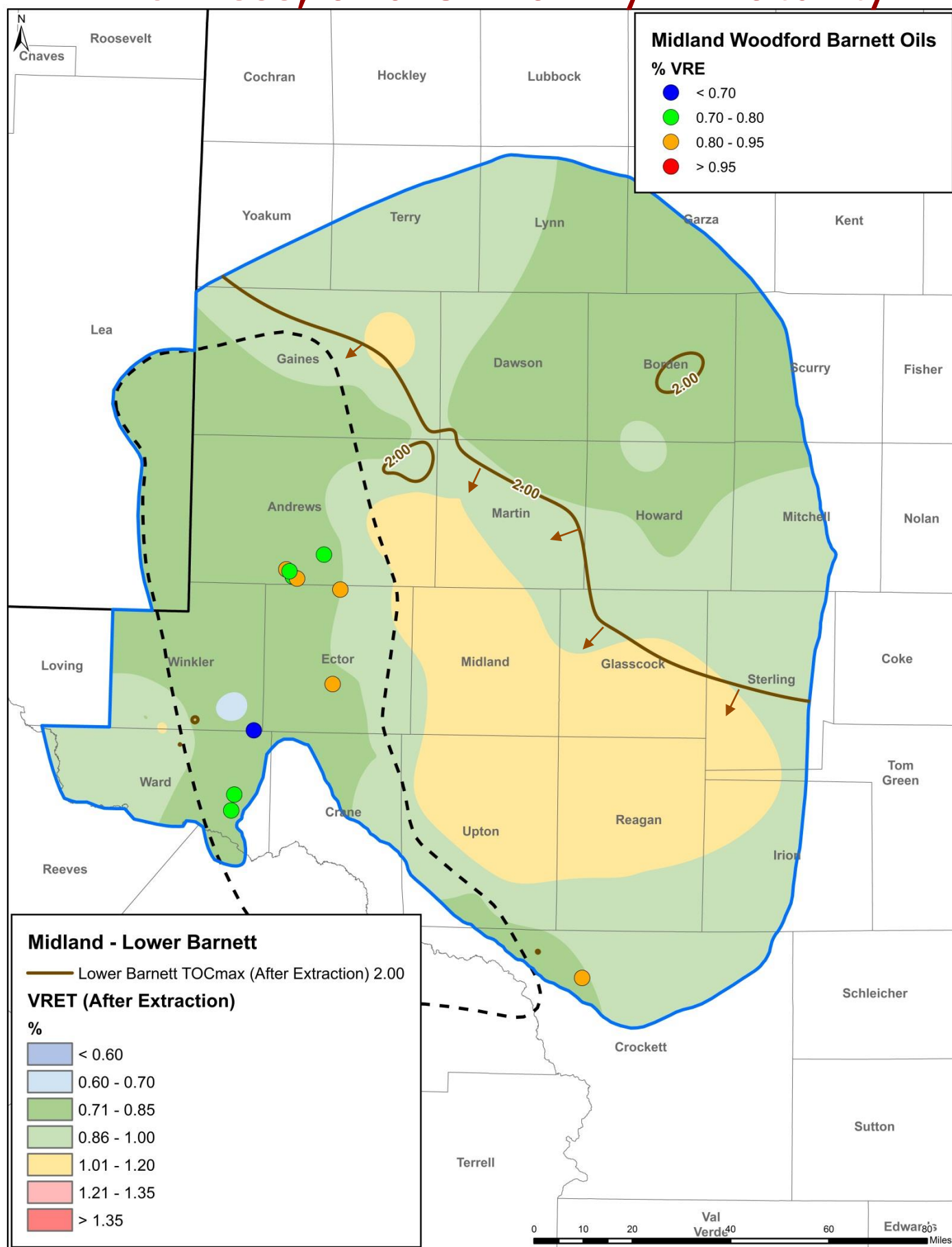
The measured Woodford %Ro map is shown in slide 84 with oil maturities and the extracted TOCpd 2% contour. The pros and cons of the two different maturity data types were discussed previously in the sediment section of this report.

These maps are very powerful as they suggest the areas of best Woodford source/reservoir unconventional potential with respect to maturity and organic richness as was highlighted in the sediment section of this report, but they also show areas where migration has moved more mature oils to areas where the Woodford is less mature. While the oils in this study are from conventional reservoirs, clearly the possibility exists for this same type of migration to occur within an unconventional Woodford play.

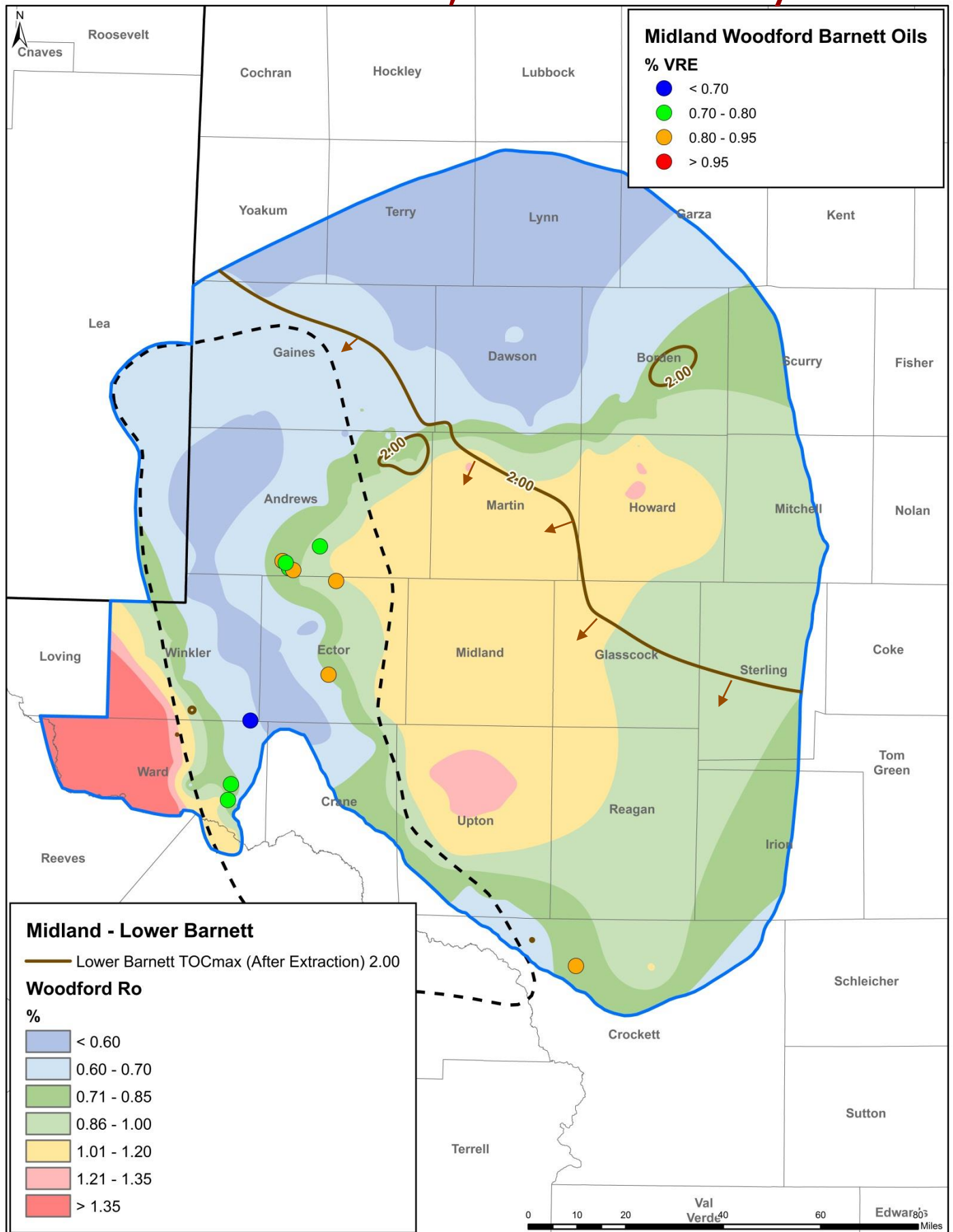
Similarly for the Lower Barnett, slide 86 shows Family 4 oil maturity plotted on the Lower Barnett extracted %VRET base with the extracted TOCpdmax 2% contour overlying. Slide 87 shows the same map as slide 86 except the base is the measured Woodford Ro map which is a reasonable maturity proxy for the Lower Barnett.

Again, similar to the Woodford, these maps also show migration of Barnett oils of higher maturity to areas of lower sediment maturity. This appears to be primarily occurring for both sources in the area of central-eastern Andrews/Ector counties.

Lower Barnett Sediment Maturity (%VRET), Richness, and Oil Family 4 Maturity



Woodford Sediment Maturity(Meas. %Ro), Lower Barnett Richness, and Oil Family 4 Maturity



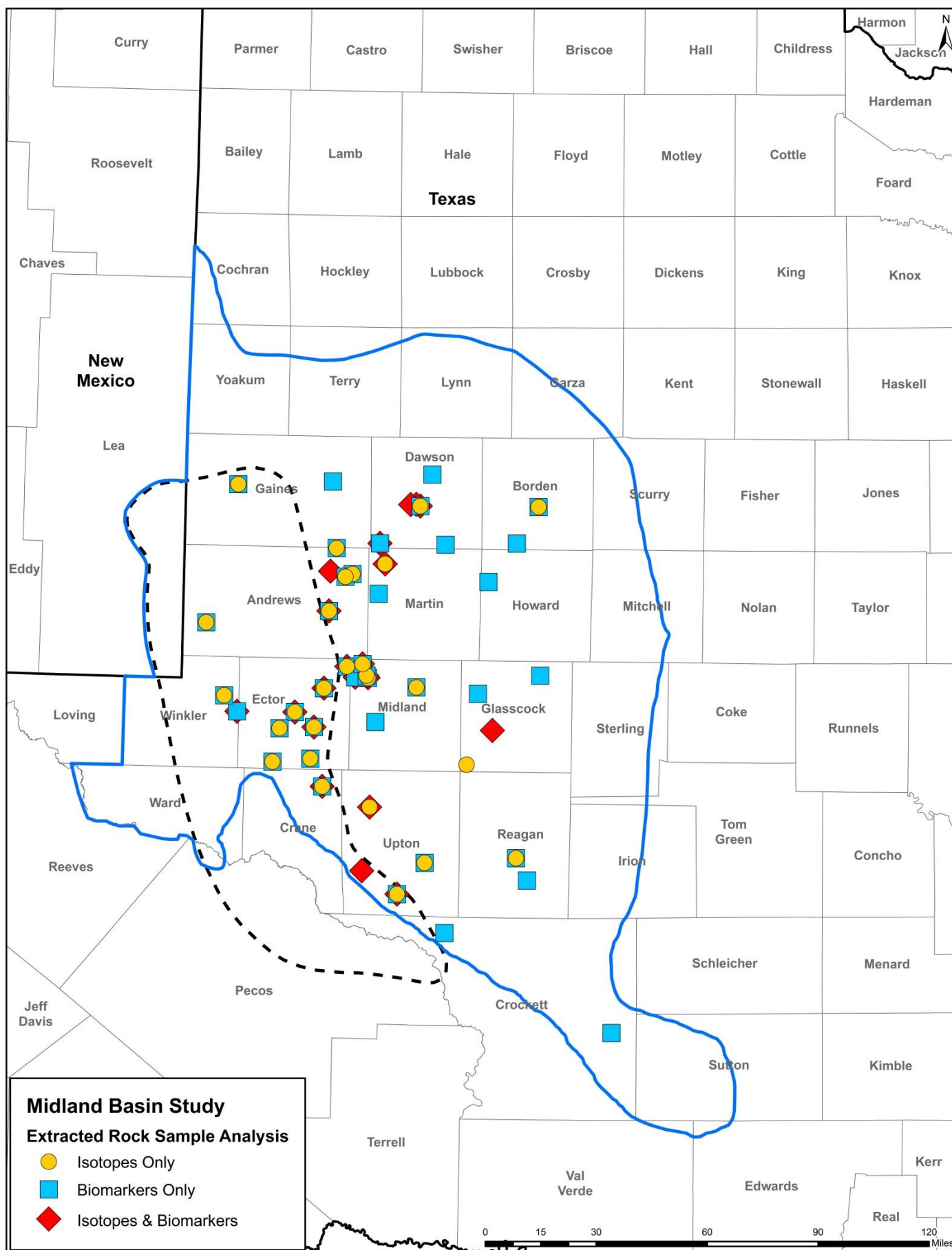
Extracts Interpretation

Over 300 cuttings sediment extracts from this study were analyzed for high resolution gas chromatography (gc), and based on their gc character, many were then separated via SARA analysis (saturates, aromatics resins/NSOs, and asphaltenes) by liquid chromatography. Subsequently, saturate and aromatic hydrocarbon fraction analyses by gc-ms and carbon isotopic compositional analysis were made. Slide 89 has a map indicating wells that had samples which were analyzed for isotopes and/or biomarkers. Those samples are also shown in the sediment interpretive Excel sheet.

The quality of the cuttings extracts caused by sample mishandling prior to their arrival at GeoMark's labs is an issue. Primarily the complicating issues are those of plasticizers and other storage related contamination. Over 100 samples were of no value due to this contamination and samples were not analyzed other than by gc.

In addition to issues of storage related contamination, a pervasive contaminant that precluded the use of many of the extracts was contribution from the Wolfcamp. Many of the analyzed Devonian and Mississippian source sediments contained the Permian age-specific biomarkers associated with generation from the Wolfcamp. It is not clear if this mixing occurred from drilling mud that went through Wolfcamp accumulations and then contaminated the deeper Barnett-Woodford sections or recycled muds, or if the Wolfcamp contamination was from Wolfcamp migrating fluids.

Cuttings Extract Samples for Isotope and/or Biomarker Analysis



Extracts Interpretation

Regardless, in most cases it made correlating the extract to one of the oil families and/or calculating maturity impossible or meaningless.

Of the over 200 extracted samples analyzed for biomarkers and isotopes, only 14 turned out to be useful for family and maturity association. A few of these 14 also showed some evidence of mixing and their interpretation is tentative. These samples and their results are summarized below and the samples are identified in the highlighted rows in the sediment interpretive Excel sheet.

Extracts Data Summary

Well	County	Formation	Ext. TOCpd	Ext Hlpd	Ext %VRET	Family	VREB	Comments
University 5-3	Andrews	Lower Barnett	1.87	145	0.87	Undiff Woodford	0.92	Some Ordovician contribution
University 5-3	Andrews	Mississippi Lime	2.07	160	0.89	2-Woodford	0.95	Some Ordovician contribution
Waddell W N 926	Crane	Lower Barnett	5.45	306	0.83	4-Lower Barnett	0.81	
Waddell W N 926	Crane	Woodford	2.03	242	0.85	2-Woodford	0.80	
Addis Frank V 1	Ector	Barnett	4.93	161	0.87	4-Barnett	1.03	
Addis Frank V 1	Ector	Woodford	1.67	139	0.89	4-Barnett	0.93	
JC Connell-Trahan 1	Ector	Woodford	3.25	524	0.72	3-Woodford	0.71	
JE Parker B 1	Ector	Mississippi Lime	1.05	141	0.83	3-Woodford	0.75	strong correlation w/Woodford sample
JE Parker B 1	Ector	Woodford	1.29	150	0.81	3-Woodford	0.77	strong correlation w/Miss.Lime sample
Superior-Ratliff 1	Ector	Woodford	1.55	83	0.99	4-Barnett	0.80	
WF Cowden 2	Ector	Lower Barnett	4.01	182	0.90	4-Barnett	0.93	
WF Cowden 2	Ector	Woodford	3.92	208	0.85	Undiff Woodford	0.93	
JP Rankin 1	Upton	Lower Barnett	5.03	79	1.08	14-Mix WF&WC	0.77	??; only saturate isotope
Rush 1	Upton	Lower Barnett	4.29	306	0.78	2/3-Woodford	0.82	

VREB=biomarker maturity

The counties that contain the usable extracts are Andrews, Ector, Crane, and Upton and the formations are mostly Woodford and Lower Barnett. These are counties where the source potential of these two formations are generally the best. The data in the table show the oil family assignments for the extracts, the extracted sediment richness and maturity (%VRET) as well as the calculated maturity (VREB) of the extracts.

Extracts Interpretation

In most cases, the oil family correlation of the extract matches the sediment age, but not always. For example the extract from the Mississippi Lime in an Ector county well correlates to the Woodford family 2 and is extremely similar (96%) to the Woodford extract in the same well.

Additionally, two Andrews County Lower Barnett and Mississippi Lime extracts correlate with the Woodford, but their gcs also have the unusual, but well documented, odd dominant n-paraffin character between C15-C20 that has been correlated with Ordovician *G.prisca* source.

Finally, comparison of VRET and VREB for some samples show a greater VREB maturity. When this occurs with a positive family correlation to the sediment age, it suggests migration within the formation from more mature down dip hydrocarbons.

Although the extracts from these sediments are generally of limited value, they do show evidence of vertical and lateral migration, indicating again that in-situ generation is not the only way to charge an unconventional play.

Conclusions

Conclusions

The Woodford and the Lower Barnett have good to excellent unconventional source potential on the basis of organic richness, kerogen quality, and thermal maturity.

The Upper Barnett and Mississippi Lime have little to no source potential and would likely not make good unconventional, self sourcing targets.

The areas of highest potential for the Lower Barnett and Woodford are:

- Central to eastern Andrews/Ector counties possibly extending south and north into Crane and Gaines counties, respectively.
- Martin/Midland/Upton counties, however, thermal maturity and therefore hydrocarbon phase is a consideration as maturity is more elevated than on the CBP.
- Ward/Winkler county areas on the CBP and to the west towards the Delaware Basin, however, the area from immature/low mature to overmature may be quite limited due to the rate that burial depth increases east to west. Maturity must be carefully evaluated.

The possibility of immature/low mature targets on the CBP and in the northern Midland may be improved by migration of more mature hydrocarbons from the deeper more mature basin.